

Faculty of Applied Medical Sciences

Nursing Department



PHARMACOLOGY FOR NURSES

Dr. Mohammed Tabash

2023-2024

Key definitions

- **Pharmacology** is a Latin word meaning drug knowledge. **Pharmaco-** means drug, **-ology** means science or knowledge.
- **Pharmacology** is the study of chemicals (drugs) effects on living tissues and how those chemicals help diagnose, treat, cure, and prevent disease or correct the pathophysiology of living tissues.
- **Pharmacopeia:** is a book containing formulas, which are standard for preparation and administration of drug. If a drug is included in this book it has met the standards of quality, purity, and strength.
- Most countries publish an authoritative lists of drugs called a pharmacopoeia such as
    - British Pharmacopoeia (BP)
    - United States Pharmacopeia (USP)
- **A drug:** is a substance or chemical capable of altering a biochemical or physiological processes in the body; these responses may be desirable (therapeutic) or undesirable (adverse).
- **A drug:** is a substance intended for use in the diagnosis, cure, mitigation, treatment, or prevention of disease.

Nursing Responsibilities:

- Registered nurses, midwives are accountable for their own acts.

The nurse must:

- Know the therapeutic uses of the medicine to be administered, its normal dosage, side-effects, precautions and contraindications
- Be certain of the identity of the patient to whom the medicine is to be administered
- Check that the prescription, or the label on the medicine dispensed by a pharmacist, is clearly written and unambiguous
- Registered nurses, midwives are accountable for their own acts.

Sources of drugs

Traditionally, drugs were derived from *natural* sources, such as:

- Plants
- Animals
- Minerals.

The earliest drug concoctions from plants used everything: the leaves, roots, bulb, stem, seeds, buds, and blossoms. As a result, harmful substances often found their way into the mixture. As the understanding of plants as drug sources became more sophisticated, researchers sought to isolate and intensify ***active components*** while avoiding harmful ones.

The active components of plants vary in character and effect:

- *Alkaloids,* the most active component in plants, react with acids to form a salt that's able to dissolve more readily in body fluids.
- The names of alkaloids and their salts usually end in "-ine"; examples include atropine, caffeine, and nicotine.
- *Glycosides* are naturally occurring active components that are found in plants and have both beneficial and toxic effects. They usually have names that end in "-in," such as digoxin.
- *Gums* give products the ability to attract and hold water. Examples include seaweed extractions and seeds with starch.
- *Resins,* of which the chief source is pine tree sap, commonly act as local irritants or as laxatives and caustic agents.
- *Oils,* thick and sometimes greasy liquids, are classified as volatile or fixed. Examples of volatile oils, which readily evaporate, include peppermint, spearmint, and juniper. Fixed oils, which aren't easily evaporated, include castor oil and olive oil.

The active components from animals

The body fluids or glands of animals are also natural drug sources.

The drugs obtained from animal sources include:

- *hormones,* such as insulin
- *oils* and *fats* (usually fixed), such as cod-liver oil

- *enzymes,* which are produced by living cells and act as catalysts, such as pancreatin and pepsin
- *vaccines,* which are suspensions of killed, modified, or attenuated microorganisms.

Minerals

Metallic and nonmetallic minerals provide various inorganic materials not available from plants or animals. Mineral sources are used as they occur in nature or they're combined with other ingredients. Examples of drugs that contain minerals are iron, iodine, and Epsom salts.

Today, most drugs are produced in laboratories. Examples of such drugs include thyroid hormone (from natural sources) and cimetidine (from synthetic sources).

Classification of drugs according to prescription

a) **Controlled drug:** are categorized based on their therapeutic use and potential for abuse. Nurses are required to keep controlled substances locked in a secure room or cabinet, administering them only to patients with valid prescriptions or physician's orders. Nurses must maintain accurate records of each dose given and the amount of each controlled substance on hand and must report any discrepancies to the proper authorities.

b) General sale list (OTC):

Medicines sold directly to a consumer without a prescription from a healthcare professional (analgesics).

Dosage forms of drugs:

A. Oral preparations

1. Liquids

a. Emulsions

Mixture of two immiscible liquids (oil and water) by means of emulsifying agent (e.g. Liquid paraffin).

b. Oral suspensions

The dispersion through a liquid of a solid in finely divided particles of a size, it tend to settle slowly and should be **chaked well** before use to provide uniform distribution of the drug.

c. Elixir

Sweetened, flavored hydro alcoholic solution containing drug

d. Syrup

Concentrated solution of sugar containing flavoring, coloring and therapeutically active substances.

2. Solids

a. Tablets

Compressed powdered dosage form into hard disks or cylinders.in addition to primary drug, it contain binders (adhesive to allow powder to stick together) and disintegrates (to promote dissolution).

b. Capsules

Shells of gelatin containing drug for oral use medication in powder or oil form. It is colored to avoid mistakes in product identification.

Notes:

- Capsules and coated tablets are used for drugs that have unpleasant taste.
- Coated gelatin capsules and tablets can resist the action of gastric juice and will not disintegrate until they reach the alkaline media of the intestine.
- Sustained release dosage are designed to release a drug at a predetermined rate in order to maintain a constant drug concentration for a specific period of time.

B. Rectal preparations

a. Suppository: solid preparations for insertion into rectum e.g. aminophylline supp.
b. Enema: liquid preparations for insertion into rectum.

C. Parenteral preparations

Sterile solution or suspension containing drug that may be dispensed in:

a. Ampules: containing single dose
b. Vials: rubber capped bottles containing a number of doses.

D. Inhalational preparations

a. Gases: administration thru special devices.
b. Volatile liquid
c. Steam inhalation
d. Aerosols: released into respiratory passages in the form of a fine mist of liquid droplets (bronchodilation).

E. Topical preparations

a. Ointments: semi solid preparations for cutaneous or mucosal applications.
b. Cream: are homogeneous, semi-solid preparations consisting of opaque emulsion systems.
c. Pastes: are homogeneous, semi-solid preparations containing high concentrations of insoluble powdered substances (usually not less than 20%) dispersed in a suitable base.
d. Ophthalmic ointment: sterile ophthalmic ointment e.g. synthomycin
e. Lotion: aqueous suspension intended for application to skin without rubbing e.g. calamine lotion.
f. Eye, Ear and Nose Drops: aqueous solution for local applications
g. Mouth washes and gargles
h. Powders
i. Vaginal douches: aqueous solution with cleansing or antiseptic properties

F. Others

- **Disposable Syringes:** containing a single dose of drug preparations are widely used.
- **Large volume intravenous solutions:** they are various amount of solution contained in glass, glass, flexible plastic or semi rigid plastic usually of 250, 500, 1000 ml e.g. 0.9% sodium chloride.
- **Intermittent IV solution:** These solutions are similar to the I.V. solutions except that they come in smaller volumes.

Dosage: it refers to the determination of size, frequency and number of doses (quantity to be administered in one time)

Loading / priming dose: A dose that quickly raises the concentration of drug in plasma to target concentration (that will produce the desired therapeutic effect).

Maintenance dose or dose rate: a dose that maintain a steady state of drug in body (dose of drug that replace the drug eliminated since proceeding dose).

Maximal tolerated dose: largest dose of a drug that can be taken safely.

Fatal dose: a dose that produces death.

What Happens After a Drug Is Administered?

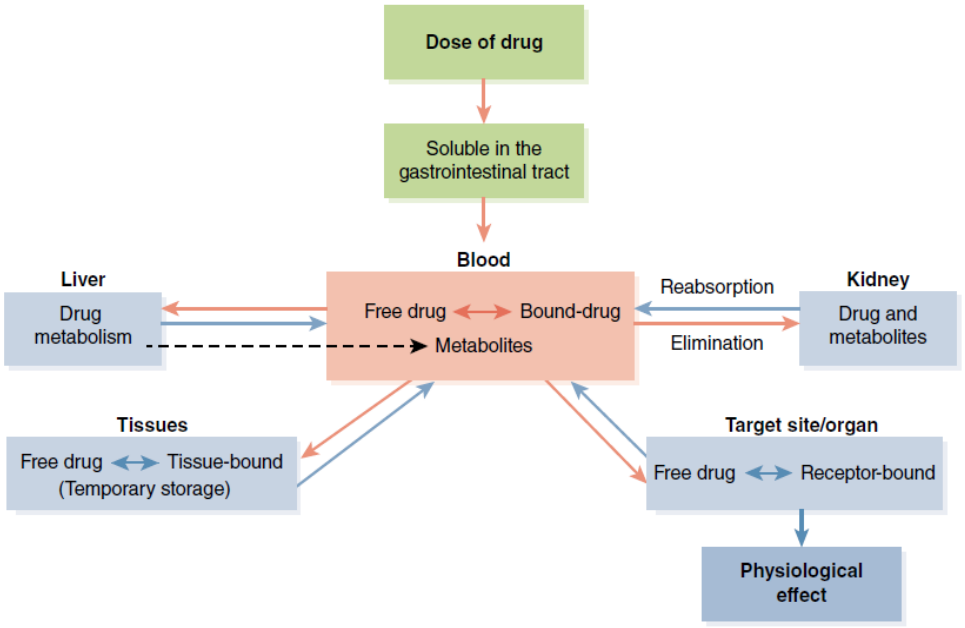
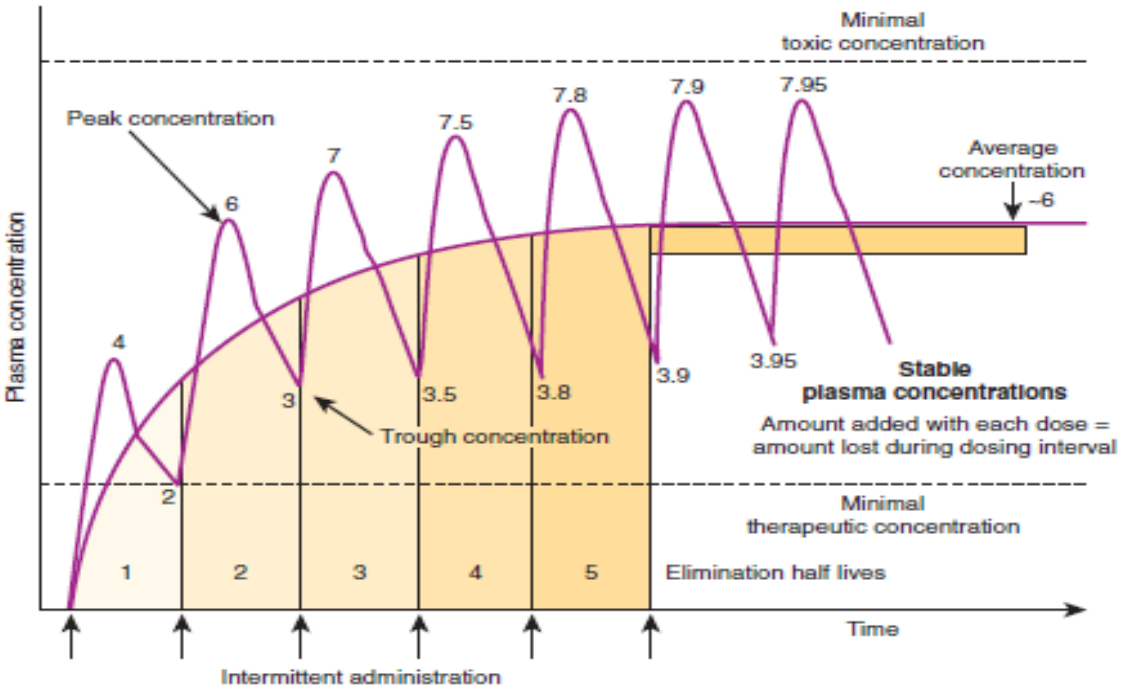


FIGURE 2-1 Pathways of a drug movement within the body (following oral administration). Note the number of sites/variables with which the drug can interact before producing a physiological effect.

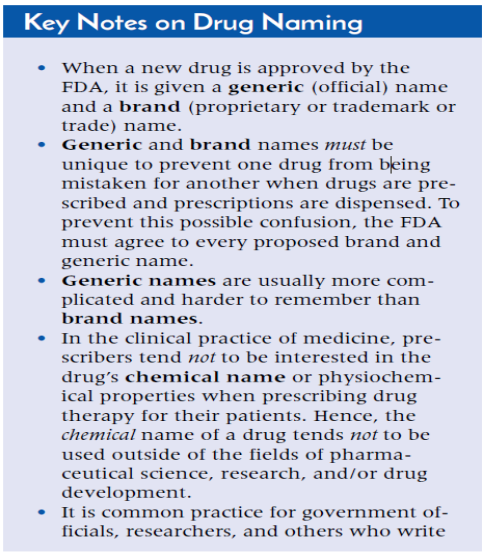
Courtesy of Diane Pacitti. © 2017



Naming and classifying drugs.

Drugs have a specific kind of nomenclature—that is, a drug can go by three different names:

- **The chemical name** is a scientific name that precisely describes the drug's atomic and molecular structure. E.g based upon chemical structure, e.g. 3- (10,11- dihydro - 5H - dibenz - azepin - 5 - yl)
- **The generic**, or nonproprietary, name is an abbreviation of the chemical name. official name that is used in pharmacopoeias, e.g. Imipramine.
- **The trade name** (also known as the brand name or proprietary name) is selected by the drug company selling the product. Trade names are protected by copyright. The symbol ® after a trade name indicates that the name is registered by and restricted to the drug manufacturer. market name that is given by pharmaceutical company, e.g. Tofranil



To avoid confusion, it's best to use a drug's generic name because any one drug can have a number of trade names. Drugs that share similar characteristics are grouped together as a *pharmacologic class* (or family). Beta-adrenergic blockers are an example of a pharmacologic class. A second type of drug grouping is the *therapeutic class*, which categorizes drugs by therapeutic use. Antihypertensive are an example of a therapeutic class.

Pharmacokinetics and pharmacodynamics are two fundamental aspects of pharmacology that deal with how drugs interact with the body and produce their effects.

I. Pharmacokinetics

The term *kinetics* refers to movement. Pharmacokinetics deals with a drug's actions as it moves through the body. Therefore, pharmacokinetics discusses how a drug is:

- Absorbed (taken into the body)
- Distributed (moved into various tissues)
- Metabolized (changed into a form that can be excreted)
- Excreted (removed from the body).

This branch of pharmacology is also concerned with a drug's onset of action, peak concentration level, and duration of action.

1) Drug Absorption

It refers to passage of a drug from its site of administration into blood stream and distributed in the body, it comes in contact with numerous membranes. Drugs pass some membranes but not others. On a cellular level, drugs are absorbed by several means— primarily through active or passive transport. Absorption process varies according to the route of administration, dosage form, and dose of the drug.

Factor affecting drug absorption:

a. Drug factors:

- **Degree of ionization**, inversely proportional to drug absorption, drugs that are acidic (aspirin) become relatively unionized in acidic environment such as stomach and can readily diffuse a cross membrane into circulation. In contrast, basic drug tend to ionized in the stomach acid and not absorbed through the gastric mucosa. The reverse occurs when the drugs are in alkaline media.
- **Solubility** (lipophilicity), the more soluble drug, the more readily to be absorbed. Parenterally administration drugs prepared in oily vehicle will be more slowly absorbed than drugs dissolved in water or isotonic sodium chloride. Drugs that are insoluble in water or lipids can not be absorbed
- **Formulation** (solution vs. tablet). Drug absorption may be altered by factors unrelated to the chemistry of the drug. For example, particle size, salt form, crystal polymorphism, enteric coatings and the presence of excipients (such as binders and dispersing agents) can influence the ease of dissolution and, therefore, alter the rate of absorption. Solution are better absorbed than tablet. The highly concentrated administered drug are tend to be more rapidly absorbed than drugs administered in low concentration.

b. Patient factors

- **The route of administration.** (oral, sublingual, rectal, inhalation) For example, the presence of food in the GI tract can dilute and decrease absorption of drug**.**
- **Blood flow**, rich blood supply (sublingual) enhances absorption whereas vascular sites (subcutaneous) delays it. For example, patient in shock may not responded to intramuscular

(IM) administration of drugs because of poor peripheral circulation while intravenous administration is placed directly into circulation system.

- **Total surface area available for absorption**: Because the intestine has surface rich in microvilli, it has a surface area about 1000-fold that of the stomach; thus, absorption of the drug across the intestine is more efficient. **NB**: the more extensive surface area, the greater absorption and more rapid effect of the drug for example capsules are absorbed more rapidly than tablet because powder inside capsules provide a large surface area than compressed tablet.
- **Contact time:** if drug moves quickly in GIT drug will not be well absorbed if transport of drug from stomach to intestine delayed absorption will delayed. If drug taken with meal the food will dilute drug and slow gastric emptying causing slow absorption.

II. Drug Distribution

Drug distribution is the process by which the drug is delivered to the tissues and fluids of the body. Distribution of an absorbed drug within the body depends on several factors, including:

- Blood Flow
- Solubility
- Protein Binding.

The following factors influence drug distribution:

1. **Membrane permeability:**
    - In order to enter an organ, a drug must permeate all membranes that separate the organ from the site of drug administration. e.g. benzodiazepines, which are very lipophilic, readily cross the gut wall, capillary wall, and blood brain barrier. Because of this, they distribute to the brain rapidly and are useful for treating anxiety and convulsions.
    - Some antibiotics are capable of passing from the gut into the blood stream, but cannot pass into the brain. These drugs cannot be used to treat infections in the brain (meningitis and encephalitis). The instillation of the drug intrathecally to bypass the BBB will provide direct effect against bacterial brain infection.
    - Poor passage of some anticancer agents across the blood-brain barrier and the blood-testes barrier results in relatively high rates of brain or testicular recurrences of some tumors.

- The blood placenta barrier prevents fetal exposure to some drugs but allows passage of others. Tissue enzymes in the placenta have the ability to metabolize some agents (e.g. catecholamine) by inactivating them as they pass from maternal circulation to the embryo.

Drug effects occur in the fetus:

- In the first trimester: One type of drug may induce aberrant development of organs & systems during the formation of these structures. This is known as a teratogenic drug, which is defined as "An agent that causes physical defects in a developing embryo".
- The second type of drug affects the second half of pregnancy as well as delivery, when respiratory depression may occur in the newborn because of its inability to bio transform & excrete the drug given to the mother.

2. Plasma protein binding:

Some drugs can bind nonspecifically and reversibly to various plasma protein e.g. albumin and globulin.

- Bound and free drug reach an equilibrium.
- Only the free drug exerts a biologic effect
- Bound drug stays in vascular apace and is not metabolized or eliminating

3. Depot storage:

Lipophilic drugs, such as the sedative thiopental, accumulate in fat. These agents are released slowly from the fat stores. Thus, an obese person might be sedated for a greater period of time than a lean person to whom the same dose of thiopental had been administered. Calcium binding drugs, such as the antibiotic tetracycline, accumulate in bone and teeth.

4. Blood flow

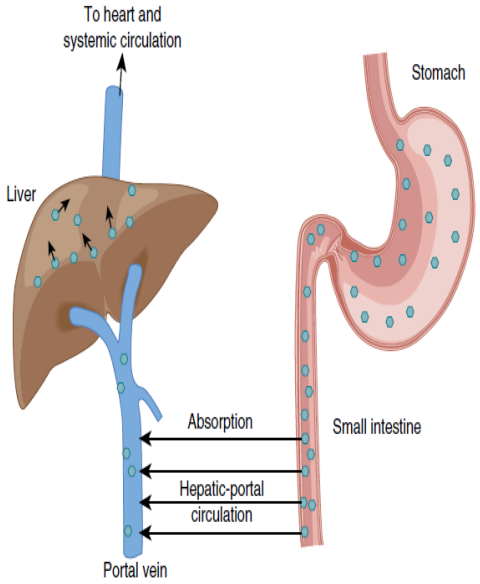
Most of drugs distributed initially to organs that have rich blood supply as the heart, liver and kidney. Delivery of the drug to the viscera, skin, bone and a dipose tissue is slower.

5. Tissue affinity:

Some tissue is localized in specific tissues which possess specific drug receptors e.g. iodine in thyroid gland and chloroquine in liver.

NB: hypoalbuminemia, as in liver disease, burn and malnutrition may affect absorption and distribution of drugs.

III. Biotransformation (Drug Metabolism)



Drug metabolism, or biotransformation, refers to the body's ability to change a drug from its dosage form to a more water-soluble form that can then be excreted. Drugs can be metabolized in several ways:

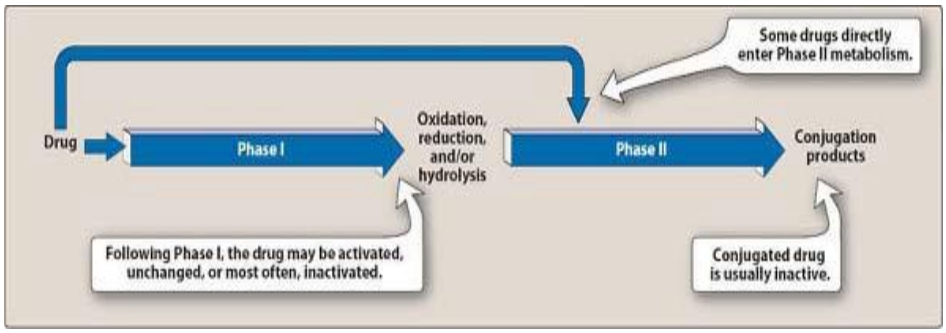
- Most commonly, a drug is metabolized into inactive metabolites (products of metabolism), which are then excreted.
- Some drugs can be converted to active metabolites, meaning they're capable of exerting their own pharmacologic action. These metabolites may undergo further metabolism or may be excreted from the body unchanged.
- Other drugs can be administered as inactive drugs, called *prodrugs,* and don't become active until they're metabolized.
- Following an oral administration, many drugs are absorbed intact from small intestine and transported via portal system to liver where they undergo extensive metabolism referred as ***first pass effect.***
- Most drugs are metabolized by enzymes in the liver; however, metabolism can also occur in the plasma, kidneys, and membranes of the intestines. Some drugs inhibit or compete for enzyme metabolism, which can cause the accumulation of drugs when they're given together. This accumulation increases the potential for an adverse reaction or drug toxicity.

Phases of biotransformation:

A. phase I reactions:

- It alter chemical reactivity and increases aqueous solubility of drugs.
- It includes oxidation (cytochrome P450), reduction and hydrologists

Metabolic reactions can transform



B. Phase II:

This phase consists of conjugation reactions.

- If the metabolite from Phase I metabolism is sufficiently polar, it can be excreted by the kidneys. However, many Phase I metabolites are too lipophilic to be retained in the kidney tubules.
- A subsequent conjugation reaction with an endogenous substrate, such as glucuronic acid, results in polar, usually more water-soluble compounds that are most often therapeutically inactive. The highly polar drug conjugates may then be excreted by the kidney or bile.

Factor affecting metabolism

a. Genetics : it may alter enzyme level
b. Chemical properties of drug: certain drugs may stimulate or inhibit the metabolism of other drug
c. Rout of administration: oral route can result in extensive first pass effect.
d. Diet: starvation depletes glycine and alter glycine conjugation.
e. Dosage: toxic dose can deplete enzyme.
f. Age: liver cannot detoxify drugs as well in neonates than adults.
g. Disease: liver and kidney disease decrease drug metabolism.

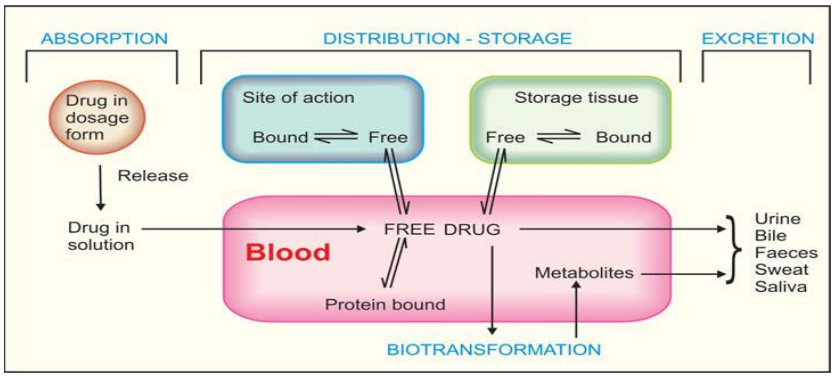
IV. Drug Elimination (Excretion)

Drug excretion refers to the elimination of drugs from the body. Most drugs are excreted by the kidneys and leave the body through urine. Drugs can also be excreted through the lungs, exocrine glands (sweat, salivary, or mammary glands), skin, and intestinal tract.

Routes of elimination:

1. Kidney: excretion of drugs and their metabolites into urine.
2. Liver: it can secrete drugs or their metabolites into bile (e.g. opioids).

3. GIT: iodides, mercury in saliva
4. Lung: gaseous and volatile general anesthetic are excreted in expired air.
5. Other routes: sweat, tears and breast milk.



Factors that can affect the excretion of drugs from the body:

- Kidney function: The kidneys are the major organ responsible for drug excretion. Reduced kidney function, as seen in kidney disease or old age, can significantly decrease the rate of drug excretion.
- Urine pH : The pH of urine impacts excretion of acidic and basic drugs. Acidic urine speeds up excretion of basic drugs, while alkaline urine does the opposite.
- Drug interactions : Some drugs can impact the excretion of other drugs by interfering with active transporters in the kidneys. For example, probenecid blocks excretion of penicillin.
- Plasma protein binding : Highly plasma protein bound drugs are not readily filtered by the kidneys. Therefore, the free unbound fraction affects excretion.
- Drug solubility : Drugs that are more water soluble tend to be more readily excreted in urine. Lipophilic drugs are more extensively reabsorbed.
- Route of administration : Drugs given orally tend to undergo more first-pass metabolism and have reduced bioavailability for excretion compared to intravenous drugs.
- Tubular secretion : Some drugs are secreted actively via transporters in the renal tubules. Competition or inhibition of these transporters can reduce excretion.
- Blood flow to kidneys - Reduced cardiac output or conditions like congestive heart failure can decrease kidney blood flow and drug excretion.

Onset, peak, and duration

In addition to absorption, distribution, metabolism, and excretion, three other factors play important roles in a drug's pharmacokinetics:

- Onset of action
- Peak concentration
- Duration of action.

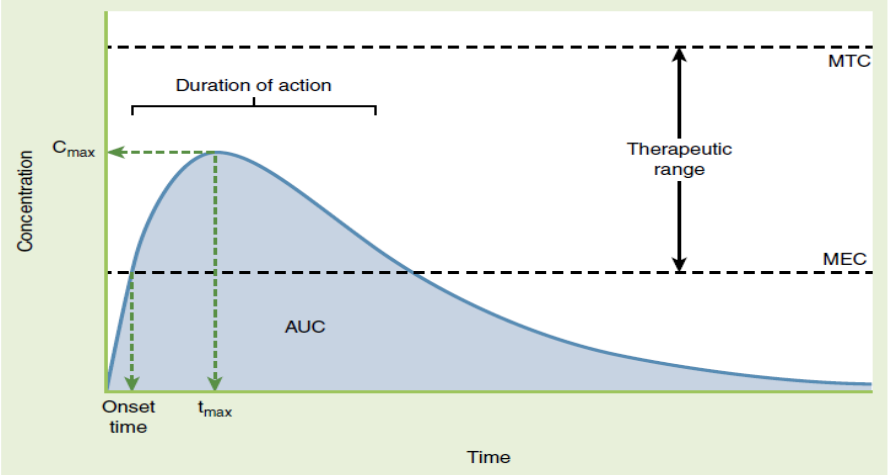


FIGURE 2-16 "Drug activity profile" (also known as the *concentration-response graph*). The drug-plasma level time curve indicates a number of pharmacodynamic parameters: the onset of action, duration of action, the therapeutic index, the minimum toxic concentration (MTC), and minimum effective concentration (MEC) for a given drug.

Onset of action: refers to the time interval that starts when the drug is administered and ends when the therapeutic effect actually begins. Rate of onset varies depending on the route of administration and other pharmacokinetic properties. As the body absorbs more drug, blood concentration levels rise. The peak concentration level is reached when the absorption rate equals the elimination rate.

The duration of action is the length of time the drug produces its therapeutic effect.

Half-life: it is the time required to change the amount of drug in body by one-half during elimination.

Note: if there is renal disease the risk of drugs accumulation and toxicity is increased.

Factor increasing drug half-life:

1. Decrease renal plasma flow e.g. cardiogenic shock, cardiac failure, hemorrhage.
2. Addition of second drug that displaces first from albumin, increasing Vd.
3. Decrease excretion ratio e.g. renal disease.
4. Decrease metabolism e.g. when another drug inhibits its biotransformation.

Steady state concentration:

A steady state plasma concentration of drug occurs when the rate of drug elimination is equal to the rate of administration.

Bioavailability:

Bioavailability is expressed as the fraction of administered drug that gains access to the systemic circulation in a chemically unchanged form. For example, if 100 mg of a drug are administered orally and 70 mg of this drug are absorbed unchanged, the bioavailability is 0.7 or seventy percent.

Factors affecting bioavailability:

1- The drug itself.
2- The metabolism of the patient.
3- The rate at which the drug is liberated (released) from its dosage form or from storage in the body.

Many drugs bind to serum protein (albumin) from which they released gradually, others stored at specific site as bone (tetracycline).

Drug isn't to be bioavailable if:

1- Bound to protein or to any other substance that makes the drug permanently or temporarily inactive.
2- Not released from its dosage form or site of administration.
3- Partially or totally degraded.

Pharmacodynamics

Pharmacodynamics is the study of the drug mechanisms that produce biochemical or physiologic changes in the body. The interaction at the cellular level between a drug and cellular components, such as the complex proteins that make up the cell membrane, enzymes, or target receptors, represents *drug action*. The response resulting from this drug action is called the *drug effect.*

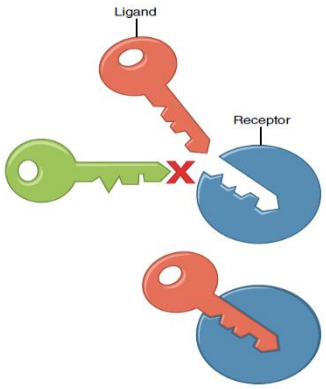
How do drugs work?

1) **Receptors:** It refers to specific drug binding sites in a cell or on its surface, which mediate the action of drug.

Nature of drug receptors

1. Regulatory proteins, which mediate the actions of endogenous chemical signals e.g. neurotransmitters, hormone.
2. Enzyme e.g. dihydrofolate reductase.
3. Transport protein e.g. Na^+ / k^+ ATPase
4. Structural proteins, e.g. tubulin.

Receptor ligands:



1. Agonist: a drug that produce some of the effects of endogenous compounds when it interacts with receptor, e.g. isoproterenol at β adrenoceptors. Drug must have affinity and intrinsic activity to be agonist.
2. Antagonist: a drug that binds to receptor without activating it, thereby blocking endogenous agonist from exerting its effect e.g. propranolol at β adrenoceptors. Drug must have affinity but no intrinsic activity to be antagonist.

   Antagonists can be competitive or noncompetitive:

   - A competitive antagonist competes with the agonist for receptor sites. Because this type of drug binds reversibly to the receptor site, administering large doses of an agonist can overcome the antagonist's effects.
   - A noncompetitive antagonist binds to receptor sites and blocks the effects of the agonist. Administering large doses of the agonist can't reverse its action.
3. Partial agonist: a drug that binds to receptor blocking access of natural agonist and also capable of a low degree of activation (ie, affinity + some intrinsic activity) e.g pindolol and oxprenolol at β adrenoceptors.
4. Inverse agonist: a drug that on binding to receptor produces effects which are specifically opposite to those of agonist, e.g. β – carboline at benzodiazepine receptors.

   NB: If a drug acts on a variety of receptors, it's said to be nonselective and can cause multiple and widespread effects.

2) Antimetabolites:

- These drugs closely resemble substances which are used by the cells for nutrition and when absorbed, the cells cannot use them and so fail to multiply e.g. Sulfonamides are used to stop the multiplication of bacteria.

- They are very similar in structure to para-aminobenzoic acid and certain bacteria can't distinguish between them, and absorb the sulfonamide so multiplication is stopped.

3) Enzyme Inhibitors:

- Enzymes are substances that speed up many chemical processes within the body.
- Some of these enzyme-activated processes are concerned with the transport of chemicals in and out of the cells. Certain drugs have the property of inhibiting their action and thus interfere with some of these processes.
- Diuretics are a good example as normally salt and water is transported out of the renal tubule back into the body, but this action requires enzymes and if they are inhibited by a diuretic, salt and water are not reabsorbed and pass out of the kidney with a resulting diuresis.

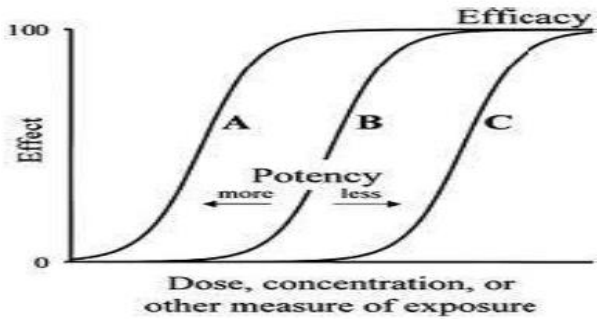
4) Action on cell membranes:

- The function of nerves and muscles depends on ions passing across the membranes surrounding these cells.
- Certain drugs interfere with movement of these ions and thus prevent nerve or muscle function as demonstrated by local anesthetics which block impulses passing up a sensory nerve.

5) Cytotoxic Effect:

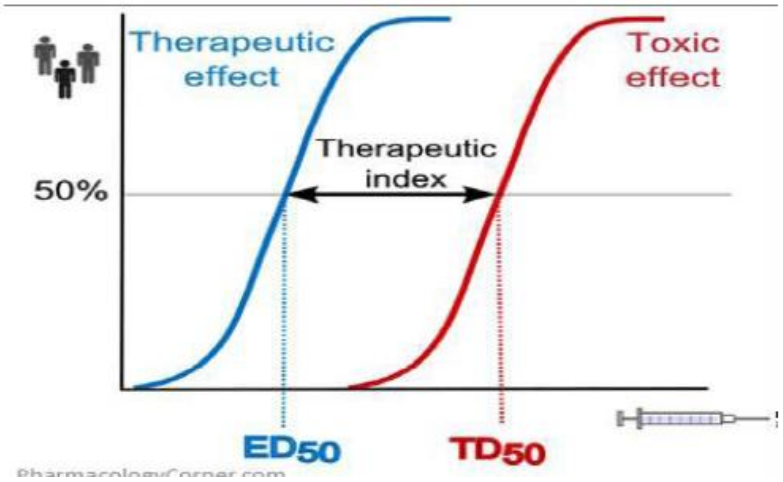
- Drugs may be used to kill bacteria or malignant cells without undue damage to the patient's cells. The way this is brought about varies between drugs.

Drug potency refers to the relative amount of a drug required to produce a desired response. Drug potency is also used to compare two drugs. If drug X produces the same response as drug Y but at a lower dose, then drug X is more potent than drug Y.



a dose-response curve is used to graphically represent the relationship between the dose of a drug and the response it produces. (See *Dose-response curve.*)

On the dose-response curve, a low dose usually corresponds with a low response. At a low dose, an increase in dose produces only a slight increase in response. With further increases in dose, there's a marked rise in drug response. After a certain point, an increase in dose yields little or no increase in response. At this point, the drug is said to have reached *maximum effectiveness.*



As the dose of drug administered is increased, pharmacological effect will also increase and at certain dose, the resulting effect will reach a maximum level.

Most drugs produce multiple effects. The relationship between a drug's desired therapeutic effects and its adverse effects is called the drug's *therapeutic index.* It's also referred to as its *margin of safety*.

The therapeutic index usually measures the difference between:

- An effective dose for 50% of the patients treated
- The minimal dose at which adverse reactions occur.

Pharmacotherapeutics is the use of drugs to treat disease. When choosing a drug to treat a particular condition, health care providers consider the drug's effectiveness as well as such factors as the type of therapy the patient will receive.

The type of therapy prescribed depends on the severity, urgency, and prognosis of the patient's condition.

Therapy types include:

- **Acute therapy**, if the patient is critically ill and requires acute intensive therapy
- **Empiric therapy**, based on practical experience rather than on pure scientific data
- **Maintenance therapy**, for patients with chronic conditions that don't resolve
- **Supplemental or replacement therapy**, to replenish or substitute for missing substances in the body
- **Supportive therapy**, which doesn't treat the cause of the disease but maintains other threatened body systems until the patient's condition resolves
- **Palliative therapy**, used for end-stage or terminal diseases to make the patient as comfortable as possible.

A patient's overall health as well as other individual factors can alter his response to a drug. Coinciding medical conditions and personal lifestyle characteristics must also be considered when selecting drug therapy.

Certain drugs have a tendency to create drug tolerance and drug dependence in patients.

Drug tolerance occurs when a patient has a decreased response to a drug over time. The patient then requires larger doses to produce the same response.

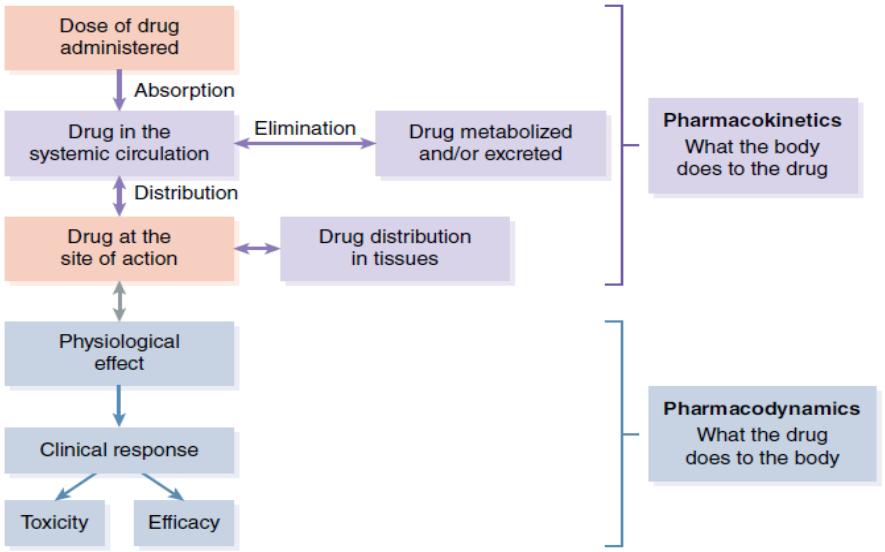


FIGURE 2-4 The interplay of pharmacokinetics and pharmacodynamics: The components, text boxes and all shapes that are colored in purple are pharmacokinetic parameters. Those same items that are colored in blue are pharmacodynamic parameters. Note the *interdependence* of the pharmacological response on the drug concentration at the site of action. This is only *one* of the components involved in the complex interplay of the drug-body interactions.

Courtesy of Diane Pacitti. © 2017

Drug interactions

Drug interactions can occur between drugs or between drugs and foods. They can interfere with the results of a laboratory test or produce physical or chemical incompatibilities. The more drugs a patient receives, the greater the chances are that a drug interaction will occur.

Enhancement of drug effect:

1. **Addition:** two drug with the same effect, when given together, produce an effect that is equal in magnitude to the sum of effects when the drugs are given individually (1+1=2) e.g. Giving two drugs together, such as two analgesics (pain relievers)
2. **Synergism:** two drugs with same effect, when given together, produce an effect that is greater in magnitude than sum of effects when the drugs are given individually (1+1›2) e.g. Carbon tetrachloride and ethanol (ethyl alcohol) are individually toxic to the liver.
3. **Potentiation:** A drug lacking an effect of its own increases the effect of a second active drug (0+1› 1) e.g. promethazine would potentiate the effect of an opioid (Codeine).
4. **Antagonism**: Drug inhibits the effect of another drug. Usually, the 1 + 1 = 0 antagonist has no inherent activity e.g. protamine sulfate is a heparin antagonist. This means that the administration of protamine sulfate completely neutralizes the effects of heparin in the body.

5. **Altering absorption:** Two drugs given together can change the absorption of one or both of the drugs. For example, drugs that change the acidity of the stomach can affect the ability of another drug to dissolve in the stomach. Other drugs can interact and form an insoluble compound that can't be absorbed. Sometimes, an absorption interaction can be avoided by separating drug administration by at least 2 hours.

Factors Influencing Drug Effects

When administering a drug to a patient, the nurse must be aware that the human factor has a tremendous influence on what actually happens to a drug when it enters the body. No two people react in exactly the same way to any given drug.

Before administering any drug, the nurse must consider a number of factors, includes:

a. Age:

- Age is a factor primarily in children and older adults. Children are not just little adults. Children metabolize many drugs differently than adults do, and they have immature systems for handling drugs.
- Many drugs come with recommended pediatric doses, and others can be converted to pediatric doses using one of several conversion formulas.
- Older adults undergo many physical changes that are a part of the aging process. Their bodies may respond very differently in all aspects of pharmacokinetics—less effective absorption, less efficient distribution because of fewer plasma proteins and less efficient perfusion, altered biotransformation or metabolism of drugs because of age-related liver changes, and less effective excretion owing to less efficient kidneys and by concurrent disease process such as C.V.D. (cardiovascular disease).
- The doses of other drugs also may need to be decreased for the older adult and it should monitor the patient closely for the desired effects. If the effects are not what would normally be expected, one should consider the need for a dose adjustment.

b. Weight:

- The greater the weight, the greater must be the dose.
- For very lean and very obese individuals, drug dosage is frequently determined on the basis of drug/kg of body weight or body surface area.

c. Sex:

- Females don't always respond to the action of drug in the same manner as do men When giving IM injections, for example, it is important to remember that men have more vascular muscles, so the effects of the drug will be seen sooner in men than in women.
- Women have more fat cells than men do, so drugs that deposit in fat may be slowly released and cause effects for a prolonged period. For example, gas anesthetics have an affinity for depositing in fat and can cause drowsiness and sedation sometimes weeks after surgery and the effect of drug will be more pronounced and prolonged.
- During pregnancy, lactation, and menstruation, many drugs are stopped:
    1) Aspirin: not used during menstruation as it increases blood fluidity.
    2) Drugs excreted in milk aren't given during lactation as penicillin.
    3) Uterine stimulant should be avoided during pregnancy as they may produce abortion such as prostaglandin.
    4) Drugs that might affect fetus and placenta should not be given during pregnancy.

d. Time of administration:

- Drug is more rapidly absorbed when the GIT is free of food, while irritating drugs are more tolerated if there is food in stomach.
- Body resistant to drug is generally greater in the early morning when the body is at its lowest point of physiologic functioning and conversely, the body is more sensitive to drugs effect during time of maximal activity.

e. Pathologic state:

- The disease can change the functioning of the chemical reactions within the body and thus change the response to the drug.
- Pathological conditions can change the basic pharmacokinetics of a drug. For example, GI disorders can affect the absorption of many oral drugs. Vascular diseases and low blood pressure alter the distribution of a drug, preventing it from being delivered to the reactive tissue, thus rendering the drug nontherapeutic.

f. Genetic Factors

- Genetic differences can sometimes explain patients' varied responses to a given drug.

- Some people lack certain enzyme systems necessary for metabolizing a drug, whereas others have overactive enzyme systems that cause drugs to be broken down more quickly.

g. Immunological Factors

- People can develop an allergy to a drug. After exposure to its proteins, a person can develop antibodies to a drug. With future exposure to the same drug, that person may experience a full-blown allergic reaction.
- Sensitivity to a drug can range from mild (e.g., dermatological reactions such as a rash) to more severe (e.g., anaphylaxis, shock, and death).

h. Psychological Factors

- The patient's attitude about a drug has been shown to have an effect on how that drug works. A drug is more likely to be effective if the patient thinks it will work than if the patient believes it will not work. This is called the **placebo effect**.

i. Environmental Factors

- Some drug effects are enhanced by a quiet, cool, non-stimulating environment. For example, sedating drugs are given to help a patient relax or to decrease tension.
- Reducing external stimuli to decrease tension and stimulation help the drug be more effective.
- Other drug effects may be influenced by temperature. For example, antihypertensive that work well during cold, winter months may become too effective in warmer environments, when natural vasodilation may lead to a release of heat that tends to lower the blood pressure. If a patient's response to a medication is not as expected, look for possible changes in environmental conditions.

j. Tolerance

- Drug tolerance refers to a phenomenon in which a person's body becomes less responsive to the effects of a drug over time. This means that, with repeated use of the drug, a person may need to take higher doses to achieve the same desired effect or experience the same level of relief from their symptoms.
- An example is morphine, an opiate used for pain relief. The longer morphine is taken, the more tolerant the body becomes to the drug, so that larger and larger doses are needed to relieve pain.

- Clinically, this situation can be avoided by giving the drug in smaller doses or in combination with other drugs that may also relieve pain. Cross-tolerance—or resistance to drugs within the same class—may also occur in some situations.

k. Accumulation

- If a drug is taken in successive doses at intervals that are shorter than recommended, or if the body is unable to eliminate a drug properly, the drug can accumulate in the body, leading to toxic levels and adverse effects. This can be avoided by following the drug regimen precisely.

l. Interactions

- When two or more drugs or substances are taken together, there is a possibility that an interaction can occur, causing unanticipated effects in the body.
- Alternative therapies, such as herbal products, act as drugs in the body and can cause these same interactions. Certain foods can interact with drugs in much the same way. Usually this is an increase or decrease in the desired therapeutic effect of one or all of the drugs or an increase in adverse effects.

Side effects and drug toxicity

Adverse effects

Adverse effects are undesired effects that may be unpleasant or even dangerous. **They can occur for many reasons, including the following:**

- The drug may have other effects on the body besides the therapeutic effect.
- The patient may be sensitive to the drug being given.
- The drug's action on the body may cause other responses that are undesirable or unpleasant.
- The patient may be taking too much or too little of the drug, leading to adverse effects.

Examples:

- Dermatologic reactions "pruritis, urticaria, alopecia, rashes, photosensitivity
- Blood dyscrasias: in certain patients, bone marrow is sensitive to certain drugs; this may result in the insufficient production of platelets, red blood cells & white blood cells.
- Patients who receive a drug that may cause bone-marrow depression are monitored closely by frequent blood counts & for early signs & symptoms of infection.
- Some forms of blood dyscrasias:

a. Agranulocytosis: (reduction in the number of granulocytes).
b. Aplastic anemia: Pancytopenia (reduction of all formed elements of blood).
c. Hemolytic anemia (reduced hemoglobin level due to lyses of RBCs).
d. Thrombocytopenia: (platelets deficiency)

- Hepatotoxicity: (Liver damage), abdominal pain & jaundice.
- Nephrotoxicity: (Kidney damages), hematuria, anuria, proteinuria, edema & uremia.
- Ototoxicity: (ear damage), damage of vestibular or/and auditory portion of the 8th cranial nerve.
- CNS toxicity: loss of judgment, loss of movement coordination, loss of consciousness, convulsions, tardive dyskinesia, reduce knee-jerk reflex, drowsiness, dizziness, …… .
- GI disturbance: nausea, vomiting, abdominal pain, diarrhea….. .
- Sexual dysfunction: change in libido.

Hypersensitivity:

- It refers to allergic or other immunologic responses to drugs. It occurs when the patient shows extreme sensitivity to an effect of the drug. The response is the usual pharmacological effect; however, the effect is intense and exaggerated. A simple decrease in the dose may be sufficient to eliminate this type of reactions.

Drug Allergies:

- A drug allergy occurs when the body forms antibodies to a particular drug, causing an immune response when the person is re-exposed to the drug.
- Allergic responses to drugs occur in some patients and not in others.
- It is an unwanted response to a drug resulting from previous exposure to that drug or one closely related to it. So drug allergy is happens only after a second or subsequent exposure to the drug.

Allergic reactions to drugs differ from drug toxicity in the following ways.

1- The allergic reaction occurs in only a fraction of the population where as toxicity will occur in all individuals if the dose is high enough.
2- The allergic response is unusual in that a small amount of unsafe drug causes a severe reaction.
3- With allergy, the reaction is different from the usual pharmacological effect of the drug.

Types of allergic responses:

1- Immediate reaction involving antigen and antibody resulting in the release of histamine.

- In mild cases, the reaction is limited to urticaria (wheals and itching of the skin).
- In severe cases (anaphylactic reaction), characterized by circulatory collapse or asphyxia due to swelling of the larynx & occlusion of the air way passages.
- Example: many patients are allergic to penicillin.

2- Delayed reaction:

- Occurring several days or even weeks after the drug has been administered. It Characterized by fever (drug fever), swelling of the joints, reaction may involves blood-forming organs and kidneys.

Treatment of anaphylactic reaction may include administration of Epinephrine, Oxygen**,** Antihistamines**,** Corticosteroids.

Drug idiosyncrasies:

Idiosyncratic reactions are defined as those reactions that occur in patients who have abnormal genes which cause this abnormal response to that drug. Reaction may be excessive and unusual that causes by:

a. Genetic variation in drugs metabolism.

b. Immunologic mechanism, including allergic reactions

Example: Succinylcholine, a muscle relaxant drug, usually is broken down rapidly by enzymes in the plasma and liver so that the effects of the drug last for only a few minutes. In few patients, a normal dose of this drug produces profound muscle relaxation and suppression of respiration, which may last few hours. Those patients have a genetic defect that produces unusual enzymes & the drug is not broken down.

Drug toxicity:

Excess dosage may be either accidental or intentional, results in an exaggerated response to this drug. It may be sever & may lead to respiratory depression. Cardio-vascular collapse and death may occur if the drug is not withdrawn and adequate treatment is started.

Routs of Administration

The route of administration is determined by:

1) Properties of the drug (solubility, ionization)

2) Therapeutic objectives (onset of action: 1. Rapid 2. long-term 3. Local)

Tow major routes of administration

Enteral	Parenteral	Others
- oral	- intravascular	-inhalation
-sublingual	-intramascular	-intranasal
-rectal	-subcutaneous	-intrathecal
		-intraventricular
		-topical
		-transdermal

1- The ORAL route: (P.O.)

- The patient usually swallows the drug, which then enters the gastrointestinal tract and absorbed from that area. This route accounts for 80% of all medications administered to patients.

Absorption Pattern:

- Variable; affected by many factors.

Advantages:

- Safest and most common, convenient, and economical route of administration.
- Doctors usually decide the amount and frequency and the patient administer the drug to himself.
- The cost of medication and therapy is low.
- Easy
- Preferred by patients
- "Slow-release" preparations may be available to extend duration of action

- There is no necessity to practice sterile technique.

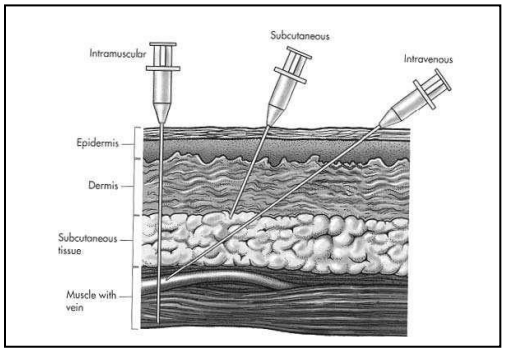
Disadvantages:

- Inconsistent, slow absorption
- Food may affect absorption
- Patent compliance is necessary
- Drugs may be metabolized before systemic absorption and degraded by stomach acid and enzymes.
- Drug irritation may causes nausea & vomiting.
- Unsuitable in patients who are uncooperative, unconscious & vomiting.

2- Parenteral Route:

Two methods of intravenous injection are used:

1) Administering the drug as a bolus.
2) Continuous infusion of a drug, called IV drip



A. Intravenous Route (IV)

- IV solutions must be completely homogenous. No solution containing particulate matter, regardless of how fine the particles should ever be injected IV because of the danger of creating EMBLOI.

Advantages:

- Dependable and reproducible effects.
- Entire administered dose reaches the systemic circulation immediately – the dose can be accurately titrated against response.
- The next most popular route of drug administration.
- Drugs that cannot be given through GIT can be given parenterally.
- Eliminate the unpredictability of absorption from GIT i.e. the amount of drug will be delivered to the patient from injection site.
- Medication can be given to an unconscious patient.

- A channel to administer drugs that irritate another sites e.g. chemotherapy.
- Less painful during drug administration.

Disadvantages:

- Need sterile technique.
- More expensive than oral preparations.
- Patient usually not able to medicate himself
- Accidental penetration of blood vessel, especially if medication is not suitable for I.V. administration (complication).
- Requires a functioning cannula
- Cannulation is distressing to some patients, especially children
- IV injection of drugs may cause local reactions
- Accidental overdose, and there is no way to withdraw the drug from circulation once it is injected.
- Thrombophlebitis from repeated injections. To minimize this phenomena (but doesn't exclude it), drug should be given by I.V. drip.

B. Subcutaneous Injection (SC):-

Absorption Pattern:

- Depends on drug diluents: Aqueous solutions: prompt Depot preparations: slow and sustained

Advantages:

- Suitable for slow-release drugs
- Ideal for some poorly soluble suspensions

Disadvantages:

- The most painful route.
- Slow rate of absorption.
- Not effective in emergency (poor blood supply).
- Unsuitable for drugs administered in large volumes.

N.B.: Drug absorption from SC injection can be reduced by:

1) Addition of vasoconstrictor e.g. epinephrine.
2) Application of cold packs (also causes vasoconstriction). Heat causes vasodilatation, which result in increased blood supply to the area and thus quicker drug absorption. This manipulation may be used to reduce the severity of toxicity from a SC injection.

C. Intramuscular Route: IM

Absorption Pattern:

- Depends on drug diluents: Aqueous solution: prompt Depot preparations: slow and sustained

Advantages:

- second most widely used route for injectable drugs.
- suitable for administration of both solutions and suspensions.
- Depending on formulation can have very long duration of action, e.g. depot antipsychotics and contraceptives (oily base).
- drug absorption is more quickly than SC route.
- Less painful (less nerve supply).
- Good absorption, especially for drugs with a low oral bioavailability
- Onset is more rapid than sc.

Disadvantages

- Invasive and painful compared to oral route.
- Risk of infection at injection site.
- Absorption can vary due to blood flow at injection site.
- Not suitable for large volumes of medication.
- Potential for nerve and tissue damage with improper injection technique.
- Depot injections have slow onset and cannot be titrated.
- Not recommended for infants due to limited muscle mass.

D. Intradermal (ID):

- ID is a route of administration in which the drug is injected just below the epidermis.
- Injection is usually used for allergy test.
- Used for Tuberculosis (TB) tests and for penicillin sensitivity test..

E. Intrathecal administration

- Intrathecal route of administration involves inserting a needle between 2 vertebrae and injecting the drug into the CSF.
- Used frequently in operating room (OR) when anesthesiologist desires to produce a spinal block and inject local anesthetic directly into the spinal cord.
- Used to administer Chemotherapy to prevent CNS metastases of some types of cancers.

3- Rectal, urethral, and vaginal suppositories:

Absorption Pattern:

- Erratic and variable

Advantages:

- Partially bypasses first-pass effect
- Bypasses destruction by stomach acid
- Ideal if drug causes vomiting
- Ideal in patients who are vomiting, or comatose

Disadvantages:

1) Erratic absorption from the rectal area result in an inaccurate predicted blood level of drug so the dosage must be adjusted.
2) Drugs may irritate the rectal mucosa
3) Not a well-accepted route.
4) Rectal route is ineffective when the patient suffers from diarrhea. *Examples:-*
   Rectal supp.: used to treat hemorrhoids.
   Vaginal supp.: used to treat vaginitis.
   Uretheral supp.: used to treat bladder conditions.

N.B.: rectal route can produce local effect or systemic effect.

4. Topical route:

- Are intended for use at the site of administration. There are exceptions e.g. Nitroglycerin ointment that is rubbed into the skin and systemic effect is achieved.
- Two topical preparations are frequently confused with each other (ointment and creams).

a) Ointments:

- Are usually made up of a petrolatum type of base.
- They are quite greasy, as they intended for prolonged contact with the skin.

- They don’t disappear or penetrate the skin, but remain on the surface.

b) Creams:

- Made of viscous water-soluble chemicals, which usually disappear after being placed to the skin.
- Patient prefers cream to ointment, as creams don’t discolor clothing.

5. Inhalation

Absorption Pattern

- Systemic absorption occur, this is not always desirable.

Advantages:

- Absorption is rapid; can have immediate effects
- Ideal for gases
- Effective for patients with respiratory problems
- Dose can be titrated
- Localized effect to target lungs: lower doses used compared to that with oral or parental administration
- Fewer systemic side effects

Disadvantages:

- Most addictive route (drug can enter the brain quickly)
- Patient may have difficulty regulating dose.
- Some patients may have difficulty using inhalers.

6. Sublingual

Absorption Pattern:

- Depends on the drugs : (for example, nitroglycerin) have rapid, direct systemic absorption Most drugs erratically or incompletely absorbed

Advantages:

- Bypasses first-pass effect
- Bypasses destruction by stomach acid
- Drug stability maintained because the pH saliva relatively neutral
- May cause immediate pharmacological effects

Disadvantages:

- Limited to certain types of drugs

- Limited to drugs that can be taken in small doses
- May lose part of the drug dose if swallowed.

Drugs orders and prescriptions

Prescription: -

- A medication order is written directions provided by a prescribing practitioner for a specific medication to be administered to an individual. Medicating a patient begins when the medication is suggested and authorized by a legal prescriber, usually a licensed physician, dentist or veterinarian.
- As an authorized provider, **you must have a medication order** in order to administer medications to individuals.

It consists of 4 parts:

1. **The superscription:**
   a) It includes the patient's name, address, date and the symbol Rx.
   b) The age of patient (infant or a child) should be written to permit the pharmacist to check the correctness of the dose.
2. **The inscription: -**
   It states the name of the drug, dosage form, and the amount.
3. **The subscription:**
   It contains the directions of the pharmacist, now usually limited to the number of the doses to be dispensed.
4. **The signature:**
   - It is abbreviated by "S" or "Sig"
   - It includes the prescriber's signature.

The prescriber's order has several elements that should be presented and identified:-

1. Patient's name and other identifying data.
2. Date that the order was written.

3. Medication name.
4. Dosage to be administered each time.
5. Route of administration.
6. Frequency of administration and special instructions.
7. Prescriber's signature.

Types of orders:

- Outpatients are free to medicate themselves with any accessible medication. Many Over the Counter (OTC) drugs are available for the public without the need for a prescription.
- Once an individual is admitted to a clinical institution, usually neither the patient nor the nurse may legally administer any medication without a written order.
- Use of variant or non-standard abbreviations should be avoided because of the danger of misinterpretation.
- The common type orders means that the drug, as ordered, is to be regularly administered until a formal discontinuation order is written, or until a specified termination date is reached.
- Automatic termination or stop may be explicit in agency policy, e.g. narcotic analgesics 3 days, Antibiotics 5 days, steroids 7 days ……
- The policies act as a stimulus to the prescriber to evaluate continued need for these drugs that require especial close attention.

Types of orders:

1. **Routine Order:** refers to a prescription or medication order provided by a healthcare provider (typically a physician) for a patient's ongoing or regularly scheduled medication therapy.
2. **Single Order:** Drug is to be administered once at the time indicated. e.g., a preoperative medication (Atropine) 0.4 mg IM on call to operating room (OR).
3. **Stat Order:** This is a single order that a drug is to be administered immediately. Stat orders are written often for emergency when the client's condition changes suddenly, e.g Hydralazine 10 mg IM stat.
4. **prn Orders:** PRN medications are given on an "as needed" basis for specific signs & symptoms. These drugs are to be administered by the nurse as necessary within the order

criteria specified by the doctor order. The decision of when to give the drug is left to the nurse's judgment.

- Medications to reduce the perception of pain make up the bulk of prn orders.
- Keeping nursing assessment of the pain is required to carry out these prn orders appropriately.

5. Telephone order and verbal order:

Usually given in emergency and should be written in the patient's chart and assigned by the nurse who receive the order and to be signed by doctor as soon as possible.

Example: 11:00 am, Pirox 20 mg IM, Phone order Dr. mohammed abu warda to Nurse Rmai Mahmoud. Then the nurse sings his/her name.

Disadvantages

1. Forgotten if not written at that time.
2. It is illegal until the order is signed for by the prescriber (depending on the hospital policy).
3. Can easily be mis-communicated, misinterpreted or not clearly heard.

General Guidelines for Handling Medications

1. When preparing or giving medicines, concentrate your attention on what you are doing.
2. Make certain that you have a written order for every medication you will administer.
3. Read the label before taking, while using, and returning the drug to its place.
4. All medicines should be labeled and the label should be clear.
5. Never give medicine from unlabeled container.
6. Never give a put a drug in a bottle that has a different label. For example, if you have Keflex that you need to put it in a bottle, and you have empty bottle that was used to contain ampeicllin, don't put the keflex in that bottle.
7. Measure quantities and calculate doses as ordered in proper way.
8. Handling tablets, capsules and pills properly and DO'NOT touch with fingers. Use the cap of the container to guide or from the container to the cup directly.
9. Avoid wastage of medicine.
10. Never administer medications prepared by another person; otherwise you will bear the responsibility of any mistake.
11. Some preparations as vaccines & suppositories need to be kept in the refrigerator "2-8º C".
12. Stay with the patient until he takes his medicine.
13. All mixtures should be made immediately before use.
14. NEVER return unused drug to a stock bottle.
15. Don't add any drug to the blood; interactions may occur without visible changes.
16. Don't use any sterile article that gets unsterile by any way.
17. Don't use clear solutions which have become cloudy or have sediment.
18. Don't use a drug that is out date.
19. Don't use a drug that has changes its color.
20. Don't use a drug which arise doubt in mind.
21. All medicines should be kept in cupboard, NARCOTICS in a locked one.
22. Narcotics have a rule controlling their ordering, giving, and their registration.
23. Many liquids should be diluted with water or other liquids. "This is especially when medicine has a bad taste".

24. Exceptions to this rule, cough medicines are not diluted, or the patient not allowed drinking water after taking cough syrup.
25. Don't leave a tray of medicines unattended. If you are in a patient's room and must leave, take the tray with you.
26. Never chart a medicine as having been given, until it has been administered.
27. Follow up the five rights
    a. Right patient.
    b. Right drug.
    c. Right dose.
    d. Right route of administration.
    e. Right time.

Common abbreviations

mg	milligram		**mcg**	Mivrogram		**Kg**	kilogram
alt. hor.	every other hour		**A.M.or a.m.**	Morning		**L**	liter
mEq	millieguivalent		**ml**	Milliliter		**Sc or SQ**	subcutaneous
NG	Nasogastric		**NPO**	nothing by mouth		**SL**	sublingnal
b.i.d.	two times a day		**p**	After		**SOS**	if necessary once only
C	with		**s**	Without		**Sol**	solution
CHF	congestive heart failure		**os**	Mouth		**Mcg**	microgram
cm	centimeter		**Caps**	Capsules		**V.O.**	verbal order
b	before		**dc**	Discontinue		**Rx**	symbol for a prescription
d.	day		**per**	by, through		**Min**	Minute
p.o.	by mouth		**P.R.**	by rectum		**Syr**	Syrup
dl	deciliter		**prn**	when necessary		**stat**	Immediately
elix.	elixir		**q**	Every		**tab.**	Tablet
g(gm)	gram		**q2h**	every 2 hours		**t.i.d.**	3 times daily
q.d.	every day		**q.h.**	every hour		**U**	Unit
gtt	a drop, drops		**qhs**	every night		**Lt**	Left
GU	genitourinary		**h.s.**	at bed time		**® or Rt**	right
H,hr	hour		**qod**	every other day		**IM**	Intramuscular
qid	4 times daily		**oz.**	Once		**I.V.**	intravenous

- **Tolerance:** Drug tolerance refers to a phenomenon in which a person's body becomes less responsive to the effects of a drug over time. This means that, with repeated use of the drug, a person may need to take higher doses to achieve the same desired effect or experience the same level of relief from their symptoms. e.g. morphine.
- **Dependence (addiction):** Dependence, often referred to as addiction, is a complex and chronic medical condition characterized by a compulsive and often uncontrollable craving for a substance (such as drugs or alcohol) or a behavior (such as gambling) despite negative consequences. **Substitute** = these are many different drugs which may use to treat a single disease. From these drugs, always present a drug of choice, if not available the others are considered as the substitute.
- **Antidote** = a drug used to antagonize the toxic effect of another drug and to neutralize its symptoms.

Weights & Measures

1 kg = 1000 gm

1 gm = 1000 mg

1 mg = 1000 Mcg

1 L = 1000 ML

1 teaspoonful = about 5 ml

1 tablespoonful = about 15 ml

1 ml = 15 gtt.

1 pound (lb.) = 454 gram

1 kg = 2.2 Libra (lb.)

1 gm = 15 grains

Risk category of drugs during pregnancy

Category		*Examples*
A	Adequate studies in pregnant women have failed to demonstrate a risk to the foetus	Inj. Mag. sulfate, thyroxine
B	Adequate human studies are lacking, but animal studies have failed to demonstrate a risk to the foetus or Adequate studies in pregnant women have failed to demonstrate a risk to the foetus, but animal studies have shown an adverse effect on the foetus	Penicillin V, amoxicillin, cefactor, erythromycin, paracetamol, lignocaine
C	No adequate studies in pregnant women and animal studies are lacking or have shown and adverse effect on foetus, but potential benefit may warrant use of the drug in pregnant women despite potential risk	Morphine, codeine, atropine, corticosteroids, adrenaline, thiopentone, bupivacaine
D	There is evidence of human foetal risk, but the potential benefits from use of the drug may be acceptable despite the potential risk	Aspirin, phenytoin, carbamazepine, valproate, lorazepam
X	Studies in animals or humans have demonstrated foetal abnormalities, and potential risk clearly outweighs possible benefit	Estrogens, isotretinoin, ergometrine

Drugs Affecting the Central Nervous System

The nervous system has two major divisions: **the central nervous system (CNS)** and the **peripheral nervous system (PNS)**. The central nervous system is made up of the brain, brain stem, spinal cord, and an intricate network of neurons. The CNS is responsible for sending, receiving, processing and interpreting sensory information from all areas of the body. It also receives information from and sends information to the peripheral nervous system (PNS). The PNS monitors and coordinates internal organ function and responds to changes in the external environment.

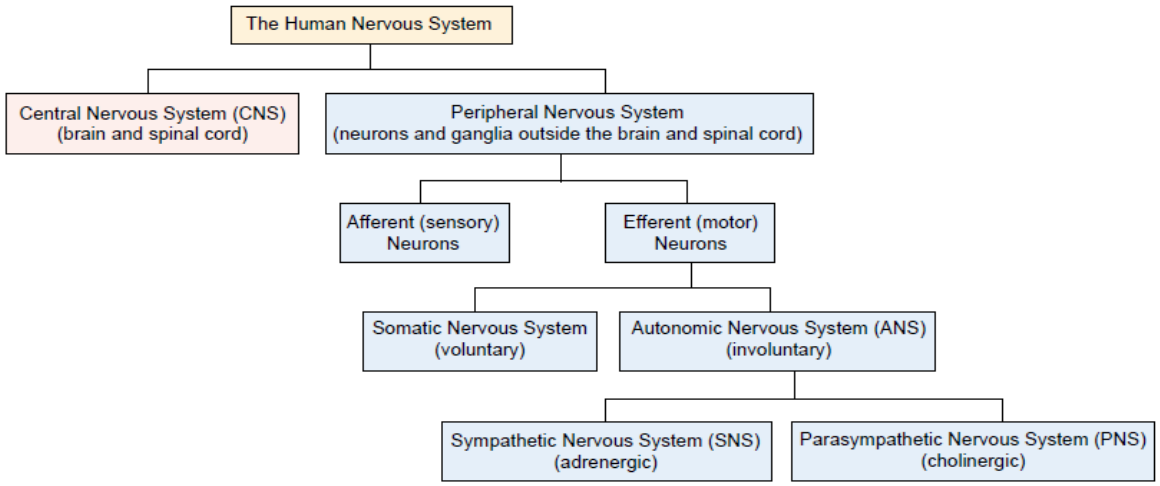
Classes of Medications for Treating CNS Disorders

CNS pharmacology will be addressed using the following system of categorization. These divisions are based on the groupings of the pharmacologic mechanisms of the drugs contained within each division.

1. Analgesics
    a. Opioids/narcotics
    b. NSAIDs (nonsteroidal anti-inflammatory drugs) and **acetaminophen**
2. Antiseizure/antiepileptic agents
3. Medications for neurodegenerative disorders
4. Muscle relaxants
5. Local anesthetics

The autonomic nervous system regulates the body's involuntary functions, including heart rate, respiratory rate, and digestion. It works through a balance of its two main components, the sympathetic and parasympathetic nervous systems. Types of drugs used to treat disorders of the autonomic nervous system include:

- cholinergic drugs
- anticholinergic drugs
- adrenergic drugs
- adrenergic blocking drugs.



There are several drugs affecting the autonomic nervous system which, for a better understanding of specific drugs, are classified into groups.

1. **Drugs acting on the sympathetic nervous system.**
   a) Sympathomimetic or adrenergic drugs: are drugs that mimic the effects of sympathetic nerve stimulation.
   b) Sympatholytic: are drugs that inhibit the activity of sympathetic nerve or that of sympathomimetic.

2. **Drugs acting on the parasympathetic nervous system**
   a) Para-sympathomimetic or cholinergic drugs: are drugs which mimic acetylcholine or the effects of parasympathetic nerve stimulation.
   b) Para-sympatholytic: are drugs that inhibit parasympathetic nervous system activity or that of cholinergic drugs.

1. Sympathomimetic (Adrenergic) Drugs

- The adrenergic drugs supplement, mimic & reinforce the message transmitted by the natural neurohormones norepinephrine & epinephrine.
- These hormones are responsible for transmitting nerve impulses of the sympathetic nervous system.

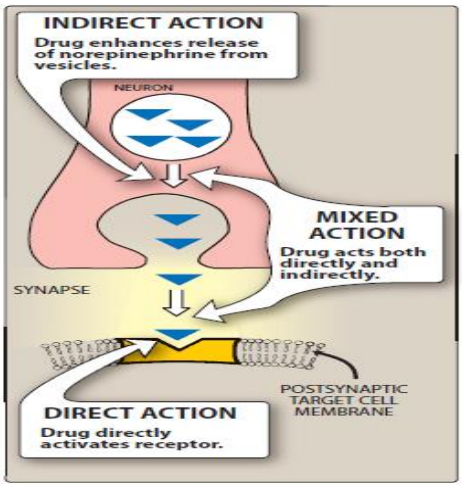
The adrenergic drugs work in two ways:

A. According to mechanism of action:

a. Direct acting agonist: by bind to adrenergic receptors without interacting with the presynaptic neuron. mimic the action of epinephrine and epinephrine .

b. Indirect acting agonist: By regulating the release of the natural neurohormones from their storage sites at the never terminals.

c. Mixed acting agonist: By induce the release of norepinephrine from presynaptic terminals, and they activate adrenergic.



B. According to receptor selectivity

These receptors are classified to receptor selectivity into:

1. Alpha selective
    - Alpha 1 selective, e.g. phenylephrine
    - Alpha 2 selective, e.g. methylnorepinephrine.
2. Beta selective
    - Beta 1 selective, e.g. dobutamine
    - Beta 2 selective, e.g. terbutaline
    - Beta1 and 2 non selective e.g. isopreterenol
3. Alpha and beta non selective e.g. epinephrine, norepinephrine.
4. Dopamine selective e.g. dopamine.

Catecholamines

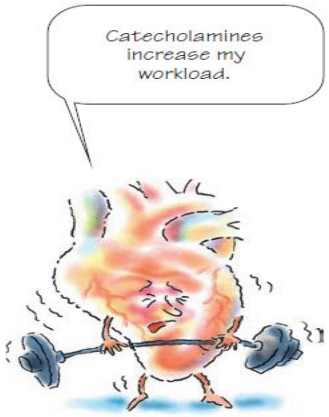
Because of their common basic chemical structure, catecholamines share certain properties—they stimulate the nervous system, constrict peripheral blood vessels, increase heart rate, and dilate the bronchi. They can be manufactured in the body or in a laboratory.

Common catecholamines include:

- dobutamine
- dopamine
- epinephrine, epinephrine bitartrate, and epinephrine hydrochloride
- norepinephrine
- isoproterenol hydrochloride and isoproterenol sulfate.

Sympathetic Nervous System (SNS):

The SNS is stimulated by physical or emotional stress, (*fight-or-flight* reaction). Specific body responses include:



1. Increased arterial blood pressure and cardiac output
2. Increased blood flow to the brain, heart, and skeletal muscles; decreased blood flow to viscera, skin, and other organs
3. Increased rate of cellular metabolism—increased oxygen consumption and carbon dioxide production
4. Increased breakdown of muscle glycogen for energy
5. Increased blood sugar
6. Increased mental activity and ability to think clearly
7. Increased muscle strength
8. Increased rate of blood coagulation
9. Increased rate and depth of respiration
10. Pupil dilation to aid vision and increased sweating.

Effects of adrenergic drugs:

1. **Heart:** increase Heart rate, increase force of contraction, increase cardiac output .
   - **Uses:** cardiogenic shock, bradycardia, resuscitation, heart block.
2. **Blood vessels:** Systemic vasoconstriction of abdominal viscera, cerebrum & skin.
   - **Uses:** Hypotension, nasal decongestion, biliary colic, nose bleeds, migraine, headache, allergic reactions.
3. **GIT and Genitourinary tracts:** Decrease glandular secretions, constriction of sphincters, decrease muscle tone & motility of GIT & urinary bladder, increase muscle tone & motility of the ureters.
   - **Uses**: Enuresis, dysmenorrhea, biliary colic.

4. **Lungs:** Relaxation of muscles of bronchial tree.
   - **Uses:** Bronchial asthma, emphysema, chronic bronchitis.
5. **Eyes:** Dilate iris, increase ocular pressure, relaxes ciliary muscle.
6. **CNS:** Excitory action, respiratory stimulation, wakefulness.
7. **Metabolism:** increase in glycogenesis (sugar metabolism), increase in lypolysis (release of fatty acids).

Pharmacotherapeutics

- Catecholamines can't be taken orally because they're destroyed by digestive enzymes. In contrast, when these drugs are given sublingually (under the tongue), they're rapidly absorbed through the mucous membranes. Any sublingual drug not completely absorbed is rapidly metabolized by swallowed saliva.
- Absorption by the sub cut route is slowed because catecholamines cause the blood vessels around the injection site to constrict. IM absorption is more rapid because less constriction of local blood vessels occurs. Catecholamines are widely distributed in the body. They're metabolized and inactivated predominantly in the liver but can also be metabolized in the GI tract, lungs, kidneys, plasma, and tissues.
- Catecholamines are excreted primarily in urine; however, a small amount of isoproterenol is excreted in feces and some epinephrine is excreted in breast milk.

Adverse reactions

Adverse reactions to catecholamines may include:

- Restlessness, anxiety, dizziness, headache, palpitations, cardiac arrhythmias, hypotension, hypertension and hypertensive crisis, stroke, angina, increased glucose levels, tissue necrosis and sloughing (if a catecholamine given IV leaks into the surrounding tissue).

Indications:

- **Helping hypotension:** Catecholamines that stimulate alpha-adrenergic receptors are used to treat low blood pressure (hypotension). As a rule, catecholamines work best when used to treat hypotension caused by relaxation of the blood vessels (also called a loss of vasomotor tone) and blood loss (such as from hemorrhage).
- **Rhythm makers:** Catecholamines that stimulate beta1-adrenergic receptors are used to treat: bradycardia, heart block, low cardiac output.

- **Increasing response:** Because they're believed to make the heart more responsive to defibrillation (using an electrical current to terminate a deadly arrhythmia), beta1-adrenergic drugs are used to treat: ventricular fibrillation (quivering of the ventricles resulting in no pulse), asystole (no electrical activity in the heart) and cardiac arrest.
- **Breathing better with beta2 activity:** Catecholamines that exert beta2-adrenergic activity are used to treat: acute and chronic bronchial asthma, emphysema, bronchitis, acute hypersensitivity (allergic) reactions to drugs.
- **More blood for the kidneys:** Dopamine, which stimulates dopaminergic receptors, is used in low doses to improve blood flow to the kidneys because it dilates the renal blood vessels.

Drug interactions

Drug interactions involving catecholamines can be serious:

- Alpha-adrenergic blockers, such as phentolamine, used with catecholamines can produce hypotension.
- Because the catecholamine epinephrine may cause hyperglycemia, diabetic patients receiving epinephrine may require an increased dose of insulin or drugs.
- Beta-adrenergic blockers, such as propranolol, used with catecholamines can lead to bronchial constriction, especially in patients with asthma.

Adding it all up

- Other adrenergic drugs can produce additive effects, such as hypertension (high blood pressure) and arrhythmias, and enhance adverse effects.
- Tricyclic antidepressants used with catecholamines can lead to hypertension.

Nursing process (Implementation)

- Before starting catecholamines, correct hypovolemia with plasma volume expanders.
- Give cardiac glycosides before catecholamines; cardiac glycosides increase atrioventricular (AV) node conduction and patients with atrial fibrillation may develop rapid ventricular rate.
- Administer drug using a central venous catheter or large peripheral vein. Adjust the infusion according to the prescriber's order and the patient's condition. Use an infusion pump.
- Dilute the concentrate for injection before administration, according to pharmacy guidelines.
- Watch for irritation and infiltration; extravasation can cause an inflammatory response.

- Don’t give catecholamines in the same IV line as other drugs; be aware of incompatibilities. For example, dobutamine is in compatible with heparin, hydrocortisone sodium succinate, cefazolin, cefamandole, cephalothin, penicillin, and ethacrynate sodium.
- Don’t mix dobutamine or dopamine with sodium bicarbonate injection or phenytoin because the drug is incompatible with alkaline solutions.
- Change IV sites regularly to avoid phlebitis.
- Provide patient teaching.

Noncatecholamine adrenergic drugs have many therapeutic uses because of the various effects these drugs can have on the body, including:

- local or systemic constriction of blood vessels (phenylephrine)
- nasal and eye decongestion and dilation of the bronchioles (albuterol, ephedrine, formoterol, isoetharine hydro chloride, metaproterenol, pirbuterol, salmeterol, and terbutaline)
- smooth-muscle relaxation (terbutaline).

Inhaled drugs, such as albuterol, are gradually absorbed from the bronchi of the lungs and result in lower drug levels in the body.

1. Epinephrine:

Trade name: Adrenaline

Class. : Direct acting-adrenergic agent.

Action: A natural hormone produced from adrenal medulla, induce marked stimulation of alpha, β1 + β2 receptors.

Actions:

- Increases cardiac output due to increased inotropy, chronotropy, dromotropy, and AV conduction (Beta-1)
- Relaxes smooth muscles of the respiratory tract (Beta-2)
- Increases systolic blood pressure due to increased cardiac output (Beta-1) and vasoconstriction (Alpha)
- Increases coronary perfusion during cardiopulmonary resuscitation (CPR) by increasing aortic diastolic pressure.

Uses: Cardiopulmonary arrest, anaphylaxis, relief of respiratory distress due to bronchospasm, rapid relief of hypersensitivity reactions, open- angle glaucoma, to prolong the action of anesthesia, topically to stop bleeding.

Contraindications:

- Narrow angle glaucoma, shock, lactation, tachycardia, during labor (it may delay the 2nd stage of labor).

Side effects:

- Anxiety, headache, tremors, pallor, sweating, nausea, vomiting,
- Hypertension, tachycardia, cardiac arrhythmias, cardiac ischemia, and intracranial hemorrhage.
- Provoke chest pain in patients with underlying ischemic heart disease due to β-mediated cardiac effects
- Urinary retention in males with enlarged prostates due to α-stimulation.
- Uterine relaxation caused by β2-receptor activity can delay the second stage of labor.

Dose:

- Adrenaline ampules of 1ml containing 1 mg can be given by I.M injection., I.V. & S.C.
- 0.2 – 0.5 mg, IM or S.C. every 20 min – 4 hr as needed. For cardiac resuscitation, 0.5 mg diluted to 10 ml with normal saline may be administered I.V. or intracrdiac to restore myocardial contractility.

Nursing considerations:

- Never administer 1 : 100 solution IV., use 1 : 1000 mg sol. For I.V. use.
- Use tuberculine (1cc) syringe to measure adrenaline.
- Administer adrenaline using piggyback set to adjust the rate of infusion.
- Administer infusion by electronic infusion device for safety & accuracy.
- Closely monitor patients receiving I.V. adrenaline infusion.
- Note the client for signs of shock "loss of consciousness, clammy, cold skin, cyanosis…. etc.).
- Briskly massage site of S.C. or I.M. injection to hasten the action of the drug.

2. Albuterol: Salbutamol

Trade name: Ventolin, Ventocare

Class.: sympathomimetic agent, bronchodilator.

Action: stimulate β 2 receptors of the bronchi

Uses: Relaxes bronchial smooth muscle (Bronchial asthma), Decreases airway resistance (Bronchospasm due to bronchitis or emphysema), Parenteral for treatment of status asthmatics.

Dosage:

- Aerosol for inhalation: 0.18 – 0.2 mg (2 inhalations) Every 4 – 8 hours.
- Solution for inhalation: 1.25 mg in 2 – 5 ml.
- Oral syrup, tablets: 2 – 6 mg tid – qid.

Side effects:

Neurological

- Tremors
- Headache/dizziness
- Sweating
- Anxiety

Cardiovascular

- Tachycardia
- Hypertension
- Dysrhythmias
- Palpitations

Gastrointestinal

- Nausea / vomiting

Nursing considerations:

- Don't exceed the recommended dose.
- The contents of the container are under pressure, don't store near heat or open flames.
- When given by neubilization, use facemask or mouth-piece.
- Compress O2 or air at 6 – 10 L\min for 5-15 minutes.
- Observe client for evidence of allergic response.
- NEVER give the solution prepared to be given as inhalation by the IV Route. It may cause severe tachycardia.

3. Dopamine Hydrochloride:

Trade names: Intropin, docard

Class.: Direct & indirect acting adrenergic agent.

N.B.: Available for hospital use only on 5 ml ampules containing 200 mg dopamine hydrochloride.

Action:

- It is the immediate precursor of epinephrine in the body.

- It produces direct stimulation of β1 receptors resulting in increasing myocardial contraction, cardiac output as well as increase renal blood flow & sodium excretion.
- It exerts little effect on diastolic B.P. & induce fewer arrhythrrias that seen on other adrenergics.
- Alpha receptors, which are stimulated by higher doses of dopamine exerts vasoconstriction effects which can override the vasodilating effect.
- At higher doses, it stimulates α receptors leading to peripheral vasoconstriction.

Indications:

- Symptomatic bradycardia persisting after prior therapies
- Cardiogenic shock with signs/symptoms of CHF.
- Hypotension due to poor cardiac output.
- Shock associated with septicemia, trauma, heart surgery, renal failure & C.H.F.
- Cardiomyopathy.
- In lower doses (1-5 Mcg/kg/min) used in renal failure.

Contraindications:

- Hypovolemia
- Pheochromocytoma, uncorrectecd tachycardia, arrhythmias.
- Safety and efficacy is not established in children.

Side effects:

Cardiovascular

- Tachycardia
- Hypertension
- Increased O2 demand
- Ventricular irritability
- Chest pain (dyspnea)
- Wide QRS complex.

Gastrointestinal

- Nausea/Vomiting

Others:

- Headache.

Dosage and administration:

It is a potent drug and must be diluted before administration to the patient.

Suggested dilution:

Transfer contents of one ampule (5ml containing 200 mg of dopamine) by aseptic technique to either 250 or 500 ml bottle of sterile I.V. solution (saline, dextrose 5% or reinger lactate). These dilutions will yield a final concentration for administration as follows.

- 250 ml dilution contains 800 mcg\ml of dopamine.
- 500 ml dilution contains 400 mcg\ml of dopamine.

N.B. :

1- Solution stable after dilution for 24 hours.

2- Don't add dopamine to NaHCO3 or other alkaline I.V. solutions since the drug is inactivated in alkaline solution.

Rate of administration.:

Through a suitable I.V. needle or a catheter & through an electronic infusion pump, rate is regulated according to required dose.

Dose:

- Renal dose: 1-5 Mcg\kg\minute.
- Cardiac dose: start initially of 5 Mcg/kg/min then increase by increments up to a rate of 20-50 Mcg/kg/min.

Nursing considerations:

- Administer only by IV INFUSION (Not IV bolus nor IM)
- Drug must be diluted before use.
- Administer drug through a central line or a big vein (vein in the anticupital fossa is preferred over those in the hand).
- Stop the drug by small increments.
- Solution is stable for 24 hrs, protect it from light.
- Monitor B.P. & ECG continuously during drug administration.
- Monitor intake & output.
- Monitor patient for occurrence of side effects.
- Check I.V. site for signs of extravasation.
- Drug should be administered through electronic infusion device.

4. Ephedrine sulfate:

Trade names: Numacin, efedron nasal "nasal decongestant".

Class. : Direct & indirect- acting adrenergic agent.

Action: Release norepinephrine from storage sites, stimulate alpha, β1 + β2 receptors bronchodilation, nasal decongestant and increase strength of skeletal muscle so it may be used on myasthenia gravis.

Uses:

- Bronchial asthma
- Topically as a nasal decongestant.
- Myasthenia gravis.

Side effects:

- Urinary retention,
- painful urination,
- dry mouth, drowsiness,
- blurring of vision.

Dose:

- As a nasal decongestant 2-3 drops at each nostril every 4 hours.
- Not recommended for children under 6 years. Not to be used for 3-4 consecutive days.

Adrenergic blocking (sympatholytic) Agents

Adrenergic blocking drugs, also called sympatholytic drugs, are used to disrupt sympathetic nervous system function. These drugs work by blocking impulse transmission (and thus sympathetic nervous system stimulation) at adrenergic neurons or adrenergic receptor sites. Their action at these sites can be exerted by:

- Interrupting the action of adrenergic drugs.
- Reducing available norepinephrine.

Adrenergic blocking drugs are classified according to their site of action as:

- Alpha-adrenergic blockers (also called *alpha blockers*).
- Beta-adrenergic blockers (also called *beta blockers*).

Alpha-adrenergic blockers work by interrupting the actions of the catecholamines epinephrine and nor epinephrine at alpha receptors.

This results in:

- Relaxation of the smooth muscle in the blood vessels
- Increased dilation of blood vessels
- Decreased blood pressure

A. alpha adrenergic blocking agent

1. Phenoxybenzamine hydrochloride

Trade name: Dibenzyline

Action:

- Potent, nonselective, irreversible α-adrenoceptor antagonist.
- Blocks catecholamine induced vasoconstriction ⟶ decrease in total peripheral resistance and BP.
- Increase cardiac output due to some blockage of alpha 2 in cardiac sympathetic nerves.

Indications:

- Hypertensive in phaeochromocytoma.
- Relieve vasospasm in Raynaud's phenomenon and peripheral vasospasm
- To relieve urinary obstruction in patient with prostatic hypertrophy.

Adverse effects:

- CNS: fatigue, sedation, convulsions following rapid intravenous infusion.
- Eye: miosis
- CVS: postural hypertension, tachycardia, idiosyncratic profound hypotension within few minutes of starting infusion,
- Respiratory tract: nasal stuffiness.
- GIT: nausea, vomiting (oral administration), dry mouth after intravenous infusion
- Reproduction: inhibition of ejaculation.

Contra-indications:

- history of cerebrovascular accident; during recovery period after myocardial infarction (usually 3‑4 weeks);
- avoid infusion in hypovolaemia
- Renal impairment
- Breast-feeding may be present in milk

Dose:

- Capsules, phenoxybenzamine hydrochloride 10 mg.
- Dose by mouth, phaeochromocytoma, intially 10 mg daily increased by 10mg daily until hypertension controlled or treatment not tolerated; usual dose 1–2 mg/kg daily in 2 divided doses.
- Injection concentrate, phenoxybenzamine hydrochloride 50 mg/mL. To be diluted before use.

Nursing considerations:

- Dose by intravenous infusion (preferably through large vein), adjunct in severe shock (but rarely used) and phaeochromocytoma,
- 1mg/kg daily over at least 2 hours; do not repeat within 24 hours.
- Caution Owing to risk of contact sensitisation healthcare professionals should avoid contamination of hands.

Prazosin, terazosin, doxazosin, alfuzosin, and tamsulosin

Trade name: Prazosin (hypotens), Terazosin (Hytrin)

- Prazosin, terazosin, doxazosin, and tamsulosin are selective competitive blockers of the alpha 1 receptor. The first three drugs are useful in the treatment of hypertension.

- Tamsulosin and alfuzosin are indicated for the treatment of benign prostatic hypertrophy (also known as benign prostatic hyperplasia or BPH).
- Metabolism leads to inactive products that are excreted in the urine except for those of doxazosin, which appear in the feces. Doxazosin is the longest acting of these drugs.

Therapeutic uses:

- Mild to moderate chronic hypertension (more effective when used in combination with a diuretic or propranolol.
- Acute congestive heart failure (prazosin).
- To relieve urineray obstruction (terazosin).

Adverse action:

- CNS: dizziness, headache.
- CVS: postural hypotension, first dose phenomenon (a precipitous drop in standing BP after the first dose that results in syncope and occure especially in patients who are salt and volume depleted.
- Respiratory tract: nasal congestion.
- GIT: GI hyper-motility.
- Kidneys: salt and fluid retention.

Dose:

1. Prazosin

- First dose, .5 mg at bed time, orally
- Then 0.5 mg BD or TDS for 3-7 days.
- Followed by 1 mg BD or TDS, for 3-7 days.
- Thereafter increases gradually as required, upto max. of 20mg/day.

2. Terazosin

- Initially at bed time,orally.
- Titrate by approximately doubling dose at weekly intervals.
- Usual maintenance dose is 2-10mg OD.

B. Beta-adrenergic blocking agents:

Action: it combines with beta-adrenergic receptors to block the response to sympathetic nerve impulses, circulating catecholamines or adrenergic drugs.

Beta-adrenergic blockers are either selective or nonselective.

Nonselective beta-adrenergic blockers affect:

- beta1-receptor sites (located mainly in the heart)
- beta2-receptor sites (located in the bronchi, blood vessels, and uterus).

Nonselective beta-adrenergic blocking drugs include carvedilol, labetalol, levobunolol, nadolol, penbutolol, pindolol, propranolol, sotalol, and timolol. (Carvedilol and labetalol also block alpha1 receptors.)

Uses: hypertension, angina pectoris, cardiac arrhythmias, myocardial infarction, prophylaxis of migraine.

Side effects:

Beta-adrenergic blockers generally cause few adverse reactions; the adverse reactions that do occur are drug- or dose-dependent and include: hypotension, bradycardia, peripheral vascular insufficiency Beta-adrenergic blockers may affect how much antidiabetic drug you need, AV block, heart failure, bronchospasm, diarrhea or constipation, nausea, vomiting, abdominal discomfort, anorexia, flatulence, rash, fever with sore throat, spasm of the larynx, respiratory distress (allergic response).

Contraindications:

- Bradycardia, C.H.F., cardiogenic shock, diabetes, thyrotoxicosis, chronic bronchitis, asthma, bronchospasm, emphysema.

Treating overdose:

- Inducing vomiting, gastric lavage.
- Artificial respiration.
- Give atropine sulfate 0.6 mg (up to 3 mg) and glycogan for the treatment of bradycardia.
- Treat hypoglycemia and hypokalemia.
- I.V fluids.
- Adrenaline or dopamine to increase Blood pressure.

Drug interactions

Many drugs can interact with beta-adrenergic blockers to cause potentially dangerous effects. Some of the most serious effects include cardiac depression, arrhythmias, respiratory depression, severe bronchospasm, and severe hypotension that can lead to vascular collapse. Here are some others:

- Increased effects or toxicity can occur when digoxin, calcium channel blockers (primarily verapamil), and cimetidine are taken with beta-adrenergic blockers.
- Decreased effects can occur when antacids, calcium salts, barbiturates, anti-inflammatories (such as indomethacin and salicylates), and rifampin are taken with beta-adrenergic blockers.
- The requirements for insulin and oral antidiabetic drugs can be altered when taken with beta-adrenergic blockers.
- Potential lidocaine toxicity can occur when it's taken with betaadrenergic blockers.
- The ability of theophylline to produce bronchodilation is impaired by nonselective beta-adrenergic blockers.
- Clonidine taken with a nonselective beta-adrenergic blocker can result in life-threatening hypertension during clonidine withdrawal.
- Sympathomimetics and nonselective beta-adrenergic blockers can cause hypertension and reflex bradycardia.

Nursing considerations:

If beta-adrenergic blocking drugs are prescribed, review these points with the patient and his caregivers**:**

- Instruct patient/family to take blood pressure and pulse.
- Provide written instructions as when to call physician (e.g. HR below 50 beat/min).
- Consult the physician before interrupting the therapy.
- Some drugs lead to blurring of vision, so that tell patients not to engage in activities need mental alertness.
- Instruct patient to dress warmly during cold weather.
- Diabetic patient should be very careful about symptoms of hypoglycemia.
- Report any asthma-like symptoms.
- Take the drug exactly as prescribed, even if signs or symptoms are relieved.

- Don’t stop the drug suddenly. This can worsen angina or precipitate myocardial infarction.
- Take the oral form of the drug with meals to enhance absorption.
- Don’t take over-the counter drugs or herbal remedies without medical consent.
- Be aware of potential adverse reactions and report unusual effects of the drug.
- Avoid performing hazardous activities until central nervous system effects of the drug are known

1. Atenolol:

Trade name: Normatin.

Classification: Beta1-selective adrenergic blocking agent**,** Antihypertensive

Uses:

- Treatment of angina pectoris due to coronary atherosclerosis
- Hypertension, as a step 1 agent, alone or with other drugs, especially diuretics
- Treatment of MI
- Unlabeled uses: Prevention of migraine headaches; alcohol withdrawal syndrome, treatment of ventricular and supraventricular arrhythmias

Adverse effects:

- Allergic reactions: Pharyngitis, respiratory distress
- CNS: Dizziness, vertigo, fatigue, emotional depression, paresthesias, sleep disturbances, hallucinations, disorientation, slurred speech.
- Dermatologic: Rash, pruritus, sweating, dry skin
- CV: Bradycardia, CHF, cardiac arrhythmias, sinoatrial or AV nodal block, tachycardia, claudication, pulmonary edema.
- Eyes : Eye irritation, dry eyes, conjunctivitis, blurred vision
- GI: Gastric pain, constipation, diarrhea, nausea, vomiting, anorexia.
- GU: Impotence, decreased libido, frequent urination.
- Musculoskeletal: Joint pain, arthralgia, muscle cramps
- Respiratory: Bronchospasm, dyspnea, cough, bronchial obstruction, nasal stuffiness, rhinitis, (less likely than with propranolol).

Contraindications

- Contraindicated with sinus bradycardia, second- or third-degree heart block, cardiogenic shock, CHF, pregnancy.

Dosage:

- Tablets: 50 mg or 100 mg daily
- Initial dose is 50 mg, if there is no response, increase dose to 100 mg daily.
- Angina pectoris: Initially, 50 mg PO daily. If optimal response is not achieved in 1 wk, increase to 100 mg daily; up to 200 mg/day may be needed.
- Acute MI: Initially, 5 mg IV given over 5 min as soon as possible after diagnosis; follow with IV injection of 5 mg 10 min later. Switch to 50 mg PO 10 min after the last IV dose; follow with 50 mg PO 12 hr later. Thereafter, administer 100 mg PO daily or 50 mg PO bid for 6–9 days or until discharge from the hospital.

Nursing considerations:

- For IV use, the drug may be diluted in sodium chloride, dextrose, on dextrose saline.
- Do not discontinue drug abruptly after long-term therapy (hypersensitivity to catecholamines may have developed, causing exacerbation of angina, MI, and ventricular arrhythmias). Taper drug gradually over 2 wk with monitoring.
- Take drug with meals if GI upset occurs.
- Do not stop taking this drug unless told to do so by a health care provider.
- Avoid driving or dangerous activities if dizziness or weakness occurs.
- You may experience these side effects: Dizziness, light-headedness, loss of appetite, nightmares, depression, sexual impotence.

2. Valolol:

Trade name: Corgard

Class: Beta-adrenergic blocking agent.

Action: manifests both beta 1 and beta 2 adrenergic blocking.

Uses: hypertension, angina pectoris.

Dose: 40-80 mg daily.

3. Propranolol hydrochloride

Trade name: Inderal, Deralin

Classification: beta-adrenergic blacking agent, antiarrhythmic.

Action: manifests both beta1 and beta 2 adrenergic blocking activity.

Indication:

1. Angina pectoris.
2. Hypertension.
3. Cardiac arrhythmias.
4. Prophylaxis of migrin.
5. Prophylaxis of MI.
6. Pheochromocytoma

Additional side effects: psoriasis-like eruptions.

Contraindications:

- Contraindicated with allergy to beta-blocking agents,
- sinus bradycardia,
- second- or third-degree heart block,
- cardiogenic shock, CHF,
- bronchial asthma, bronchospasm, COPD,
- pregnancy (neonatal bradycardia, hypoglycemia, and apnea, and low birth weight with long-term use during pregnancy),
- lactation.

Dosage:

- Tablets: initial dose of 40 mg bid, then 120-240 mg in 2-3 divided doses.

4. Timolol maleate:

Trade name: Blocadren, Timpotic.

Classification: Ophthalmic agent, beta-adrenergic agent.

Action: both beta1 and beta 2 receptors blocking activity.

Uses:

- Tablets: for hypertension.
- Ophthalmic solution: chronic open-angle glaucoma

Para sympathomimetic (cholinergic) Drugs

There are two major classes of cholinergic drugs:

- *Cholinergic agonists* mimic the action of the neurotransmitter acetylcholine.
- *Anticholinesterase drugs* work by inhibiting the destruction of acetylcholine at cholinergic receptor sites.

The parasympathetic maintains essential bodily functions, such as digestive processes and elimination of wastes, and is required for life. It usually acts to oppose or balance the actions of the sympathetic division and is generally dominant over the sympathetic system in rest and digest.

The receptors of parasympathetic nervous system are classified into:

a) muscarinic type b) nicotinic type.

Cholinergic drugs can be divided into two classes:

1- Directly acting drugs that mimic the action of acetylcholine e.g. muscarine, pilocarpine, carbachol.

2- Indirectly acting drugs that increase the concentration of acetylcholine, usually by inhibiting acetylcholinesterase, e.g neostigmine, physostigmine.

Physiological Responses to Cholinergic Stimulation:

Eye	contraction - miosis contraction - near vision
heart	decrease in HR decrease in contractility decrease in conduction ± block ± decrease in contractility
Blood vessels	Vasodilation
lung	constriction increased secretion
GIT	increased motility & tone sphincter relaxation increased secretion
Bladder	Detrusor contraction Trigone & internal sphincter relaxation
Pancreas Salivary Lacrimal Pharyngeal	increased secretion
Sweat Glands	increased secretion
Sexual organs	erection (male)

Cholinergic agonists are used to:

- Treat atonic (weak) bladder conditions and postoperative and postpartum urine retention
    - Treat GI disorders, such as postoperative abdominal distention and GI atony
    - Reduce eye pressure in patients with glaucoma and during eye surgery
    - Treat salivary gland hypofunction caused by radiation therapy

Adverse reactions

- Nausea and vomiting
- Cramps and diarrhea
- Blurred vision
- Decreased heart rate and low blood pressure
- Shortness of breath
- Urinary frequency
- Increased salivation and sweating.

The cholinergic agonists are rarely administered by IM or IV injection because they're almost immediately broken down by cholinesterase in the interstitial spaces between tissues and inside the blood vessels. Moreover, they begin to work rapidly and can cause a *cholinergic crisis* (a drug overdose resulting in extreme muscle weakness and possible paralysis of the muscles used in respiration).

1. Pyridostigmine Bromide:

Trade name: Mestinon.

Class. : Indirectly acting, cholinergic – acetylcholinesterase inhibitor.

Action:

- By inhibiting the enzyme cholinestrase, that lead to increase the concentration of acetylcholine at the myoneural junction which facilitate the transmission of nerve impulse across the myoneural junction, that lead to increase muscle strength in myasthenia gravis.
- It has a slower onset, longer duration of action & fewer side effects than neostigmine.
- Atropine may be given to control side effects.

Uses: Myasthenia Gravis.

Contraindication: Hypersensitivity, Bradycardia, Hypotension, Astha, Hyperthyroidism GI obstruction peptic ulcer.

Side effects: Skin rash, thrombophlebitis after I.V. use, Nausea, vomiting, diarrhea, Bradycardia, Hypotension, headache, dizziness, Urinary frequency (Incontinence) , bronchospasm

Dose: Tab. 600 mg daily.

2. Neostigmine Bromide:

Trade name: Prostigmin**.**

Dose: Tab. 150 mg daily.

All information as pyridostigmine

3. Ophthalmic cholinergic (Miotic) Agents:

- Commonly used for the treatment of glaucoma.

Action:

- These agents inhibit the enzyme cholinestrase which lead to accumulation of acetylcholine & stimulate the ciliary muscles & increases contraction of the iris sphincter muscle. This opens the angle of the eye & results in increased the outflow of aqueous humor & consequently in decreased of intraocular pressure .

Uses: Glaucoma, Diagnosis & treatment of esotropia.

- Antidote against harmful effects of atropine-like drugs in patients suffering from glaucoma.

Contraindications: hypertension, inflammatory eye diseases (critis), asthma, history of retinal detachment, bradycardia, peptic ulcer.

Side effects: Pain in eye, blurring of vision, Headache, Failure to accommodate to darkness, retinal detachment. Diarrhea, hypotension, salivation, bronchial constriction.

Nursing considerations:

- Have adrenaline & atropine available for emergency treatment.
- Stress the importance of taking eye drops exactly as prescribed
- Minimize side effects by taking medication at bed-time.
- Advice client not to drive a car for 1-2 hrs after administering eye drops.
- Notify the physician, if side effects occurred.
- Cold compresses for painful eye spasms.
- Frequent eye examinations.

1. **Carbachol:**

   Trade name: Isopto carbachol.

   Uses: glaucoma.

   Dose: 1 gtt of the solution in the conjunctiva 1-2 times daily.

2. **Physostigmine salicylate:**

   Trade name: Eserine salicylate.

   Uses: glaucoma.

   Dose: 1 gtt of the 0.25 or 0.5 % solution bid-tid.

3. **Pilocarpine nitrate:**

   Uses: glaucoma

   Dose: 1 gtt of 1-4 % solution gid.

Cholinergic Blocking (Parasympatholytic) Drugs Action

These agents prevent the neurotransmitter acetylcholine from combining with receptors on the muscarinic site & nicotonic site.

The major anticholinergic drugs are the belladonna alkaloids:

- Atropine (see *Anticholinergic drugs: Atropine*)
- Homatropine
- Ipratropium
- Scopolamine.

The main effects:

- Reduce spasm of smooth muscle such as spasm of the urinary bladder or intestines.
- To block vagal impulses to the heart which will increase heart rate & conductivity.
- To suppress or decrease gastric secretions, perspiration, salivation and secretion of bronchial mucus.
- To relax the sphincter muscles of the iris & cause pupillary dilation (mydriasis) & loss of accommodation for near vision.
    - Act on CNS producing such reactions as depression (scopolamine) or stimulation (toxic dose of atropine) to produce ant parkinsonism effect.

Contraindications:

- Glaucoma, tachycardia, myocadial ischemia
- Prostate hypertrophy, myasthenia gravis, paralytic ileus, Mental impairment, lactation, hepatic disease.

Side effects:

- Nausea, vomiting, dry mouth, constipation, heartburn, dizziness, drowsiness, headache, insomnia, blurring of vision, photophobia, flashing, euphoria, hallucination flushing of the skin.

1) Atropine sulfate:

Class.: Cholinergic blocking agent.

Action: It is a parasympatholytic agent, which cause relaxation of smooth muscles & inhibition of secretary glands.

Uses:

- Symptomatic bradycardia
- Preoperative reduction of secretions and blockage of cardiac vagal reflexes
- Adjunct treatment of peptic ulcer disease
- Functional GI disorders, irritable bowel syndrome.
- Treatment of spastic disorders of biliary tract .
- During anesthesia to control salivation & bronchial secretions.
- Parkinsonism.
- Anti-arrhythmic (prophylaxis).
- Prophylaxis and treatment of toxicity due to cholinestrase inhibitor including organophosphate pesticides.
- Ophthalmologic treatment of uveitis.

Contraindications: See parasympatholytics

Side effects:

- Dry mouth
- Reduced bronchial secretions
- Increased heart rate
- Blurred vision
- Decreased sweating.

Dose:

- Tablets: 0.3 – 1.2 mg Q 4-6 hr
- Available in 1ml-ampoule containing 1 mg atropine."
- IM, I.V. & S.C. 0.4 – 0.6 mg Q 4-6 hour for anticholinergic action.

N.B. : For treatment of toxicity from cholinestrase inhibitor " organophosphorus poisoning", give 2-4 mg IV initially then 2 mg every 5-10 minutes until muscarinic symptoms disappear and signs of atropine toxicity begins to appear like dilation of pupils, flushing of face & tachycardia.

Nursing considerations:

- Monitor for adverse reactions, such as headache, tachycardia, restlessness, dizziness, blurred vision, dry mouth, urinary hesitancy, and constipation.
- Monitor vital signs, cardiac rhythm, urine output, and vision for signs of impending toxicity.
- Provide stool softeners or bulk laxatives as ordered for constipation.
- Check dosage & measure the drug exactly.
- Assess for history of asthma, glaucoma, ulcer .. etc.
- Frequent mouth care.
- Assess client for change in pulse rate.
- In case of blurring of vision, assist on ambulating & give safety measures.

2) Scopalamine Hydrobromide :

Trade name: Hyoscine.

Class.: cholinergic blocking agent.

Action: It is a parasympatholytic agent, depress the cerebral cortex, especially the motor area, act as a powerful hypnotic.

Uses:

- Motion sickness (prevention and control of nausea and vomiting).
- Preanesthetic.
- Antiarrhythemic.
- Mydriatic and cycloplegic.
- Adjunctive with other drugs to treat GIT ulcers.
- With other narcotics to treat biliary colic.

Contraindications:

- Hypersensitivity.
- Glaucoma
- Bronchial asthma
- Cardiac arrhythmias
- Pregnancy
- Lactation.

Dose:

- Oral: O.25 mg I hour before travel (for motion sickness).
- Parenteral: 0.32-0.6 mg SC or IM.

Side effects:

- Pupil dilation, photophobia, blurred vesion, headache, drowziness. dry mouth, constipation, nausea, vomiting, tachycardia, arrhythmia, suppression of lactation, flushing, nasal congestion.

Nursing considerations: As atropine.

3) Ipratropium bromide:

Trade name: atrovent, irovent

Class.: cholinergic blocking agent, antimuscarinic drug.

Action: It is a parasympatholytic agent, chemically related to atropine antagonized the action of acetylcholine. It causes bronchodilation and inhibits secretion from nasal mucosa gland.

Uses:

- Bronchodilator for maintenance treatment of bronchospasm associated with COPD, chronic bronchitis, and emphysema (solution, aerosol).
- Nasal spray: symptomatic relief of rhinorrhea with perennial rhinitis, common cold.

Contraindications : Hypersensitivity to atropine or its derivative, soy bean or peanut allergy

Dosage:

- Aerosol inhalation, ipratropium bromide 20 micrograms/ metered inhalation. two inhalations qid. Do not exceed 12 inhalations/ 24 hr.
- Nebulizer solution, ipratropium bromide 250 micrograms/ mL.

- By inhalation of nebulized solution, reversible airways obstruction in chronic obstructive pulmonary disease, 250–500 micrograms 3–4 times daily.
- Acute bronchospasm; 500 micrograms repeated as necessary; CHILD under 5 years 125–250 micrograms, max. 1mg daily; 6–12 years 250 micrograms, max. 1mg daily.

Adverse effects:

- **CNS**: headache, dizziness, fatigue, insomnia, blurred vision.
- **GI**: nausea, dry mouth.
- **Respiratory**: bronchitis, dyspnea, cough.

Nursing considerations:

- Protect solution for inhalation from light and store unused vials in foil pouch.
- Use nebulizer mouthpiece instead of face mask to avoid blurred vision or aggravation of narrow angle glaucoma.
- Can mix albuterol in nebulizer for up to 1 hr.
- Ensure adequate hydration, control temperature to avoid hyperpyrexia.\
- Teach patient proper use of inhalator.

Anti – infectives

Classification of Antimicrobials:

1. Antibacterial agents.
2. Antiviral agents.
3. Antiprotozoal agents.
4. Antifungal agents.
5. Anthelmintic agents.

- Many new bacteria appear each year, and researchers are challenged to develop new antibiotics to deal with each new threat.
- As newer drugs are introduced, our ability to fight infections increases even more. However, despite impressive advances, continued progress is needed.
- There remain organisms that respond poorly to available drugs; there are effective drugs whose use is limited by toxicity; and there is, because of evolving microbial resistance, the constant threat that currently effective antibiotics will be rendered useless.
- **The beginning of modem medicine is generally related to 2 events:**
- The proof by Pasteur that many diseases are caused by microorganisms.
- The discovery of effective anti-infective drugs.
    - Penicillin in 1929 by Fleming
    - Sulfonamides in 1938 by Domagk

Antibiotics: Antibiotic is a chemical substance produced by a *microorganism that inhibits the growth of or* kills other microorganisms.

Broad-spectrum antibiotics are designed to target a wide range of different bacteria. They work against both Gram-positive and Gram-negative bacteria, which have different cell wall structures. Examples of broad-spectrum antibiotics include tetracyclines, aminoglycosides, cephalosporins, and fluoroquinolones. These target multiple bacterial processes so are effective against many types.

Narrow-spectrum antibiotics: target specific types of bacteria. They may only work against certain Gram-positive or Gram-negative bacteria. Examples of narrow-spectrum antibiotics include penicillins, macrolides, and sulfonamides. Penicillins, for instance, disrupt cell wall synthesis in Gram-positive bacteria.

Extended-spectrum antibiotics: are a subset of broad-spectrum antibiotics that cover an even wider range of bacteria. Like broad-spectrum antibiotics, they target both Gram-positive and

Gram-negative bacteria. However, they are also effective against a wider variety of multi-drug resistant bacteria compared to traditional broad-spectrum antibiotics. Examples include third and fourth generation cephalosporins, carbapenems, and some fluoroquinolones. These expanded-spectrum antibiotics were developed to combat newly emerged resistant bacteria like MRSA and drug-resistant pneumococci.

Antimicrobial Activity:

Bacteriostatic: drugs arrest the growth of bacteria so it limits the spread of infections

Bactericidal: drugs that kill bacteria

Choice of an antimicrobial agent

The choice is depending on patient, infecting organism and drug.

1. Patient factors:
   - Age
   - Renal & hepatic function
   - Drug allergy
   - Pregnancy & lactation
2. Organism related considerations:
   - Clinical diagnosis.
   - Culture
3. Drug factors:
   - Spectrum of activity
   - Site of infection
   - Safety and efficacy
   - Cost

Most patients require empiric therapy (immediate administration of drug(s) prior to bacterial identification and susceptibility testing).

4. Risk factors for multidrug-resistant organisms.
   - Infections with multi drug-resistant pathogens need broader antibiotic coverage when initiating empiric therapy.
   - Common risk factors for infection with these pathogens include prior antimicrobial therapy in the preceding 90 days.
5. Route of drug administrations:

- The oral route of administration is appropriate for mild infections that can be treated on an outpatient basis.
- Parenteral is used for drugs that are poorly absorbed from the Gl tract and for patients with sever infections who need of higher concentrations of antimicrobial agents.

Which member of antibiotic class to use first?

The general view of antibiotic usage is that:

- if a choice of antibiotics within a drug class is available less active members of the class should be used first.
- The wisdom behind this is that: when resistance to this drug emerges, then the more active drug can be introduced to overcome it.
- The flaw in this strategy is that the development of resistance to the less effective member of a drug class often results in decreased susceptibility or resistance to the more active drugs within that class.

Antimicrobial Activity

Bacteriostatic; only arrest the growth and replication of bacteria at drug levels achievable in the patient.

Bactericidal; effectively kill >99.9% within 18 to 24 hours of incubation under specific laboratory conditions.

Bactericidal:	**Bacteriostatic**
β-Lactatm	Erythromycin
Aminoglycosides	Tetracyclines
Co-trimexazole	Chloramphinicol
Quinolones	Sulfonamides, Trimethoprim

Combinations of Antimicrobial Drugs

It is therapeutically advisable to treat patients with a single agent that is most specific to the infecting organism.

This strategy:

- Reduces the possibility of superinfections.
- Decreases the emergence of resistant organisms.
- Minimizes toxicity. However, in some situations, combinations of antimicrobial drugs are advantageous or even required.

Hypersensitivity

- Hypersensitivity reactions to antimicrobial frequently occur.

For example, the **penicillins** and **cephalosporins** can cause serious hypersensitivity problems, ranging from urticaria (hives) to anaphylactic shock.

Some reactions may be related to the rate of infusion, such as "Red man syndrome" seen with rapid infusion of *vancomycin.*

Patients with a documented history of Stevens-Johnson syndrome (SJS) or toxic epidermal necrolysis reaction, to an antibiotic should never be rechallenged, not even for antibiotic desensitization.

Antiepileptic drugs
Carbamazepine
Phenytoin
Phenobarbital
Antibiotic sulfonamide drugs
Sulfamethoxazole
Sulfadiazine
Sulfapyridine
Sulfamethizole
B- Lactam antibiotics
Cephalosporins
Carbapenems
Penicillins
Drugs induce SJS

Classification Antimicrobial

Antimicrobial drugs can be classified according to:

1. Chemical structure (β-lactams or aminoglycosides).
2. Mechanism of action (for example, cell wall synthesis inhibitors, protein synthesis inhibitors).
3. Activity Against particular types of organisms (for example, bacteria, fungi, or viruses).

Drug resistance

It refers to unresponsiveness of a microorganism to an AMA, and is similar to the phenomenon of tolerance seen in higher organisms (What the pathogen does to the drug).

Natural resistance: Some microbes have always been resistant to certain AMAs. They lack the metabolic process or the target site which is affected by the particular drug. This is generally a

group or species characteristic, e.g. gram-negative bacilli are normally unaffected by penicillin G, or *M. tuberculosis* is insensitive to tetracyclines. This type of resistance does not pose a significant clinical problem.

Acquired resistance: It is the development of resistance by an organism (which was sensitive before) due to the use of an AMA over a period of time. This can happen with any microbe and is a major clinical problem. However, development of resistance is dependent on the microorganism as well as the drug. Some bacteria are notorious for rapid acquisition of resistance, e.g. staphylococci, coliforms, tubercle bacilli. Others like *Strep. pyogenes* and spirochetes have not developed significant resistance to penicillin despite its widespread use for > 50 years. Gonococci quickly developed resistance to sulfonamides, but only slowly and low-grade resistance to penicillin. However, in the past 30 years, highly penicillin resistant gonococci producing penicillinase have appeared**.**

Resistance may be developed by mutation or gene transfer.

- **Mutation:** It is a stable and heritable genetic change that occurs spontaneously and randomly among microorganisms. It is not induced by the AMA.
- **Gene transfer:** infectious resistance from one organism to another.

Prevention of drug resistance:

It is of utmost clinical importance to control development of drug resistance.

Measures are:

a. Selective and adequate or correctly use of AMAs should be made. For acute localized infections in otherwise healthy patients, symptom determined shorter courses of AMAs are being advocated now.
b. Prefer rapidly acting and selective (narrowspectrum) AMAs whenever possible; broad-spectrum drugs should be used only when a specific one cannot be determined or is not suitable.
c. Use combination of AMAs whenever prolonged therapy is undertaken, e.g. tuberculosis.
d. Infection by organisms that have high ability for developing resistance, e.g. *Staph. aureus, E. coli, M. tuberculosis, Proteus,* etc. must be treated intensively.

Superinfection

This refers to the appearance of a new infection as a result of antimicrobial therapy.

- Use of most AMAs causes some alteration in the normal microbial flora of the body. The normal flora contributes to host defence by elaborating substances called *bacteriocins* which inhibit pathogenic organisms
- Complete the suppression of body flora, greater are the chances of developing superinfection. Thus, it is commonly associated with the use of broad/extended-spectrum antibiotics, such as tetracyclines, chloramphenicol, ampicillin, newer cephalosporins; especially when combinations of these are employed. Tetracyclines are more prone than chloramphenicol and ampicillin is more prone than amoxicillin to cause superinfection diarrhoeas because of incomplete absorption—higher amounts reach the lower bowel and cause greater suppression of colonic bacteria.
- Superinfections are more common when host defence is compromised.
- Sites involved in superinfection are those that normally harbor commensals, i.e. oropharynx; intestinal, respiratory and genitourinary tracts; occasionally skin.
- Superinfections are generally more difficult to treat.

Conditions predisposing to superinfections

- Corticosteroid therapy
- Leukaemias and other malignancies, especially when treated with anticancer drugs (these are also immunosuppressants and decrease WBC count)
- Acquired immunodeficiency syndrome (AIDS)
- Agranulocytosis
- Diabetes**,** disseminated lupus erythematosus

Nutritional deficiencies

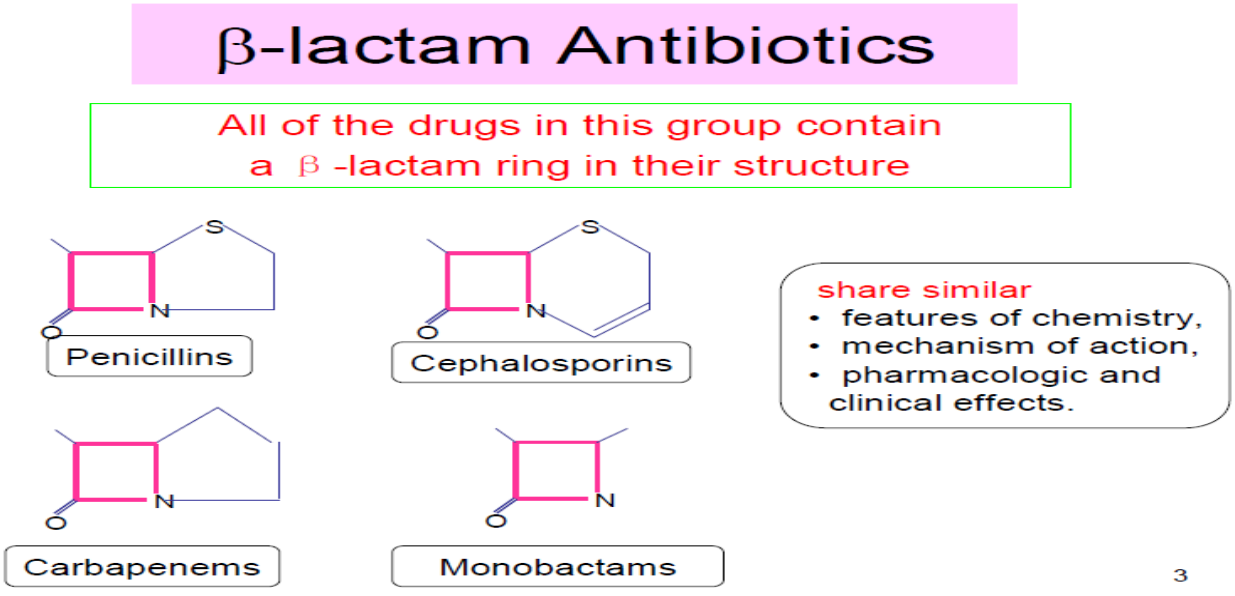
Some of the B complex group of vitamins and vit K synthesized by the intestinal flora is utilized by man. Prolonged use of antimicrobials which alter this flora may result in vitamin deficiencies. e.g, Neomycin causes morphological abnormalities in the intestinal mucosa—steatorrhoea and malabsorption syndrome can occur.

INHIBIT		CLASIFICATION		ANTIBIOTICS			
Cell Wall Synthesis	Beta Lactams	Penicillins		***Penicinillase – Sensible***			
			Natural Penicillins (narrow spectrum)	Penicillin G: Na, K, Procainic, Benzathine (IV, IM) Penicillin V: VO			
			Aminopenicillins (broad spectrum)	Ampicillin Amoxicillin			
			Penicinillase – Resistant (very narrow spectrum)				
			Nafcillin	Oxacillin	Dicloxacillin		
			Antipseudomonal (extended spectrum)				
			Carboxipenicillins	Ticarcillin Carbenicillin			
			Ureidopenicillins	Piperacillin Azlocillin Mezlocillin			
		Cephalosporins	1° Generation	Cefazolin	Cephalexine	Cephapirin	
				Cefadroxil	Cephadrine	Cephalotin	
			2° Generation	Cefuroxime	Cefamandole	Cefprozil	
				Cefoxitin	Cefonicid	Cefmetazole	
				Cefotetan	Cefaclor		
			3° Generation	Cefoperazone	Ceftriaxone	Ceftazidime	
				Cefpodoxime	Ceftizoxime	Cefotaxime	
				Cefdinir	Ceftibuten	Cefixime	
				Cefditoren			
			4° Generation	Cefepime	Cefpirome *		
			5° Generation	Ceftaroline			
		Carbapenems		Meropenem	Ertapenem	Doripenem	Imipenem + Cylastatine
		Monobactams		Aztreonam			
		**** Beta-lactamase inhib.		Sulbactam	Tazobactam	Clavulanic Acid	
	No lactam	Glycopeptides		Vancomycin	Bacitracin		
				Teicoplanin	Polymyxin B		
Protein Synthesis		30S	Amino-glycosides	Gentamycin	Neomycin	Streptomycin	
				Amikacin	Tobramycin		
			Tetracyclins	Doxycycline	Demeclocylin *	Minocycline	
				Tetracyclin	Tigecyclin		
		50S	Oxazolidonones	Linezolid			
			Streptogramins	Quinupristin/Dalfopristin			
			Cloramphenicol				
			Macrolides	Erythromycin	Azithromycin	Clarithromycin	
			Lincosamides	Clindamycin	Lincomycin		
DNA topoisomerases		Fluorquinolones		Ciprofloxacin	Norfloxacin	Levofloxacin	Ofloxacin
				Sparfloxacin	Moxifloxacin	Gemifloxacin	Enofloxacin
		Quinolones		Nalidixic Acid			
Folic Acid Synthesis		Sulfonamides		Sulfamethoxazole (SMX)	Ag Sulfadiazine	Sulfasalazine	Sulfisoxazole
		DHFR inhibitors		Trimethroprim (TMP)	Pirymethamine		
DNA (damage)		Metronidazole					
mRNA synth.		Rifampim					

Penicillins

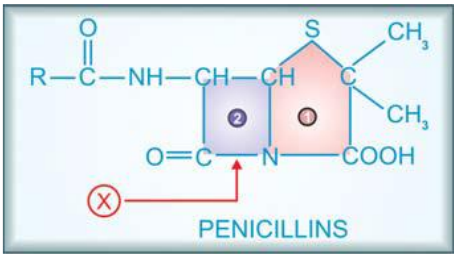
Penicillins remain one of the most important and useful antibacterial drugs, despite the availability of numerous others. The penicillins can be divided into four groups:

- Natural penicillins (penicillin G benzathine, penicillin G potassium, penicillin G procaine, penicillin G sodium, penicillin V potassium)
- Penicillinase-resistant penicillins (dicloxacillin sodium, cloxacillin sodium, nafcillin sodium)
- Aminopenicillins (amoxicillin, ampicillin, amoxicillin– clavulanate potassium)
- Extended-spectrum penicillins (carbenicillin indanyl sodium, ticarcillin disodium).
- Most penicillins should be given on an empty stomach (1 hour before or 2 hours after a meal) to enhance absorption. Penicillins that can be given without regard to meals include amoxicillin, penicillin V, and amoxicillin–clavulanate potassium.
- Penicillins are distributed widely to most areas of the body, including the lungs, liver, kidneys, muscle, bone, and placenta. High concentrations also appear in urine, making penicillins useful in treating utis.
- Penicillins are metabolized to a limited extent in the liver to inactive metabolites. Most penicillins are excreted 60% unchanged by the kidneys. Nafcillin and oxacillin, however, are excreted in bile



Penicillins

Penicillin antibiotics were among the first medications to be effective against many bacterial infections caused by staphylococci and streptococci. Penicillins are still widely used today, though many types of bacteria have developed resistance following extensive use. All penicillins are β-lactam antibiotics.



Chemical structure of penicillins. (1) Thiazolidine ring; (2) ®-lactam ring; (X) Bond

Classification of Penicillins:

1. Natural Penicillin:
   - Penicillin G (benzyl penicillin).
   - Procaine Penicillin G IM 12-24 hrs
   - benzathine Penicillin G IM weeks
   - penicillin V (phenoxymethyl penicillin).
2. Anti-staphylococcal Penicillins:
   - Cloxacillin, dicloxacillin, oxacillin and methicillin
3. broad spectrum Penicillins:
   - Amoxicillin and Ampicillin
4. Antipseudomonal penicillin:
   - Piperacillin, Ticaracillin, Carbenicillin, Aziocillin and mezocillin
5. β-lactamase inhibitors:
   - Clavulanic acid, sulbactam and tazobactam.
6. Combination penicillin:
   - amoxicillin + Clavulanic acid
   - ampicillin + sulbactam

Action of Penicillin: Inhibit cell wall synthesis; they inhibit cell division & growth of bacteria.

Uses:

Advantages of Penicillin

- safe

disadvantages of Penicillin

- Resistance (Pencillineas or B-lactmase enzyme)
- Short action

- Narrow+ve only
- Penicilline G strong hydrolysis with HCL

Clinical uses of Penicillin

Uses of penicillin G & V:

- Syphilis ,gonorrhea , drug of choice in gas Gangrene
- Endocarditis, meningitis and pneumonia Staphylococci
- Respiratory tract streptococci

Uses of broad spectrum Penicillin

- Urinary tract infection. E. coli
- Lower respiratory tract infection.
- Salmonella, shigella H. pylori and H. infulanza
- Gas gangrene

Uses of Antipseudomonase penicillin:

- Serious infection such as bacteremia and pneumonia.
- wounds, burns and UTI

Prophylactic uses of penicillins:

- Rheumatic fever recurrence, streptococci
- surgical procedures in patient with valvular disease , dental extraction, tonsillectomy, intestinal operation and bacteremia.

Contraindications:

- Allergy to penicillins & cephalosporins.
- Use with caution in patients with renal disease (lowered doses are necessary because excretion is reduced)

NB: The penicillins should not be given by intrathecal injection as they can cause encephalopathy, which can be fatal.

Side effects:

1. Hypersensitivity reactions. Approximately 10% of patients self-report allergy to penicillin Cross-allergic reactions occur among the β- lactam antibiotics.
2. Secondary infections like oral candidiasis.
3. GI upset: Nausea, vomiting, Pseudomembranous colitis (diarrhea).

4. Nephritis: Penicillins, particularly methicillin, have the potential to cause acute interstitial nephritis. [Note: Methicillin is therefore no longer used clinically.]
5. Hematologic toxicities: Decreased coagulation may be observed with high doses of piperacillin and nafcillin

Penicillins may interact with various drugs:

a. Probenecid increases the plasma concentration of penicillins.
b. Penicillins reduce tubular secretion of methotrexate in the kidneys, increasing the risk of methotrexate toxicity.
c. Tetracyclines and chloramphenicol reduce the bactericidal action of penicillins.
d. Neomycin decreases the absorption of penicillin V.
e. The effectiveness of hormonal contraceptives is reduced when they're taken with penicillin V or ampicillin

Dosage: Individualized.

Nursing considerations for penicillin:

a. Monitor the patient for adverse effects, such as seizures, anaphylaxis, leukopenia, and thrombocytopenia.
b. Obtain a specimen for culture and sensitivity tests before giving the first dose.
c. Therapy may begin pending results
d. These antibiotics should be injected slowly to minimize local irritation.
e. I.M. injections are made deeply into the gluteal muscle.
f. Assess regularly for allergic reactions. If reaction occurs the drug must be discontinued immediately, Epinephrine, O2, antihistamines, & corticosteroids must be immediately available.
g. Used only for I.M. use & do not massage after injection because rate of absorption should not be increased.
h. Take oral penicillin 1 hr before or 2-3 hr after meals.
i. Complete entire prescribed course of therapy.

Penicillin G: Procaine

PnG is a narrow spectrum antibiotic; activity is limited primarily to gram-positive bacteria and few others.

Trade name: Bepen, Duplo-penicillin, procaine benzyl penicillin. .

Class.: Antibiotic, long acting penicillin.

Action: inhibit bacterial cell wall and causing cell death.

Uses: Streptococcal infections (without bacteremia) of:

- Upper respiratory tract such as Tonsillitis, pharyngitis.
  - Otitis media
  - Skin and soft tissue infections.
  - Scarlet fever.
  - Acute glomerulonephrites.
  - Gonorrhea and syphilis.
  - Rheumatic fever.
  - Pneumococcal infections

Dosage form:

- Vials containing 1.2 MU, 2 MU.
- Penicillin G is inherently rapid in action while procaine penicillin G provides penicillin G in a prolonged action.

Contraindications: Hypersensitivity to penicillin.

Side effects: nausea, vomiting , stomatitis, skin rash, anaphylaxis.

Dosage: I.M. ONLY

- Adult: 1,200,000 – 2,400,000 units " 10 days for streptoceccal inflections"
- Pediatric: 600,000 – 1,200,000 units.

Nursing considerations:

- Shake multiple-dose vial vigorously before withdrawing the desired dose.
- Use a 21- gauge needle & don't allow medication to remain in the syringe & needle for long periods of time before administration because the needle may become plugged & the syringe "frozen".
- Inject slowly into the muscle & don't massage the injection site.
- Before injection of medication, aspirate needle to ascertain that needle is not in a vein.
- Rotate site of injections.
- **Don't administer IV.**
- If dose is large and the available muscle is small, divide the dose into 2 injection sites.

Penicillin G Benzathine

Benzathine Penicillin G, Parenteral.

Trade name: Bicillin L-A (long acting penicillin), Depocillin .

Class. : Antibiotic, penicillin.

The parenteral product is long-acting form of penicillin in an aqueous vehicle. It is administered as a sterile suspension.

Uses:

- Most gram +ve (streptococci , staphylococci, pneumoeocci) & some gram -ve (gonococci) organisms .
- Syphillis
- Prophylaxis of glomerulonephritis & rheumatic fever.
- Surgical infections.
- Secondary infections following tooth extraction + tonsillectomy.

Contraindications: Hypersensitivity to penicillin.

Dosage form: vial containing 600,000 unit – 1200000 unit.

Dosage: 200,000 – 600,000 unit q 4-6 hr.

- Parenteral suspension: 1,200,000 units as single dose for adults and for pediatric 300,000 – 900,000 units as a single dose.
- Early syphilis: 2,400,000 units q 7 days for 3 weeks.
- Prophylaxis of Rheumatic fever: 1,200,000 units as a single dose every 4 weeks.
- Children: half the adult dose.

Side effects and Nursing considerations: See the previous drug.

Benzyl penicillin.

Trade name: Crystapen, crystaline penicillin .

Class: Antibiotic , penicillin.

- The low cost of penicillin G still makes it the first choice for treatment of many infections.
- Rapid onset makes it especially suitable for fulminating infections.
- Destroyed by acid & penicillinase.

Add side effects: Rapid I.V. administration may cause hyperkalemia & cardiac arrhythmias.

Uses: Infections due to penicillin- sensitive organisms.

N.B.: Penicillin G potassium is indicated as on alternative to penicillin G sodium in those patients in whom intake of sodium must be restricted.

Contraindications: Hypersensitivity.

Dosage forms: Penicillin G potassium (injection) vial of 1 MU and Penicillin G sodium vials of 1,5,10 Mu.

Dosage: I.M. , continuous I.V. infusion 300,000 – 30 million unit depending on the use.

- Pediatric: 100,000 – 250,000 units \kg daily in divided dose.
- Pediatric: 25,000 – 90,000 units \kg daily in divided doses.

Some β-lactamase inhibitors have been developed which themselves are not antibacterial, but augment the activity of penicillins against β-lactamase producing organisms.

Nursing considerations:

- I.M. is preferred, minimize discomfort by using solution of up to 100,000 units \ml.
- Monitor intake & output (I & O).
- Solution may be stored at room temperature for 24 hr or in refrigerator for 1 week.
- Use 1% - 2% lidocaine as a dilutent for I.M. use to decrease pain at injection site.
- Note the penicillin G should not be mixed during I.V. administration with the following drugs: aminophylline, gentamycin, heparin, vancomycin & sodium bicarbonate.

Penicillin V Potassium:

Trade name: Rafapen, Bepen vk.

Class.: Antibiotic, penicillin.

- Is acid stable and resist inactivation by gastric secretions.
- Well absorbed from the GIT & not affected by food.

Dosage forms:

- Tab.: 250 mg (400,000 units), 500 mg (800,000 units).
- Syrup:125 mg\5ml (200,000 units), 250 mg (400,000 units).

Dosage: according to causative agent, 125 mg-500mg q 6-8 hr for 10 days.

Aminopenicillins:

Ampicillin:

Trade name: Penbritin, Ampipharm.

Class: Antibiotic, penicillin.

Action: bactericidal, inhibit bacterial cell wall and causing cell death.

- Destroyed by penicillinase (lactamase enzymes)
- 30-60% absorbed from GIT after oral use.
- Broad – spectrum antibiotic.

Uses: treatment of infections caused by susceptible strains e.g. salmonella, E. coli, H. influenza, enterococci etc. particularly it is recommended in respiratory, urinary & GI tract infections & meningitis caused by Neisseria meningitis.

Side effect:

The aminopenicillins and extended-spectrum penicillins can produce pseudomembranous colitis (diarrhea caused by a change in the flora of the colon or an overgrowth of a toxin-producing strain of *Clostridium difficile*).

Dosage forms:

- Capsules 250 mg, suspension 125 mg\5ml, vials 250, 500, 1000 mg.
- Ampicillin forte capsules 500 mg, ampicillin forte suspension 250 mg\5ml.

Dosage:

- P.O., I.V. or I.M. 250 mg – 500 mg q 6 hr (adult).
- Children 50 mg\kg 1 day in 4 divided doses.
- For bacterial meningitis
    - Adults: 12gm are given in divided doses (3 gm q 6 hr).
    - Children: up to 400 mg\kg daily in divided doses q 4 hr.

Specific nursing considerations:

- Culture infected area before treatment
- Infuse an IV drug continuously or intermittently (over 30 minutes). Rotate the infusion site every 48 hours. An intermittent IV infusion may be diluted in 50 to 100 mL of sterile water, normal saline solution, dextrose 5% in water (D5W), D5W annormal saline solution, or lactated Ringer's solution.
- Monitor the patient continuously for possible allergic reactions.

- A patient who has never had a penicillin hypersensitivity reaction may still have future allergic reactions.
- Monitor patients (especially elderly patients, debilitated patients, and patients receiving immunosuppressants or radiation) receiving long-term therapy for possible superinfection.
- Monitor the patient's hydration status if adverse GI reactions occur.
- Administered oral drug on an empty stomach or 1-2 hr. after meal.

Amoxycillin:

Trade name: Moxypen, Amoxitid, Amoxicare, Moxepharm.

Class: Antibiotic, penicillin.

Action: bactericidal, inhibit bacterial cell membrane and causing cell death.

- Inactivated by penicillinase, including those produced by staphylococcus aureus and gram negative baccili.
- Acid stable
- Broad-spectrum antibiotic.
- Absorbed well from the GIT.

Uses:

- To treat many different types of infection caused by bacteria, such as tonsillitis, bronchitis, pneumonia, gonorrhea, and infections of the ear, nose, throat, skin, or urinary tract.
- Used together with another antibiotic called clarithromycin to treat stomach ulcers caused by Helicobacter pylori infection.

Dosage Forms:

- Caps. 250 mg, suspension 125 mg \5 ml
- Moyxpen forte caps. 500 mg, Moxypen forte suspension, 250 mg\5ml

Dosage: 250-500 mg q 8 hr.

Side effects:

- GI: stomatitis, gastritis, sore mouth, nausea, vomiting, diarrhea, abdominal pain, bloody diarrhea etc.
- Hematologic: anemia, thrombocytopenia, leukopenia etc.
- Hypersensitivity: rash, fever, wheezing, anaphylaxis.
- Other: super-infection, vaginitis.

Specific nursing consideration:

- Reconstituted suspension is stable for 1 week at room température & for 2 weeks at 2- 8º Centigrade degrees.
- Teach patient about side effects specially on GI.

Amoxycillin & Potassium Clavulanate:

Trade name: Augmentin, , moclav, clamoxin, ogmin.

Class: Antibiotic, penicillin.

Note: Potassium clavulanate inactivates beta-lactamase enzymes which are responsible for resistance to penicillin.

Dose: tablets 500mg q8 hr. or suspension 125, 250 mg q 8 hr.

Cloxacillin:

Trade name: Orbenin

Class. : Antibiotic, penicillin. resistant to penicillinase.

Action: inhibit bacterial cell membrane and causing cell death.

Uses: Infections caused by penicillinase- producing staphylococci, streptococci, pneumococci that are resistant to Benzyl penicillin such as abscesses (skin and soft tissue, Osteomylitis (bone and joints), infected wounds & burns, Septic arthritis, pneumonia.

Dosage forms:

- Vials containing 250 mg, 1g .
- Capsules 250 mg - 500 mg.
- Syrup 125 mg\ 5 ml, 250 mg\5ml.

Dose:

- Adult: 250 – 500 mg q 6 hr.
- Children: 50 mg\kg per day in 4 divided doses.

Nursing considerations:

- Administer on an empty stomach.
- Refrigerate reconstituted solution & discard remaining amount after 14 days.
- N.B.: To prepare oral suspension, add amount of water stated on label and shake well.
- Shake the bottle well before each use.

Classification, Route of administration of various penicillins

Type and Generic Name	Route	Trade Names
1. Natural Penicillins		
Penicillin G Sodium (Benzyl-penicillin Na)	IV/IM	Crystalline Penicillin
Penicillin G Potassium (Benzyl-penicillin K)	PO/IM/IV	Megacillin
Penicillin G procaine or Procaine benzyl-penicillin	IM	Duracillin
Penicillin G Benzathine or Benzathine benzyl-penicillin	IM	Bicillin L-A
Penicillin G Benzathine and Procaine Combined	IM	Bicillin C-R
Penicillin V or (Phenoxymethyl penicillin)	PO	Rafapen
2. Penicillinase-Resistant Penicillins		
Cloxacillin	PO/ IV/IM	Orbenin
3. Amino-penicillin		
Ampicillin	IV, IM, PO	Penbertin
Amoxicillin	PO	Moxypen,
4. Broad-Spectrum Penicillins (Anti-Pseudomonas Penicillin)		
Piperacillin IV		Pipracil, Pipril
5. ß-Lactamase Inhibitor Combinations		
Amoxicillin-Clavulanic acid	PO	Augmentin, Amoxiclav
Ampicillin-Sulbactam	IV	Unasyn
Piperacillin-Tazobactam	IV	Zosyn

Notes:

- **Clavulanic acid, Tazobactam and Sulbactam** resembles the structure of beta lactam ring bind and inactivate the beta-lactamase enzymes produced by many bacteria and allow for the antibacterial effect penicillins.
- **Procaine penicillin and benzathine penicillin** have the same antibacterial activity as benzylpenicillin but act for a longer period. They are sustained released preparation of benzylpenicillin with slow absorption and steady blood concentration (12-24 hrs for **Procaine penicillin** and few weeks for **benzathine penicillin**).

Cephalosporins

- Cephalosporins share with penicillins a common mechanism, susceptibility to B-Iactamases (penkillioases, cephalosporinases) and similar side effects.
- Like penicillins, the original "generation" of cephalosporins was most effective against Gram-positive organisms. Chemical modifications increased the effectiveness of subsequent generations against Cram negatives and reduced their ability to fight Gram positive infections.
- Are semisynthetic antibiotics that resemble the penicillin both chemically and pharmacologically.

First generation cephalosporins	
Parenteral	*Oral*
Cephalothin*	Cephalexin
Cefazolin	Cephradine
	Cefadroxil
Second generation cephalosporins	
Parenteral	*Oral*
Cefuroxime	Cefaclor
Cefoxitin*	Cefuroxime axetil
Third generation cephalosporins	
Parenteral	*Oral*
Cefotaxime	Cefixime
Ceftizoxime	Cefpodoxime proxetil
Ceftriaxone	Cefdinir
Ceftazidime	Ceftibuten
Cefoperazone	Ceftamet pivoxil
Fourth generation cephalosporins	
Parenteral	
Cefepime	
Cefpirome	

- Are absorbed rapidly from the GI tract and quickly reach effective concentrations in the urinary, GI and respiratory tracts except in patients with pernicious anemia or obstructive jaundice & eliminated rapidly in patients with normal renal function.
- Absorption from GIT is decreased in presence of food except Cefuroxime (Zinnat).
- They are broad- spectrum antibiotics that have been classified as first, second and third generation drugs. The difference among generations is based on antibacterial spectra.
- Third generation cephalosporins have more activity against gram negative organisms and resistant organisms & less activity against gram positive organisms than first generation drugs. They are also stable against beta-lactemase enzymes.
- The cost increases from 1st to 3rd generation cephalosporins.
- Cefuroxime (second-generation) and the third-generation drugs cefotaxime, ceftizoxime, ceftriaxone, and ceftazidime cross the blood-brain barrier. Cefepime (fourth-generation) also crosses the blood-brain barrier, but the extent to which it does isn't known.
- All cephalosporins are excreted primarily unchanged by the kidneys with the exception of cefoperazone and ceftriaxone, which are excreted in feces via bile.

Action:

- They interfere with the final step in the formation of the bacterial cell wall resulting in unstable cell membranes that undergo lysis, also cell division & growth are inhibited.

They are classified to five generations according to their spectrum of antimicrobial activity & route of administration and penetration to CNS.

First generation:(cefadroxil, cephalexin, Cephradin and cefazolin)

- They are active against Gr+ bacteria infections, and used orally Except Cefazolin is used parentally.
- Cefazolin is a drug of choice for surgical prophylaxis.

Second generation: (cefoxitin, cefuroxime and cefaclor)

- They are less active against Gr+ bacteria, but have an extended Gr -ve activity like: Klebsiella and H.influenza,
- They are used for: otitis, sinusitis and lower RTI

Third generation:

- **Parentally** third generation cephalosporines: (ceftriaxone, cefotaxime, Cefoperazone and ceftazidime)
- They are active against Gr- bacterial infections (gonorrhea, meningitis and pneumonia parenterally).
- Third generation that can be given **orally**: (Cefixime, Cefpodoxime and Ceftibuten).

Uses of 3^{ed} generation:

- They are used to treat a wide variety of serious infections caused by organisms that are resistant to most other drugs.
- Ceftriaxone and cefotaxime are approved for treatment of meningitis.

Fourth generation: (Cefepime and cefpirome)

- More resist to hydrolysis by chromosomal B-lactamase produced by enterobacter, so it is active against them.

Fifth generation: (Ceftaroline)

- Ceftaroline is a broad-spectrum cephalosporin antimicrobial with extended gram-positive bacteria coverage.
- It exhibits greater binding affinity than ceftriaxone.

Uses: was approved by the FDA for the treatment of skin and skin structure infections (SSSI), including MRSA and community-acquired pneumonia (CAP).

- Ceftaroline is a broad-spectrum cephalosporin antimicrobial with extended gram+ve bacteria coverage.

First generation	***Second generation-***	***Third generation***	***Forth generation***
More active against gram positive organism	***more selective against gram positive and gram negative organisms***	***Highly active against gram negative otganisms***	***simillar antibacterial activity as that 0f third generation but highly resistent to beta lactamases***
Parenteral- *Cephalothin* *Cefazolin* *Cephaloridine* ***Oral-*** *Cephalexin* *Cephadine* *Cefadroxil*	***Parenteral*** *Cefuroxim* *Cefoxitin* ***Oral*** *Cefaclor* *Cefuroxim*	***Parenteral-*** *Cefotaxim* *Ceftizoxime* *Ceftraxone* *Cefoperazone* ***Oral-*** Cefixime *Cefpodoxime*	***Parenteral-*** *Cefepime* *Cefiperome*

General uses of cephalosporins:

- Ear infections pneumonia
- Skin infections
- Kidney infections
- Std such as gonorrhea
- Throat infections
- Meningitis.

Side effects:

- Hypersensitivity & cross allergy with Penicillin.
- Local pain and irritation with IM injection.
- Nephrotoxicity.

- biliary stone. Ceftriaxone
- Blood abnormalities.
- Stomach discomfort.
- Nausea or vomiting.
- Superinfection. (↓ normal flora ↑ Clostridium difficilli ↑ Candida)

Contraindications: Hypersensitivity to cephalosporins or Penicillin, renal failure, Pregnancy, Lactation.

Side effects:

- Confusion
- Seizures
- Bleeding (vit K deficiency)
- Nausea
- Vomiting
- Diarrhea.
- Nephrotoxicity

Hypersensitivity reactions are the most common systemic adverse reactions to cephalosporins and include:

- Hives
- Itching
- Rash that appears like the measles
- Serum sickness (reaction after injection of a foreign serum characterized by edema, fever, hives, and inflammation of the blood vessels and joints)
- Anaphylaxis (in rare cases).

Drug interaction: Probenecid and sulfinpyrazone, can reduce kidney excretion of some cephalosporins

Nursing considerations:

- Refrigerate oral suspensions (stable for 14 days); shake well before administering to ensure the correct dosage.
- Follow the manufacturers' directions for reconstitution, dilution, and storage of drugs; check expiration dates.

- Administer an IM dose deep into a large muscle mass (gluteal or midlateral thigh); rotate injection sites to minimize tissue injury.
- Don't add or mix other drugs with IV infusions, particularly aminoglycosides, which are inactivated if mixed with cephalosporins.
- If other drugs must be given IV, temporarily stop infusion of the primary drug.
- Ensure adequate dilution of the IV infusion and rotate injection sites every 48 hours to help minimize local vein irritation; using a small-gauge needle in a larger available vein may be helpful.
- Monitor renal function studies; dosages of certain cephalosporins must be lowered in patients with severe renal impairment.
- In patients with decreased renal function, monitor blood urea nitrogen (BUN) and creatinine levels and urine output for significant changes.
- Monitor PT and platelet counts, and assess the patient for signs of hypoprothrombinemia, which may occur (with or without bleeding) during therapy with cefoperazone, cefotetan, or ceftriaxone.
- Monitor patients on long-term therapy for possible bacterial and fungal superinfection, especially elderly and debilitated patients and those receiving immunosuppressants or radiation therapy.
- Administer cephalosporins at least 1 hour before bacteriostatic antibacterial drugs (tetracyclines, erythromycins, and chloramphenicol).

Examples:

Cephalexin monohydrate:

Trade names: Keflex , Jeflex.

Class.: antibiotic, cephalosporin (first generation).

Uses: infections of respiratory tract, skin, bones, GU and otitis media.

Additional Side-effects: Nephrotoxicity, jaundice.

Dosage:

- Caps. 250-500 mg q 6 hr.
- Suspension 125-250mg q 6 hr.

Cephalothin sodium:

Trade name: Keflin.

Class: Broad-spectrum antibiotic, cephalosporin (first generation).

N.B.: poorly absorbed from GI tract & must be given parenterally.

Dosage: Deep I.M. or I.V. 500-1000 mg a 4-6 hr.

Uses: see Keflex.

Cefaclor:

Trade name: Ceclor

Classification: antibiotic, cephalosporin " second generation"

Uses: infections of upper & lower respiratory tract. UTI & dermatological infections and otitis media.

Dosage: tablet (500,750 mg), oral suspension. 250 mg every 8 hr & may be doubled in case of severe infection (for adults). Children 20 mg/kg/day in 3 divided doses. Maximum 1 gm daily.

Specific drug considerations:

- The suspension should be refrigerated after reconstitution and discarded after 14 days.
- Give drug with meals or food to decrease GI disturbance.
- Discontinue drug if hypersensitivity reaction occurs.

Cefadroxil:-

Trade name: cefadrox.

Class: Cephalosporin (first generation), antibiotic.

Dose: 0.5–1 g BID.

- A close congener of cephalexin;
- has good tissue penetration—
- exerts more sustained action at the site of infection;
- It is excreted unchanged in urine, but dose need be reduced only if creatinine clearance is < 50 ml/min.
- The antibacterial activity of cefadroxil and indications are similar to those of cephalexin.

Cefuroxime Sodium:

Trade names: Zinacef, Zinnat.

Class: Antibiotic, cephalosporin (second generation).

Uses:

- P.O. :pharyngitis, tonsillitis, UTI, bronchitis.
- I.M. or I.V.: pneumonia, UTI, osteomyelitis, meningitis, septicemia, pre-op. Prophylaxis.

Add. Side effects: decrease in Hb & HCT (hematocrit).

Dosage:

- Tablets 250-500 mg q 12 hr (adult).
- 125- 250 mg bid (children).
- I.M. –I.V. 750 mg q 8 hr

Cefixime, oral:

Trade name: suprax, supran .

Classification: antibiotic, cephalosporin (third generation).

- The third generation drug can be given orally.
- Stable in the presence of beta-lactamase enzymes.

Uses: Uncomplicated UTI, pharyngitis, tonsillitis & acute bronchitis.

Additional side effects: Flatulence, alkaline phosphatase levels.

Dosage: oral suspension, tablets. Adults either 400 mg once daily or 200 mg every12 hr, children 8 mg/kg / day.

Specific drug consideration:

- Once reconstituted, the suspension should be kept at a room temp, where it maintains potency for 14 days.
- Therapy should be taken once a day at the same time each day.

Cefotaxime Sodium:

Trade name: claforan.

Classification: Antibiotic, cephalosporin (third generation).

- N.B.: Treatment should be continued for a minimum of 10 days for group A beta-hemolytic streptococcal infections to minimize the risk of glomerulonephritis or rheumatic fever. The I.V. route is preferable for patients with severe or life – threatening infections.

Uses: Infections in respiratory tract (pneumonia), gu tract infections, meningitis, peritonitis, septicemia, pelvic cellulitis, endometritis.

- N.B. used with aminoglycosides when causative agent has not been identified.

Dosage:

- I.V., I.M. 1 gm q 12 hr (adult).
- Moderate to severe infection. 1-2 g every 6-8 hr.
- Pre-op. Prophylaxis, 1 gm 30-90 min prior to surgery.

Drug considerations: -

- Should not be mixed with aminoglycosides. (each should be given separately)
- For I.V. use, should be mixed with 10 ml sterile water & administer over 3-5 minutes.
- Observe for signs and symptoms of anaphylaxis during first dose; CBC with differential specially in long course treatment.

Ceftriaxone Soduim:-

Trade name: Rocephin, cefixon.

Class: Cephalosporin (third generation), antibiotic.

Uses: lower respiratory tract (pneumonia), UTI, infections of skin, bone & abdomen. Meningitis, bacterial septicemia, pre-op. prophylaxis.

Dosage: I.V., I.M. adult 1-2g daily in single or divided doses every 12 hr.

- Pediatric: 50-75 mg\kg \day (other than meningitis) meningitis 100 mg\kg 1 day.

Drug considerations: -

- I.M. injection should be deep into the body of large muscle.
- I.V. injection should be diluted.
- For stability of solution the package insert should be checked carefully.
- Dosage should be maintained for at least 2 days after symptoms of infection have disappeared (usual course is 4-14 days).

Cefpodoxime proxetil

Trade name: Cefpo

Class: It is the orally active ester prodrug of 3rd generation cephalosporin In addition to being highly active against Enterobacteriaceae and streptococci, it inhibits *Staph. aureus*.

Uses: It is used mainly for respiratory, urinary, skin and soft tissue infections.

Dose:

- Tablet 200 mg BID (max 800 mg/day)
- Susp. 50 mg/5 ml.

Other B- lactam drugs:

Monobactam: (Aztreonam)

- It has monocyclic beta-lactam ring and resists beta-lactamase.
- Active against Gr-ve bacteria but it lacks activity against gram +ve organisms.
- Uses: in variuos types of infections caused by E. coli, Klebsiella, pneumoniae, H. influenzae, P. aeruginosa, Enterobacter spp., and Citrobacter spp.
- **Side Effect:**
    - skin rash
    - no cross-reactivity with penicillins or cephalosporins for hypersensitivity reactions.

Carbapenems

Carbapenems are a class of beta-lactam antibacterial that includes:

Imipenem- cilastatin sodium (a combination drug), Meropenem, Ertapenem, Doripenem.

- The antibacterial spectrum of activity for imipenem-cilastatin is broader than that of other antibacterial drugs studied to date. Because of this broad spectrum of activity, it's used for serious or life-threatening infection, especially gram-positive and gram- negative nosocomial infections.
- Imipenem is given with cilastatin because imipenem alone is rapidly metabolized in the tubules of the kidneys, rendering it ineffective. After parenteral administration, imipenem-cilastatin sodium is distributed widely. It's metabolized by several mechanisms and excreted primarily in urine.
- After parenteral administration, meropenem is distributed widely, including to the CNS. Metabolism is insignificant; 70% of the drug is excreted unchanged in urine.

Action: Imipenem-cilastatin sodium, ertapenem, and meropenem are usually bactericidal. They exert antibacterial activity by inhibiting bacterial cell-wall synthesis.

Indications:

- Imipenem-cilastatin sodium has the broadest spectrum of activity of currently available beta-lactam antibacterial drugs:
    - It's effective against aerobic gram-positive species, such as Streptococcus, Staphylococcus aureus, and Staphylococcus epidermidis.

- It inhibits most Enterobacter species.
- It inhibits P. aeruginosa (including strains resistant to piperacillin and ceftazidime) and most anaerobic species, including B. fragilis.
- It may be used to treat serious health care–associated infections and infections in immunocompromised patients caused by mixed aerobic and anaerobic organisms.
- Meropenem is indicated for treatment of intra-abdominal infections as well as for management of bacterial meningitis caused by susceptible organisms.

Adverse effects:

Common adverse reactions to imipenem-cilastatin sodium, ertapenem, and meropenem include:

- nausea and vomiting
- diarrhea
- hypersensitivity reactions such as rashes (particularly in patients with known hypersensitivity to penicillins).

Ertapenem may also cause:

- seizures
- hypotension
- hyperkalemia
- respiratory distress
- death.

Nursing consideration:

- Assess the patient's infection before therapy and regularly throughout therapy
- Don't give the drug by direct IV bolus injection; reconstitute it as directed. Infuse over 40 to 60 minutes. If nausea occurs, slow the infusion.
- When reconstituting the powder form, shake it until the solution is clear. Solutions may range from colorless to yellow; variations of color within this range don't affect the drug's potency. After reconstitution, the solution is stable for 10 hours at room temperature and for 48 hours when refrigerated.
- Reconstitute the drug for an IM injection with 1% lidocaine hydrochloride (without epinephrine) as directed.
- If seizures develop and persist despite using anticonvulsants, notify the prescriber, who may discontinue the drug and institute seizure precautions and protocols.

- Monitor the patient's hydration status if adverse GI reactions occur.
- In a patient with decreased or impaired renal function, the dosage of carbapenems may need to be adjusted.

Vancomycin Hydrochloride:

Vancomycin hydrochloride is used increasingly to treat methicillin-resistant *S. aureus.* As a rule, it should be used only when culture and sensitivity test results confirm the need for it.

Trade name: vancocin.

Class: Antibiotic, miscellaneous.

Action:

- It inhibits bacterial cell wall synthesis
- It is bactericidal for most organisms and bactereostatic for enterococci.
- It is poorly absorbed from GIT so it should be administered parenteral only for life threatening infections.

Uses:

- Drug should be used for treatment of life threatening infections when other treatment has been ineffective.
- The IV form is the therapy of choice for Patients with severe staphylococcal infections resistant or allergic to penicillins or cephalosporinssuch as: Endocarditits, Osteomylitis, Pneumonia, Septicemia.

Contraindications:

- Hypersensitivity

Side effects:

- Ototoxicity (deafness).
- Nephrotoxicity (uremia).
- Red-Neck (man) syndrome: "chills, erythema of neck and back fever".
- Skin rashes, Drug fever
- Hypotension (due to rapid administration).
- Thrombophlebitis at site of injection.

Drug interactions:

- Vancomycin may increase the risk of toxicity when administered with other drugs toxic to the kidneys and organs of hearing, such as aminoglycosides, amphotericin B, cisplatin, bacitracin, colistin, and polymyxin B.

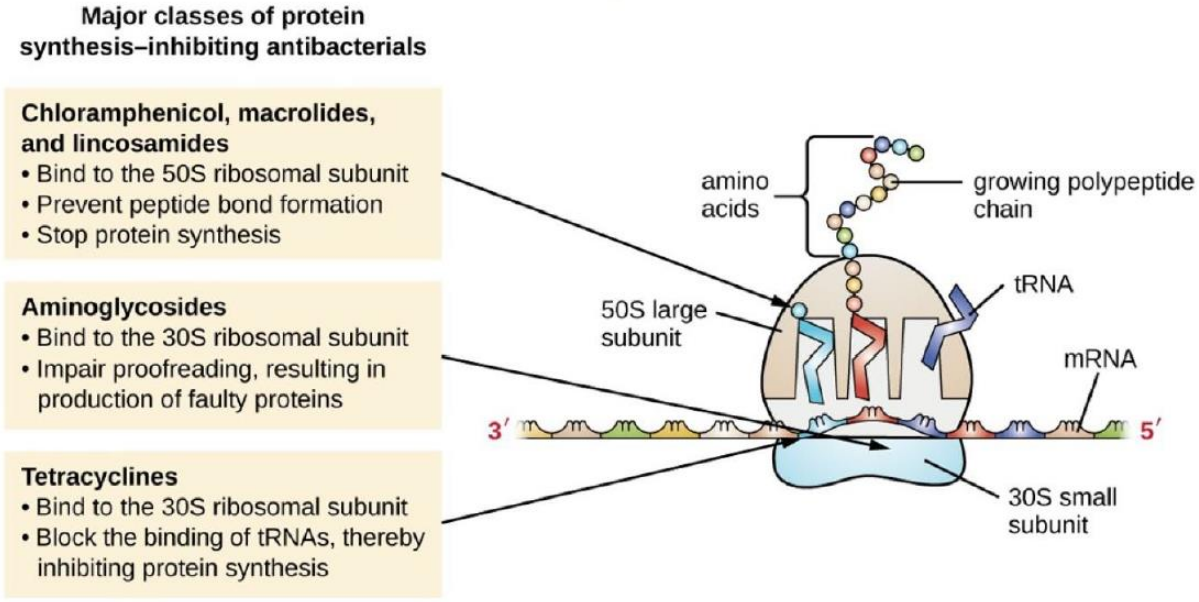
Dosage: Caps., syrup, or I.V. 0.5 g q 6 hr or 1g q 12 hr.

Nursing considerations:

- Assess the patient's infection before therapy and regularly throughout therapy
- Mix as indicated on package insert.
- Intermittent infusion is the preferred route.
- Avoid extravastion during injections.
- Severe hypotension may occur with rapid IV administration of vancomycin and may be accompanied by a red rash with flat and raised lesions on the face, neck, chest, and arms (red man syndrome). Doses of less than or equal to 1 g should be given over 1 hour and doses greater than 1 g, over 11/2 to 2 hours.
- If red neck or red man syndrome occurs because the drug is infused too rapidly, stop the infusion and report this finding to the prescriber.

Protein synthesis inhibitor

- Bacterial ribosomes and inhibiting bacterial protein synthesis. Bacterial ribosomes differ structurally from mammalian cytoplasmic ribosomes and are composed of 30S and 50S subunits (mammalian ribosomes have 40S and 60S subunits).
- In general, selectivity for bacterial ribosomes minimizes potential adverse consequences encountered with the disruption of protein synthesis in mammalian host cells. However, high concentrations of drugs such as *chloramphenicol* or the tetracyclines may cause toxic effects as a result of interaction with mitochondrial mammalian ribosomes, since the structure of mitochondrial ribosomes more closely resembles bacterial ribosomes.



Tetracyclines

Tetracyclines are broad-spectrum antibacterial drugs. They may be classified as:

- intermediate-acting compounds, such as demeclocycline hydrochloride and tetracycline hydrochloride
- long-acting compounds, such as doxycycline and minocycline hydrochloride.

Action:

- The drugs bind reversibly to the 30S subunit of the bacterial ribosome. This action prevents binding of tRNA to the mRNA–ribosome complex, thereby inhibiting bacterial
- protein synthesis.

- Bacteriostatic antibiotics effective against a wide variety of organisms, including gram-positive and gram-negative bacteria, protozoa, spirochetes and mycobacteria.
- Well absorbed from the stomach & upper GIT. Moreover it can pass through the placental barrier.
- Aluminum, calcium, and magnesium antacids reduce the absorption of oral tetracyclines. Iron salts, bismuth subsalicylate, and zinc sulfate reduce the absorption of doxycycline and tetracycline. Reduced absorption can be prevented by separating doses of tetracyclines and these agents by 2 to 3 hours.
- They are deposited in the fetal skeleton and calcifying teeth.
- Elimination: Tetracycline is not hepatically metabolized. Tetracycline is primarily eliminated unchanged in the urine, whereas minocycline undergoes hepatic metabolism and is eliminated to a lesser extent via the kidney. In renally compromised patients, doxycycline is preferred, as it is primarily eliminated via the bile into the feces.

Uses:

Tetracyclines provide a broad spectrum of activity, which means they cover a wide range of organisms, including:

- Tetracyclines are drug of choice for rickettsia, chlamydial infections.
- Tetracycline+ aminoglycosides for plague and brucellosis.
- Sometimes is used in protozoal infections (Entamoeba and P. falciparm).
- Acne, pneumonia and UTI.
- Cholera. (Drug of choice is doxycycline single dose)
- Peptic ulcer (Tetracycline) combination with others
- Brucellosis (Malta fever, or Mediterranean fever) doxycycline
- Alternative treatment to penicillin for uncomplicated gonorrhea and syphilis
- Plague (yersina -ve bacteria)

Contraindications:

- Hypersensitivity.
- During tooth development stage:
    - Last trimester of pregnancy
    - During breast-feeding.
    - Neonatal period and during childhood up to 8 years.

Side effects:

- GIT disturbances, NVD
- Affects bone structures (bone deformity and teeth discoloration).
- Liver & kidney toxicity. (Doxycycline is preferred in patients with renal dysfunction, as it is primarily eliminated via the bile into the feces.
- Intramuscular injection produces painful local irritation and should be avoided.
- Photosensitivity.
- Dental: Discoloration of nails, infants' & children's teeth.

Superinfection Tetracyclines are the most common antibiotics responsible for superinfections, because they cause marked suppression of the resident flora. Though mouth, skin or vagina may be involved, intestinal superinfection by *Candida albicans* is most prominent.

Fanconi like syndrome: it results from ingestion of outdated and degraded tetracycline and characterized by nausea, vomiting, acidosis, proteinuria, hypokalemia, polyuria & polydipsia.

Drug interactions

- Tetracyclines can reduce the effectiveness of hormonal contraceptives, which may result in breakthrough bleeding or ineffective contraception.
- Tetracyclines may also decrease the bactericidal action of penicillin.

Nursing considerations:

- Take the drug exactly as prescribed for as long as prescribed.
- Don't use outdated or deteriorated drugs. Discard unused medications.
- Administer on an empty stomach (1hr before or 2 hr after meals) except doxycycline.
- If the client is female & pregnant, consider period of pregnancy.
- Report side effects to physician.
- Avoid direct sunlight, which can cause severe sunburn-like reaction.

Examples:

1- Tetracycline Hcl: Brimocyclin

Dose: cap. 250, 500mg 1-2 gm every 6 hr

NB: all doses given on empty stomach.

Recycline 1% as eye oint.

2- Chlortetracycline HCl

- Ophthalmic & topical 3% on affected area 1-5 times daily.

3- Doxycycline :

Trade names: Doxylin , Doxypal , Doxypharm, doxacylin.

Class: Antibiotic, tetracycline.

Dosage form: Caps., film-coated tab. 100 mg

Dose: The usual dose for adults.

- First-day 100 mg q 12hr followed by 1 tab. daily.
- In severe cases: 1 tablets daily in divided doses q 12 hr.

Side effects: nausea, vomiting, diarrhea, headache, hypertension, colitis, hemolytic anemia.

4- Oxytetracycline: oxycin

Trade name: Terramycin

Class: antibiotic , tetracycline.

Dose: ointment 5mg , vaginal tablet with polymyxine B sulphate

5- Minocycline: Minocycline, Minocin

Indications: add. meningococcal carrier state; acne vulgaris

Contra-indications: see notes above

Side-effects: see notes above; also dizziness and vertigo (more common in women); rarely anorexia, tinnitus, impaired hearing, acute renal failure, pigmentation (sometimes irreversible), alopecia, discoloration of conjunctiva, tears, and sweat

Dosage form: tablet, capsules 100mg

Dose: 50**,** 100 mg twice daily (initial 200mg followed by 100mg every 12 hr.

6- Tigecycline: Tygacil

A derivative of *minocycline*, is the first available of the glycylcycline antimicrobial class

Action: It exhibits bacteriostatic action by reversibly binding to the 30S ribosomal subunit and inhibiting protein synthesis.

Indications:

- It exhibits a broad-spectrum activity.
- it is active against Gram-positive and Gram-negative bacteria, including tetracycline-resistant organisms, and some anaerobes. It is also active against meticillin-resistant Staphylococcus aureus and vancomycin-resistant enterococci
- Tigecycline was developed to overcome the recent emergence of tetracycline class–resistant organisms.

- It should be reserved for the treatment of complicated skin and soft-tissue infections and complicated abdominal infections caused by multiple-antibacterial resistant organisms when other antibacterials cannot be used; it is not recommended for the treatment of foot infections in patients with diabetes.

Adverse effects

- Significant nausea and vomiting.
- Acute pancreatitis, including fatality, has been reported with therapy.
- Photosensitivity, pseudotumor cerebri, discoloration of permanent teeth when used during tooth development, and fetal harm when administered in pregnancy.

Drug interaction:

- It may decrease the clearance of warfarin and increase prothrombin time. Therefore, the international normalized ratio should be monitored closely when it is coadministered with warfarin.

Dosage form: Following IV infusion,

- The primary route of elimination is biliary/fecal. No dosage adjustments are necessary for patients with renal impairment. However, a dose reduction is recommended in severe hepatic dysfunction.

Dose: Adult over 18 years, initially 100 mg, then 50mg every 12 hours for 5–14 days.

Chloramphenicol:

Trade names: chloromycetin , chloram, synthomycin

Action:

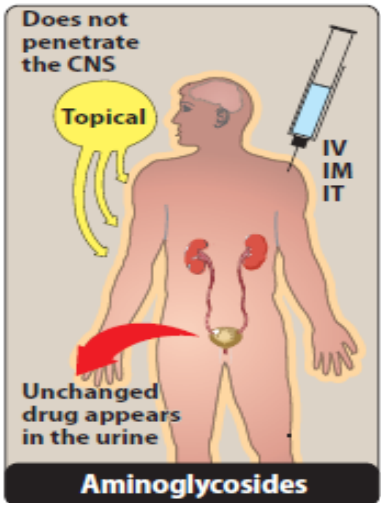
- It inhibits protein synthesis in bacteria by binding to ribosomes (50S).
- Chloramphenicol is primarily bacteriostatic, though high concentrations have been shown to exert cidal effect on some bacteria, e.g. *H. influenzae*. It is a broad-spectrum antibiotic, active against nearly the same range of organisms.

Uses : Because of * potential toxicity, * bacterial resistance, and the *availability of many other effective alternatives, chloramphenicol rarely used and limited to topical treatment for susceptible infections.

Formulation: Eye drops, chloramphenicol 0.5%, Eye ointment, chloramphenicol 1%, Skin ointment chloramphenicol 3%.

Aminoglycosides

- They are more toxic than penicillin and cephalosporins and must be maintain within a certain therapeutic range.
- They are distributed in extracellular fluids, cross the placenta barriers but not the BBB because it isn't lipid soluble.
- They are poorly absorbed from the GI tract, therefore are usually administered parenterally, not orally.
- Are excreted largely in the urine, so it is suitable for urinary tract infections.
- Are powerful antibiotics that can induce serious side effects (nephrotoxicity, ototoxicity).



Aminoglycosides are bactericidal. They're effective against:

- Gram-negative bacilli
- Some aerobic gram-positive bacteria
- Mycobacteria
- Some protozoa.

Aminoglycosides currently in use include:

- Amikacin Sulfate, Gentamicin Sulfate, Kanamycin Sulfate, Neomycin Sulfate, Netilmicin Sulfate, Paromomycin Sulfate, Streptomycin Sulfate, Tobramycin Sulfate.

Action: They are inhibit protein synthesis by binding irreversibly to ribosome which leads to production of nonfunctional protein.

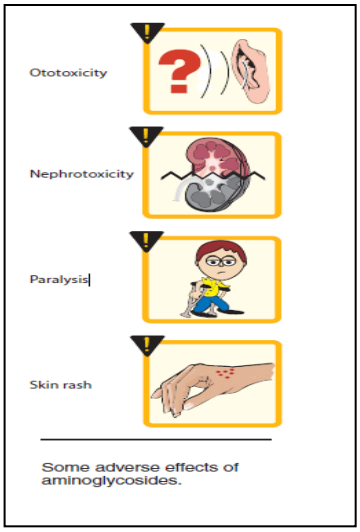
Uses: Gram negative bacteria causing: bone and joint infections, respiratory tract infections, septicemia (including neonatal sepsis), urinary tract infection (UIT), post operative infections, intra–abdominal infections (as peritonitis), skin infections (including burns).

- It should be used for gram +ve bacteria only when other less toxic drugs either ineffective or contraindicated.

Contraindications:-

- Hypersensitivity to aminoglycosides.
- Long–term therapy "except streptomycin for tuberculosis".
- For patients with impaired renal function or pre-existing hearing impairment.

Side effects



- Ototoxicity: tinnitis , hearing impairment, ataxia & dizziness.
- Renal impairment (Nephrotoxicity) hematuria, proteinuria
- Neurotoxicity: headache, tremor, lethargy, numbness, burning of face.
- Others: nausea, vomiting , skin rash & superinfection.
- Allergic reactions: Contact dermatitis is a common reaction to topically applied *neomycin.*

Precautions and interactions

1. Avoid aminoglycosides during pregnancy: risk of foetal ototoxicity.
2. Avoid concurrent use of other ototoxic drugs, e.g. high ceiling diuretics, minocycline.
3. Avoid concurrent use of other nephrotoxic drugs, e.g. amphotericin B, vancomycin, cyclosporine and cisplatin.
4. Cautious use in patients past middle age and in those with kidney damage.
5. **Gentamycin:** is the cheapest, **Tobramycin:** seems to be the best for pseudomonas infection, **Amikacin:** has the widest antibacterial activity, but its more ototoxic than other drugs and the most expensive and **Gentamycin and Amikacin** are more nephrotoxic than Tobramycin and netilmicin.

Nursing considerations:

- IM administration inject deep into muscle to minimize pain.
- Administer for only 7-10 days.
- Assess history of hypersensitivity.
- Obtain laboratory studies for renal function.
- Continue to monitor for ototoxicity.
- The importance of administered medications at the appropriate prescribed time intervals.

Amikacin Sulfate:

Trade name: Amikin

- it has the widest spectrum of activity, including many organisms resistant to other aminoglycosides. However, relatively higher doses are needed for Pseudomonas, Proteus and Staph. infections.
- It is recommended as a reserve drug for hospital acquired gram-negative bacillary infections where gentamicin/tobramycin resistance is high.

Classification: Antibiotic, aminoglycoside .

Uses: Serious infections caused by pseudomonas, proteus, serratia and gram positive bacilli including bone infections, respiratory tract infections, endocarditis, and septicemia.

Dosage: IM (preferred) I.V. adults, children 15 mg/ kg / day in 2-3 equally divided doses every 8-12 hr for 7-10 days. Newborns loading dose, 10 mg/ kg followed by 7.5 mg\ kg every 12 hr.

Gentamycin Sulfate:

Trade name: Garamyein.

Class: antibiotic, aminoglycosides.

Indication:

- Serious infections caused by susceptible organisms
- Prevention of endocarditis during GI and genitourinary procedures or surgery
- Infections caused by gram-negative bacilli
- Serious nosocomial (hospital-acquired) infections, such as gram-negative bacteremia (abnormal presence of microorganisms in the bloodstream), peritonitis (inflammation of the peritoneum, the membrane that lines the abdominal cavity), and pneumonia in critically ill patients
- Urinary tract infections (UTIs) caused by enteric bacilli that are resistant to less toxic antibacterial drugs, such as penicillins and cephalosporins
- Infections of the central nervous system (CNS) and the eye (treated with local instillation).

Side effects:

- Alopecia
- CNS: ototoxicity, neurotoxicity, tinnitus, dizziness, ringing in the ears, vertigo.
- GI: nausea, vomiting, anorexia, weight loss, increased salivation.

- C.V.: palpitation, hypotension or hypertension.
- Hematologic: Decrease number of blood cells.
- GU: nephrotoxicity
- Local: Pain and irritation at IM injection site.

Dosage form:

- Ampules 20, 40, 80 mg
- Vial 240 mg per vial inj.
- 0.3% eye/ear drops,
- 0.1% skin cream.

Dose:

- IM (usual) . I.V. adults 3 mg\kg q 8 hr up to 5 mg\kg daily.
- Children 2-2.5 mg\kg every 8 hr.
- Newborns 2.5 mg\ kg every 12 hr.
- Ophthalmic solution 0.3% , 1-2 drops every 15-30 minutes.
- Topical ointment 0.1% , 1-5 times daily to the affected area.
- N.B.: should not be mixed with other drugs for parenteral use.

Nursing Consideration:

- When given IM, give it slowly and deep in a large muscle.
- Dilute dose when given IV.
- Monitor for kidney function tests (BUN & Creatinine), complete blood count when used for more than 7 days. Consult with Dr. accordingly.
- The daily dose of gentamicin (and other aminoglycosides) should be reduced in patients with impaired renal function according to measured creatinine clearance.
- Monitor the patient for adverse effects, such as ototoxicity, nephrotoxicity, anaphylaxis, thrombocytopenia, and agranulocytosis.
- Peak gentamicin levels occur 1 hour after IM injection and 30 minutes after IV infusion; check trough levels before the next dose.

Drug interactions

- Carbenicillin and ticarcillin reduce the effects of amikacin, gentamicin, kanamycin, neomycin, streptomycin, and tobramycin. This is especially true if the penicillin and

aminoglycoside are mixed in the same container or IV line. When giving penicillin with aminoglycosides, penicillin should be given first.

Toxicity

- Toxicity to the kidneys may result in renal failure, and toxicity to the neurologic system results in peripheral neuropathy with numbness and tingling of the extremities. The risk of renal toxicity also increases when amikacin, gentamicin, kanamycin, or tobramycin is taken with cyclosporine, amphotericin B, or acyclovir.
- Don't give other potentially nephrotoxic drugs such as vancomycin.

Neomycin Sulfate:

Class.: antibiotic, aminoglycoside.

Indications (uses):

a) P.O

- Diarrhea due to enteropathogenic E.coli.
- Suppression of intestinal bacteria for short-term therapy prior to intestinal surgery.
- Suppression of the intestinal flora for long-term therapy in hepatic coma. Neomycin attacks ammonia- forming bacteria in the intestinal tract causing reduction in blood ammonia and resulting in neurological improvement.

b) Topical:

- Inflamed skin conditions, burns.

Additional Contraindication: intestinal obstruction, pregnancy.

Additional Side effects: steatorrhea & electrolytes imbalance (mal-absorption).

Formulation: Neomycin Sulphate

- 350, 500 mg tab,
- 0.3% skin oint,
- 0.5% skin cream, eye oint.

Streptomycin Sulfate:

- It is the oldest aminoglycoside antibiotic.
- It used extensively in the past, but now practically restricted to treatment of tuberculosis.
- It is less potent (MICs are higher) than other aminoglycosides. The antimicrobial spectrum of streptomycin is relatively narrow: active primarily against aerobic gram-negative bacilli, but potency is low.

- Sensitive organisms are, Brucella, granulomatis, M. tuberculosis. Only few strains of E. coli, H. influenzae, V. cholerae, Shigella, Klebsiella, enterococci and some gram-positive cocci are now inhibited, that too at higher concentrations. All other organisms including Pseudomonas are unaffected.
- Streptomycin is highly ionized. It is neither absorbed nor destroyed in the GIT However, absorption from injection site in muscles is rapid.
- Streptomycin is not metabolized—excreted unchanged in urine.
- Half-life is prolonged and accumulation occurs in patients with renal insufficiency, in the elderly and neonates who have low GFR Reduction in dose or increase in dose-interval is essential in these situations.

Class: antibiotic, aminoglycosides.

Additional uses: Tuberculosis (T.B.), plague, cholera, bacterial endocarditis caused by hemophillus influenza.

Additional Contraindications: contact dermatitis, myasthenia gravis.

Dosage:

- 0.75, 1 g dry powder per vial for inj.
- Acute infections: 1 g (0.75 g in those above 50 yr age) i.m. BD for 7–10 days.
- Tuberculosis: 1 g or 0.75 g i.m. twice weekly for 30–60 days.
-

Macrolides and Ketolides

- Erythromycin was the first of these drugs to find clinical application,
- Used as a drug of first choice and as an alternative to *penicillin* in individuals with an allergy to β-lactam antibiotics.
- *Clarithromycin* (a methylated form of *erythromycin*) and *azithromycin* (having a larger lactone ring) have some features in common with, and others that improve upon, *erythromycin.*
- *Telithromycin*, a semisynthetic derivative of *erythromycin*, is the first "ketolide" antimicrobial agent.
- The macrolides bind irreversibly to a site on the 50S subunit of the bacterial ribosome, thus inhibiting translocation steps of protein.

- Erythromycin and azithromycin are primarily concentrated and excreted in the bile as active drugs. In contrast, clarithromycin and its metabolites are eliminated by the kidney as well as the liver. The dosage of this drug should be adjusted in patients with renal impairment.

Erythromycin:

Trade names: Erthrocare, Erthrolet, Erthrotab, Erthroteva

Action:

- Erythromycin acts by inhibiting bacterial protein synthesis. It combines with 50S ribosome subunits and interferes with 'translocation.
- Bacteriostatic and bactericidal in high doses.
- Absorbed well from the upper GIT.
- Enteric or coated forms to prevent destruction by gastric acid.
- Diffuse poorly to C.S.F. & primarily excreted in bile.

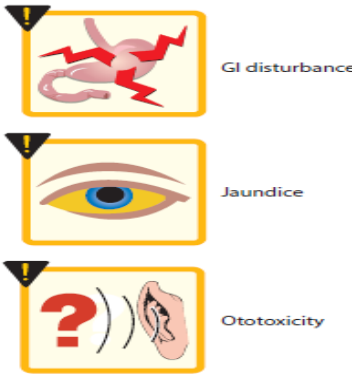
Uses:

- The drug of choice to treat respiratory tract infections.
- Intestinal amebiasis.
- Acne.
- An alternative drug to treat the following conditions in patients who are allergic to penicillin.
    - Infections due to group A- beta hemolytic streptococci.
    - To prevent bacterial endocarditis prior to dental procedures.
    - Gonorrhea or syphilis.
    - Conjunctivitis of the newborn.
    - Long term prophylaxes in rheumatic fever.

Contraindication: Hypersensitivity.

Side effects:

- GI: Nausea, vomiting, diarrhea, skin rashes, pseudomembranous colitis, Hepatotoxicity, confusion, I.V. uses: thrombophlebitis, CVS: reversible hearing loss in patients receiving high doses or with renal function impairment.



Some adverse effects of macrolide antibiotics.

Formulation: Tablets 250, 500 mg , susp, 125 mg, 200 mg, 400mg.

Dosage:

- The usual adult dose is 250 mg q 6 hr (can be doubled in severe infections).
- Children, 30-50 mg\kg in divided doses.

Drug Consideration:

- Should be taken on an empty stomach.
- In the treatment of sterptococcal infections, should be administered for at least 10 days.
- Don't administer with fruit juice or other acidic drinks because it may decrease the activity of the drug.
- Erythromycin inhibits hepatic oxidation of many drugs. The clinically significant interactions are—rise in plasma levels of theophylline, carbamazepine, valproate, ergotamine and warfarin.

Clarithromycin

Trade name: Klaricare, klacid

- The antimicrobial spectrum of clarithromycin is similar to erythromycin.
- It is more active against sensitive strains of gram-positive cocci, Moraxella, Legionella, Mycoplasma pneumonia and Helicobacter pylori.
- Bacteria that have developed resistance to erythromycin are resistant to clarithromycin also.
- Clarithromycin is more acid-stable than erythromycin, and is rapidly absorbed; oral bioavailability is ~50% due to first pass metabolism; food delays but does not decrease absorption.
- It has slightly greater tissue distribution than erythromycin and is metabolized by saturation kinetics—t½ is prolonged from 3–6 hours at lower doses to 6–9 hours at higher doses.
- An active metabolite is produced. About 1/3 of oral dose is excreted unchanged in urine, but no dose modification is needed in liver disease or in mild to- moderate kidney failure.

Uses:

- Indicated in upper and lower respiratory tract infections, sinusitis, otitis media, whooping cough, atypical pneumonia, skin and skin structure infections due to Strep, pyogenes and some Staph. Aureus, used as a eradicates H. pylori in 1–2 weeks.

Side effects:

- Similar to erythromycin, but gastric tolerance is better.
- High doses can cause reversible hearing loss.
- Few cases of pseudomembranous enterocolitis, hepatic dysfunction .
- Its safety in pregnancy and lactation is not known.
- The drug interaction potential is also similar to erythromycin.

Dose:

- 250, 500 mg BD for 7 days; severe cases 500 mg BD up to 14 days.
- 125, 250 mg/5 ml dry syr;

Azithromycin:

Trade name: Azicare

- This new azalide congener of erythromycin has an expanded spectrum,
- Improved pharmacokinetics, better tolerability and drug interaction profiles.
- It is more active than other macrolides against H. influenzae, but less active against gram-positive cocci.
- High activity is exerted on respiratory pathogens—Mycoplasma, Chlamydia pneumoniae, Legionella, Moraxella and on others like Campylobacter, granulomatis, N. gonorrhoeae. However, it is not active against erythromycin resistant bacteria. Penicillinase producing Staph. aureus are inhibited but not MRSA.
- Because of higher efficacy, better gastric tolerance and convenient once a day dosing, azithromycin is now preferred over erythromycin as **first choice** drug for infections such as: Legionnaires' pneumonia, Chlamydia trachomatis, urethritis.
- The other indications of azithromycin are pharyngitis, tonsillitis, sinusitis, otitis media, pneumonias, acute exacerbations of chronic bronchitis, streptococcal and some staphylococcal skin and soft tissue infections.
- Other potential uses are in typhoid, toxoplasmosis and malaria.

Dose:

- 250, 500 mg cap and 250 mg per 5 ml dry once daily 1 hour before or 2 hours after food (food decreases bioavailability);

LINCOSAMIDE ANTIBIOTICS

Lincomycin and Clindamycin

Clindamycin:

Clindamycin is a derivative of another drug, lincomycin. Because of its high potential for causing serious adverse effects, clindamycin is prescribed only when there's no therapeutic alternative. It's used for various gram-positive and anaerobic organisms.

Trade name: Dalacin C,

- This potent lincosamide antibiotic is similar in mechanism of action and spectrum of activity to erythromycin.
- Oral absorption of clindamycin is good. It penetrates into most skeletal and soft tissues, but not in brain and CSF.
- It is largely metabolized and metabolites are excreted in urine and bile.
- Side effects are rashes, urticaria, abdominal pain, but the major problem is diarrhoea and pseudomembranous enterocolitis due to Clostridium difficile superinfection which is potentially fatal.
- The drug should be promptly stopped and metronidazole (alternatively vancomycin) given to treat it. Because of potential toxicity, use of clindamycin is restricted to anaerobic and mixed infections, especially by Bact. fragilis causing abdominal, pelvic and lung abscesses.
- It is generally combined with an aminoglycoside or cephalosporin.
- Metronidazole and chloramphenicol are the alternatives to clindamycin for covering the anaerobes. It has also been employed for prophylaxis of endocarditis in penicillin allergic patients with valvular defects who undergo dental surgery, as well as to prevent surgical site infection in colorectal/pelvic surgery.
- Topically it can be used for infected acne vulgaris.

Dose:

- 150–300 mg cap. QID oral;
- 200–600 mg i.v. 8 hourly;
- Vaginal cream 2%, 1 application intravaginal at bed time for 7 consecutive days.
- Topical solution: 10mg /ml twice daily on affected area.

Spiramycin (Rovamycine)^{R}

- Spiramycin is a macrolide antimicrobial agent with activity against +ve organisms
- Oral infections with Metronidazole

Uses:

- It is used to treat many kinds of infections. It is often used to treat toxoplasmosis protozoa in pregnant women.
- To treat gonorrhea in patients who are allergic to Penicillin

Side effects:

- Skin rash and itch
- Bloody stools
- Chest pain
- fever
- heartburn
- nausea
- GIT upset

Other drugs: Telithromycin**,** Roxithromycin

Fusidic Acid

MOA: protein synthesis inhibitor

Uses:

1. G+ve bacteria such as Staphylococcus, Streptococcus topically.
2. eye drops.
3. gauze.

Quinolones, Fluoroquinolones

Ciprofloxacin, Ofloxacin, Norfloxacin ,Lomfloxacin, Levofloxacin and Moxifloxacin, Gemifloxacin and fluorinated analogs of Natidixic acid which now used infrequently.



Clinical uses of quinolones :

- Ciprofloxacin, ofloxacin, levofloxacin, and lomefloxacin are highly active against gram -ve bacteria and moderately active against gram +ve bacteria.
- Moxifloxacin and Gemifloxacin have even greater activity against gram +ve.
- Ciprofloxacin is used to treat lower respiratory tract infections, infectious diarrhea, and skin, bone, and joint infections.
- Levofloxacin is indicated for treatment of lower respiratory tract infections, skin infections, and UTIs.
- Moxifloxacin is used to treat acute bacterial sinusitis and mild to moderate community-acquired pneumonia.
- Norfloxacin is used to treat UTIs and prostatitis (inflammation of the prostate gland).
- Ofloxacin is used to treat selected sexually transmitted diseases, lower respiratory tract infections, skin and skin-structure infections, and prostatitis.
- ofloxacin is used to treat infection of the middle ear, and ophthalmic ofloxacin solution (drops) is used to treat conjunctivitis.

Nalidixic acid

Trade name: Negram, Ugram

- It is active against gram-negative bacteria, especially coliforms: E. coli, Proteus, Klebsiella, Enterobacter, Shigella but not Pseudomonas.

Action:

- It acts by inhibiting bacterial DNA gyrase, urinary germicide.
- It is bactericidal.
- Resistance to nalidixic acid develops rather rapidly.

Pharmacokinetics:

- Nalidixic acid is absorbed orally,
- highly plasma protein bound and
- partly metabolized in liver: one of the metabolites is active.
- It is excreted in urine.

NB. Concentration of the free drug in plasma and most tissues attained with the usual doses is nontherapeutic for systemic infections. However, high concentration attained in urine (20–50 times that in plasma) is lethal to the common urinary pathogens.

Uses:

- Nalidixic acid is primarily used as a urinary antiseptic, generally as a second line drug in recurrent cases or on the basis of sensitivity reports.
- Nitrofurantoin should not be given concurrently— antagonism occurs.
- It has also been employed in diarrhoea caused by Proteus, E. coli, Shigella or Salmonella, and has a special place in ampicillin resistant Shigella enteritis.

Contraindications:

- To be administered with caution in patients with: Liver disease, impaired kidney function, epilepsy, pregnancy, lactation, children, and cerebral arteriosclesois.

Side effect: Nausea, vomiting, diarrhea, seizure, headache dizziness, leukopnea, thrombocytopnea, hemolytic anemia due to G-6-PD deficiency.

Dose: 0.5–1 g TDS or QID 0.5 g tab, 0.3 g/5 ml syrup for 1-2 weeks.

Ciprofloxacin Hydrochloride:

It is the most potent first generation FQ active against a broad range of bacteria, the most susceptible ones are the aerobic gram-negative bacilli, especially the Enterobacteriaceae and *Neisseria.*

Pharmacokinetics:

- Ciprofloxacin is rapidly absorbed orally, but food delays absorption, and first pass metabolism occurs.
- It is high tissue penetrability: concentration in lung, sputum, muscle, bone, prostate and phagocytes exceeds that in plasma, but CSF and aqueous levels are lower.
- It is excreted primarily in urine, both by glomerular filtration and tubular secretion. Urinary and biliary concentrations are 10–50 fold higher than plasma.

Trade name: cipro, ciproxin, ciprocare.

Class. Antibacterial , quinolone derivative .

Action: it has broad- spectrum bactericidal activity, inhibits the synthesis of bacterial DNA by inhibiting the enzyme DNA gyrase.

Uses: Acute otitis media, UTI, infectious diarrhea, infection of lower respiratory tract, bone, joints & skin.

Contraindications:

- Hypersensitivity, Pregnancy, Children, Lactation.

Side effects: Headache, dizziness, insomnia, fatigue, blurred vision, Nausea vomiting, dysphagia, crystalluria, hematuria. Because gut anaerobes are not affected diarrhea is infrequent, Rashes, bad taste, GI bleeding, Headache, insomnia.

Dosage: 250-500 mg q 12 hr for 7-14 days.

Nursing considerations:

- Arrange for culture and sensitivity test before beginning therapy.
- Give medication 2 hr after meals.
- Stress importance of drinking increased amounts of fluids to keep urine acidic & to minimize the risk of crystalluria.
- Plasma concentration of theophylline, caffeine and warfarin are increased by ciprofloxacin (also by norfloxacin and pefloxacin) due to inhibition of metabolism: toxicity of these drugs can occur.

Ofloxacin

- This FQ is intermediate between ciprofloxacin and norfloxacin in activity against gram-negative bacteria, but it is comparable to or more potent than ciprofloxacin for gram-positive organisms and certain anaerobes.
- Good activity against Chlamydia and Mycoplasma has been noted:
- It is an alternative drug for nonspecific urethritis, cervicitis and atypical pneumonia.
- It also inhibits M. tuberculosis; can be used in place of ciprofloxacin.
- It is highly active against M. leprae: is being used in alternative multidrug therapy regimens.

Trade name: Taracin, oflox

Class. Antibacterial , quinolone derivative .

Action: it has broad- spectrum bactericidal activity, inhibits the synthesis of bacterial DNA by inhibiting the enzyme DNA gyrase.

Pharmacokinetics:

- Ofloxacin is relatively lipid soluble; oral bioavailability is high: attains higher plasma concentrations.
- Food does not interfere with its absorption.
- It is excreted largely unchanged in urine; dose needs to be reduced in renal failure.

Indications:

- Ofloxacin is comparable to ciprofloxacin in the therapy of systemic and mixed infections.
- It is particularly suitable for chronic bronchitis and other respiratory or ENT infections.
- Inhibition of theophylline metabolism is less marked.
- Gonorrhoea has been treated with a single 200 mg dose.
- It is also useful in nongonococcal urethritis.

Dose:

- 100, 200, 400 mg tab; 200 mg/100 ml i.v. infusion, 50 mg/5 ml susp. 0.3% eye drops.

Norfloxacin

- It is less potent than ciprofloxacin: MIC values for most gram-negative bacteria are 2–4 times higher. Many Pseudomonas and grampositive organisms are not inhibited at clinically attained concentrations. Moreover, it attains lower concentration in tissues. It is metabolized as well as excreted unchanged in urine.

Trade name: Apirol

Class. Antibacterial , quinolone derivative .

Action: it has broad- spectrum bactericidal activity, inhibits the synthesis of bacterial DNA by inhibiting the enzyme DNA gyrase.

Indications:

- Norfloxacin is primarily used for urinary and genital tract infections.
- It is also good for bacterial diarrhoeas, because high concentrations are present in the gut and anaerobic flora is not disturbed.
- Norfloxacin is not recommended for respiratory and other systemic infections, particularly where gram-positive cocci are involved.

Dose: 200, 400, 800 mg tab, 3 mg/ ml eye drops, 100 mg/5 ml syr.

Levofloxacin

Trade name: levox

Class. Antibacterial , quinolone derivative .

Action: it has broad- spectrum bactericidal activity, inhibits the synthesis of bacterial DNA by inhibiting the enzyme DNA gyrase.

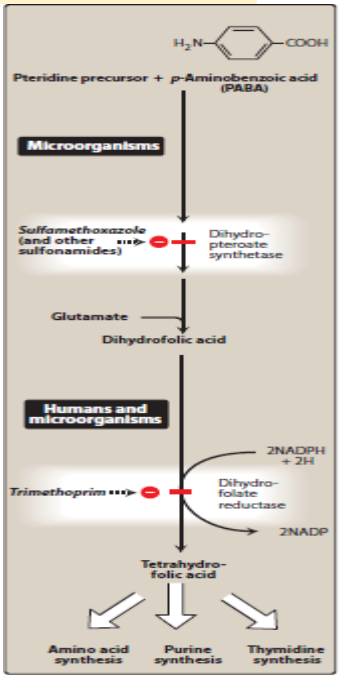
Indications:

- The primary indication of levofloxacin is community acquired pneumonia and exacerbations of chronic bronchitis in which 87–96% cure rate has been obtained.
- High cure rates have been noted in sinusitis, enteric fevers, and pyelonephritis and skin/soft tissue infections as well.
- It is the levoisomer of ofloxacin having improved activity against Strep. Pneumonia and some other gram-positive and gramnegative bacteria.
- Anaerobes are moderately susceptible.
- Oral bioavailability of levofloxacin is nearly 100%; oral and i.v. doses are similar.
- It is mainly excreted unchanged and a single daily dose is sufficient because of slower elimination.
- Theophylline, warfarin, cyclosporine and zidovudine pharmacokinetics has been found to remain unchanged during levofloxacin treatment.

Dose: 500 mg tab, 500 mg/100 ml inj.

Inhibitors of Bacterial metabolism

Sulfonamides



Sulfonamides are bacteriostatic drugs that prevent the growth of microorganisms by inhibiting folic acid production. The decreased folic acid synthesis decreases the number of bacterial nucleotides and inhibits bacterial growth.

Action: sulfonamides are structurally related to para-aminobenzoic acid; the bacteria take it instead of these acids. Therefore, tetrahydrofolic acid is not formed which is necessary for synthesis of DNA; therefore, there will be no cell multiplication.

- Sulfonamides were the first antimicrobial agents (AMAs) effective against pyogenic bacterial infections. because of rapid emergence of bacterial resistance and the availability of many safer and more effective antibiotics, their current utility is limited, except in combination with trimethoprim (as cotrimoxazole)
- Sulfonamides are primarily bacteriostatic against many gram-positive and gram-negative bacteria. However, bactericidal concentrations may be attained in urine.
- Relatively inexpensive drugs and excreted mainly via kidneys.

Classification of sulfonamides

1. Oral absorbable agents:
   - Sulfamethoxazole are used in UTI.
   - Sulfadiazine + Pyrimethamine combination is the first choice in ttt. Toxoplasmosis .
2. Oral non-absorbable agents:
   - Sulfasalazine is used in Ulcerative colitis, enteritis, and other inflammatory bowel disease
3. Topical Agents
   - Sodium sulfacetamide ophthalmic solution or ointment is effective in the treatment of bacterial conjunctivitis and as adjunctive therapy for trachoma.
   - Silver sulfadiazine (Silvadene) Dose: Topical application to burned area, once or twice daily, in a thin layer.

Uses:

- Less useful because of bacterial resistance & development of other antibiotics.
- UTI, e.g. E. coli, proteus & staph. Aureus. “cephalosporines are given now”.
- To eliminate meningococci from nasopharynx (carriers).
- As alternative to penicillin for prophylaxis from rheumatic fever.
- As alternative to tetracycline in treatment of chlamydia, trachoma & conjunctivitis.
- In combination with penicillin to treat otitis media.

Contraindications:

- Newborns and infants less than 2 months.
- Pregnancy except for treatment of toxoplasmosis.
- Used with caution in patients with impaired liver & kidney function & blood disorders.

Side effects

- Nausea, vomiting, stomatitis, abdominal pain, headache, dizziness, ataxia, photosensitivity, rash.
- Allergic reactions: Steven –Johnson syndrome (rhinitis, conjunctivitis, fever, stomatites, & rash).
- Renal damage: crystaluria, hematuria, proteinuria. Adequate hydration and alkalinization of urine can prevent the problem by reducing the concentration of drug and promoting its ionization.
- Hemolytic anemia is encountered in patients with glucose-6-phosphate dehydrogenase (G6PD) deficiency. Super-infection.
- Kernicterus: This disorder may occur in newborns, because sulfa drugs displace bilirubin from binding sites on serum albumin. The bilirubin is then free to pass into the CNS, because the blood–brain barrier is not fully developed.

Nursing consideration:

- Obtain a complete nursing and drug history.
- If the client is pregnant, the physician should be informed so that, another type of medication not harmful to a developing fetus may be used.
- With large dosages or prolonged treatment, monitor the patient for superinfections.

- During drug therapy, assess client for any of the following: reactions that may require withdrawal of the drugs:
    - Skin rashes, abdominal pain & anorexia.
    - Jaundice (hepatic involvement).
    - Renal colic, oliguria, anuria, proteinuria (renal involvement).
    - Jaundice, pallor, weakness (blood dyscrasias).
    - Rhinitis, stomatitis, fever, headache, conjunctivitis (Steven- Johnson syndrome)
    - Hemorrhage, due to vitamin K deficiency.

Silver sulfadiazine:

Trade name: Silvadene, Silverol cream**.**

Uses:

- Topically 1% cream for prevention and treatment of sepsis in second and third degree burns.
- Minor bacterial skin infections and dermal ulcers. it is not good for treating established infection

Dosage: Once or twice daily topically by sterile gloves over a clean or debrided burn.

Contraindications:

Pregnancy, lactation, infant below 2 months.

Mafenide Acetate:

Trade name: Sulfamylon

Class: Topical sulfonamide.

Uses:

- For prevention of infection in 2nd and 3rd degree burns.
- Limitation is that mafenide produces burning sensation and severe pain when applied to raw surface.

Contraindications:

- Not to be used for already established infection
- Used with caution in pregnancy & lactation.

Dosage: As silver sulfadiazine.

Sulfacetamide sodium:

Trade name: optizole eye drops, Ramasulf.

Class: sulfonamide.

Uses: Topically for treatment of ophthalmic infections including trachoma.

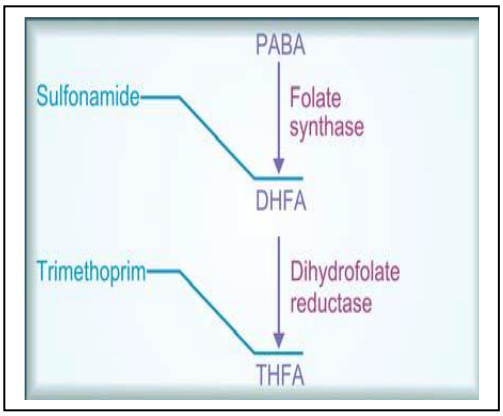
TRIMETHOPRIM

- A potent inhibitor of bacterial dihydrofolate reductase,
- Exhibits an antibacterial spectrum similar to that of the sulfonamides.
- Trimethoprim is most often compounded with sulfamethoxazole, producing the combination called cotrimoxazole.

Cotrimoxazole

Sulfamethoxazole and trimethoprim:

- The combination of trimethoprim with sulfamethoxazole, called cotrimoxazole , shows greater antimicrobial activity than equivalent quantities of either drug used alone.



Mechanism of action

- The synergistic antimicrobial activity of cotrimoxazole results from its inhibition of two sequential steps in the synthesis of tetrahydrofolic acid. Sulfamethoxazole inhibits the incorporation of PABA into dihydrofolicacid precursors, and trimethoprim prevents reduction of dihydrofolate to tetrahydrofolate

Trade name: Resprim, pathoprim, Bactil, septin, ultrasept

Dosage forms:

- Oral suspension: Sulfamethexazole 200 mg + trimethoprin 40 mg \5ml.
- Tablets: Sulfamethexazole 400 mg & trimethoprin 80/tab.
- Double strength tab. (forte): Sulfamethexazole 800 mg + trimethoprin 160 mg\tab.

Uses: UTI, otitis media , enterites (shigella), bronchitis (adult)

Additional Contraindications:

- Megaloblastic anemia due to folate deficiency, Infants less than one month of age.

Dose: one or 2 tablets q 12 hours/adults, or 4 teaspoonful 12 hrs for 10-14 days.

Nursing consideration: see above

DNA-binding agents

Nitrofurantoin:

MOA: Nitrofurantoin causes bacterial DNA damage by an unknown mechanism;

- It is bacteriostatic.
- Nitrofurantoin is concentrated in urine and is used solely as a urinary tract antiseptic against E. coli. Other urinary tract gram-negative bacteria are often resistant.

Adverse effects:

- NV.
- headache,
- hemolytic anemia in G6PD, and acute pneumonitis;
- this agent turns urine brown.

Contraindications:

- Anuria, ologuria, impaired renal function. pregnancy, lactation, infant less than 1 month of age.

Special concerns:

- Used with caution in patients with anemia, DM, electrolyte imbalance, avitaminosis, debilitating disease.
- Give the drug with food or milk to minimize GI distress

Dose: 100 mg QID with food for one week. Taking the drug with food enhances its bioavailability.

Drug interactions

Nitrofurantoin has few significant interactions:

- Probenecid and sulfinpyrazone inhibit the excretion of nitrofurantoin by the kidneys, reducing its efficacy and increasing its toxic potential.
- Magnesium salts and antacids containing magnesium can decrease the extent and rate of nitrofurantoin absorption.
- Nitrofurantoin may decrease the antibacterial activity of norfloxacin and nalidixic acid.

Inhibit cell membrane.

- Daptomycin is a cyclic lipopeptide antibiotic that is an alternative to other agents, such as linezolid and quinupristin/dalfopristin, for treating infections caused by resistant gram+ve organisms, including MRSA and vancomycin-resistant enterococci (VRE).

Mode of action:

Active only against gram (+). Works at cell membrane, not really cell wall.

Uses:

- Gram+ve bacteria (broad-spectrum activity)
- Multidrug-resistant (MDR) gram +ve bacteria
- MRSA&VRSA.
- Vancomycin-resistant enterococci.

Side effects:

- Myopathy
- Constipation, nausea, headache, and insomnia.

Colistin:

The polymyxin group of polypeptide antibiotics, discovered in the 1940s

- Polymyxin E (colistin) and polymyxin B are the main antibiotics of this group.
- Following reports on nephrotoxicity and neurotoxicity in the 1970s, they were largely replaced by other antibiotics.
- Because the resistant of new antibiotics resistant has led to their reconsideration as a therapeutic option.

Mechanism of Action:

- causing disruption of the membrane, The result of this process is an increase in the permeability of the cell envelope, leakage of cell contents.

Uses:

- For the treatment of acute or chronic infections due to sensitive strains of certain gram-negative bacilli, particularly Pseudomonas aeruginosa (ESBL).

Side effects:

- Nephrotoxicity and Neurotoxicity

Urinary Germicides/Analgesics

Phenazopyridine: sedural

Phenazopyridine hydrochloride, an azo dye used in commercial coloring, produces a local analgesic effect on the urinary tract, usually within 24 to 48 hours after the start of therapy. It's used to relieve the pain, burning, urgency, and frequency associated with urinary tract infections.

Class: Urinary analgesic.

Action: Azo dye has local anesthetic effect on UTI.

Uses: pain relief in chronic UTI, irritation, trauma instrumentation. (Not For Treatment of UTI).

Contraindications: Renal insufficiency.

Side effects: hemolytic anemia (G6 PD), nephrotoxicity, hepatotoxicity, nausea. yellow-brownish discoloration of urine.

Note: Yellow discoloration of skin and sclera indicates accumulation as a result of renal insufficiency.

Dose: 200 mg t.i.d with or after meal.

- Treatment of overdose: methylene blue IV or ascorbic acid P.O.

Nursing consideration:

- Inform patient that drug may turn urine color into deep yellow or orange color.
- Renal function test is necessary to avoid toxicity. Encourage fluid intake.

Antiviral Drugs

- Viruses are obligate intracellular parasites.
- They lack both a cell wall and a cell membrane, and they do not carry out metabolic processes.
- Viruses use much of the host's metabolic machinery, and few drugs are selective enough to prevent viral replication without injury to the infected host cells.
- Viral diseases are further complicated by the fact that the clinical symptoms appear late in the course of the disease, at a time when most of the virus particles have replicated. At this stage of viral infection, administration of drugs that block viral replication has limited effectiveness. However, some antiviral agents are useful as prophylactic agents.
- Synthetic nucleosides are a group of drugs used to treat various viral syndromes that can occur in immunocompromised patients, including herpes simplex virus (HSV) and cytomegalovirus (CMV).

Drugs in this class include: Acyclovir**,** Famciclovir**,** Ganciclovir**,** Valacyclovir hydrochloride

Antiviral drugs are a class of medication used specifically for treating viral infections, Vaccines have been widely used to prevent certain viral infections e.g measles, small pox, polio etc.

Most of the antiviral drugs now available are designed to help deal with HIV, herpes viruses, the hepatitis B and C viruses, and influenza A and B viruses.

Classification:

1. Anti-Herpes virus

   Idoxuridine, Acyclovir, Valacyclovir, Famciclovir

2. Anti-Retrovirus

   a. Nucleoside reverse transcriptase inhibitors (NRTIs):
   Zidovudine (AZT), Didanosine, Zalcitabine, Stavudine, Lamivudine, Abacavir

   b. Nonnucleoside reverse transcriptase inhibitors (NNRTIs): Nevirapine, Efavirenz,

   c. Protease inhibitors: Ritonavir, Indinavir, Nelfinavir, Saquinavir, Amprenavir, Lopinavir

3. Anti-Influenza virus

   Amantadine, Rimantadine

4. Nonselective antiviral drugs

   Ribavirin, Lamivudine, Adefovir dipivoxil, Interferon α

1. Acyclovir : Zovirax, Virax

Class. : Antiviral , anti-infective

Action:

- Drug is converted to acyclovir triphosphate which interferes with herpes simplex virus DNA polymerase and therefore, inhibit DNA replication.

Uses:

- P.O.: for initial and recurrent herpes infection in immunocompromised & nonimmunocompromised patients.
- Parenteral: initial therapy of severe genital herpes, varicella zoster infection in immunocompromised patients.
- Herpes simplex encephalitis.
- Topical: it decreases duration of healing in limited nonlife threatening infection.

Contraindications:

- Hypersensitivity to this product,
- Precautions: Pregnancy (B), breastfeeding, renal/hepatic/neurologic disease,
- electrolyte imbalance, dehydration.

Side effects:

- P.O: nausea, vomiting, anorexia, sore throat.
- Parenteral: phlebitis, hypotension , skin rash.
- Topical: Burning pain.

Dose:

- 200, 400, 800 mg tab.
- Caps. & tabs: Initial 200 mg /4 hrs for a total of 5 caps\day for 10 days.
- Susp. 200mg/5ml
- 3% eye oint, 5% skin cream;

Nursing considerations

- Signs of infection, anemia
- Toxicity: any patient with compromised renal system because product is excreted slowly with poor renal system function; toxicity may occur rapidly
- Hepatic, renal studies: AST, ALT; urinalysis, protein, BUN, creatinine, CCr, watch for increasing BUN and serum creatinine or decreased CCr; I&O ratio; report hematuria,

oliguria, fatigue, weakness; may indicate nephrotoxicity; check for protein in urine during treatment

- Bowel pattern before, during treatment; if severe abdominal pain with bleeding occurs, product should be discontinued
- Skin eruptions: rash, urticaria, itching
- Allergies before treatment, reaction of each medication; place allergies on chart in bright red letters
- Neurologic status with herpes encephalitis
- Provide adequate intake of fluids (2 L) to prevent deposits in kidneys, more likely to occur with rapid administration or in dehydration

2. Amantadine: Symmetrel

Class: Antiviral, antiparkinson, anti-influenza virus drug

Action: Prevent penetration of the virus into cell, may be by preventing uncoating of RNA.

Uses:

- Influenza A viral infection.
- Symptomatic treatment of idiopathic Parkinson.
- Parkinsonism syndrome postencephalitis.

Contraindications:

- Hypersensitivity, epilepsy and other CNS disease, gastric ulcer, pregnancy.

Side effects:

- Nausea, vomiting, constipation, anorexia, CHF, leukopnea, dizziness, nightmares, lack of mental concentration

Dose: 100 mg BD, elderly—half dose, children 5 mg/kg/ day.

3. Idoxuridine:

Trade name: virusan oint. , iduridine eye drops.

Class: ophthalmic antiviral.

Uses: Herpes simplex keratitis.

Contraindications: Hypersensitivity, deep ulceration, and lactation.

Side effects: localized to eye (visual haze , irritation, pain , edema of eye lids, photosensitivity) .

4. Interferon α: intron A

- Interferons are low molecular weight glycoprotein cytokines produced by host cells in response to viral infections and some other inducers. They have nonspecific antiviral as well as other complex effects on immunity and cell proliferation.
- Interferons bind to specific cell surface receptors and affect viral replication at multiple steps: viral penetration, synthesis of viral mRNA, assembly of viral particles and their release, but the most widespread effect is direct or indirect suppression of viral protein synthesis.
- Three types of human interferons (α, β and γ) have been produced by recombinant DNA technology.

Uses:

- Chronic hepatitis B and C
- AIDS-related Kaposi's sarcoma (but not to treat HIV as such).
- H. simplex, H. zoster and CMV infections in immunocompromised patients
- Rhinoviral cold: intranasal interferon is prophylactic, but not beneficial in those already having cold.
- Interferon is not effective orally. Clinical utility of s.c. or i.m. injected interferon is limited by substantial adverse effects.

Adverse effects

- Flu-like symptoms—fatigue, aches and pains, malaise, fever, dizziness, anorexia, taste and visual disturbances: develop a few hours after each injection, but become milder later.
- Neurotoxicity—numbness, neuropathy, tremor, sleepiness, rarely convulsions.
- neutropenia, thrombocytopenia.
- Thyroid dysfunction (hypo as well as hyper).
- Hypotension, transient arrhythmias, alopecia and liver dysfunction.

Dose: Interferon α_2B 3MU, 5MU vials for inj.

5. **Oseltamivir (Tamiflu)** is the **only** antiviral agent effective in treatment of avian & swine flu

Anti-tubercular Agents

Antitubercular drugs are used to treat tuberculosis (TB), which is caused by *Mycobacterium tuberculosis.* Unlike most antibacterials, antitubercular drugs may need to be administered over many months. This creates such problems as patient noncompliance, the development of bacterial resistance, and drug toxicity

- Take more than one drug.
- Every 3-4 months' drugs are changed to prevent resistance.
- The long duration of drug therapy increases the risk for complications.
- Drugs of T.B. are categorized as 1st line and 2nd line depending on their efficacy, activity & incidence of side effects.
- 1st line drugs: Ethambutol, Isoniazide, Rifampin, Streptomycin.
- 2nd line drugs: Capreomycin, Kanamyein, PAS.

Nursing considerations:

1- Anticipate that more than one drug will be given at the same time.
2- Don't administer concomitantly antilubercular agents that are highly toxic.
3- Conduct laboratory tests for nephrotoxicity, ototoxixity and hepatotoxicity caused by most of antilubercular agents.
4- Incorporate safety precautions to protect the client manifesting vestibular difficulties during ambulation to prevent falls and injury.
5- Teach patients about the nature of T.B. infection & how to protect others from contracting the disease.
6- Stress the importance of taking prescribed drugs as ordered and of reporting the monthly check-ups and lab studies.

Ethambutol Hcl:

Trade name: Myambutol.

Class: 1st line antitubercular agent.

Action:

- Arrest multiplication of rapidly dividing tubercle bacilli
- Absorbed from GIT & widely spread in body tissues except CSF.
- 65% of metabolized & unchanged drug execrated in urine & accumulate in patients with renal insufficiency.

Uses: Pulmonary T.B. in combination therapy.

Contraindications: Hypersensitivity, less than 13 years of age & preexisting optic neuritis.

Special concerns: pregnancy, impaired renal function, gout.

Side effects:

- Optic neuritis, loss of color (green) discrimination.
- Loss of vision (temporary).
- Peripheral neuritis (tingling), Hyperuricemia.

Note: side effects appear in early months, and then disappear.

Isoniazide: INH "Is the most effective tuberculostatic agent"

Class: 1st line antilubercular agent.

Action:

- It interfere with lipid & nucleic acid metabolism of growing bacteria, resulting in alteration of bacterial wall.
- Absorbed after oral or IM administration & is widely spread in body tissues.

Uses: T.B. caused by human, BCG., & Prophylaxis of T.B.

Contraindications: Hypersensitivity.

Side effects: peripheral neuritis, ataxia, jaundice optic neuritis and thrombocytopnea.

Note:

- Vitamin B-complex or pyridoxine 10-15mg \day may be given to CNS side effects.
- Liver function & ophthalmic test are recommended periodically.

Nursing considerations

- Monitor the patient for adverse effects, such as peripheral neuropathy, seizures, hematologic disorders, hepatitis, and hypersensitivity reactions.
- Always give isoniazid with other antitubercular drugs to prevent the development of resistant organisms.

Streptomycin sulfate:

Discussed before

Rifampin: Rifadin, Rimactane

Class: 1st line antitubercular agent.

Action: suppress RNA synthesis.

- Both bacteriostatic + bactericidal
- Well absorbed from GIT & widely distributed in body tissues.
- Metabolized in liver & execrated in feces.

Uses: Pulmonary T.B. in combination therapy.

In treatment of meningococcal carrier in Niesseria meningitis infection.

Contraindication: Hypersensitivity.

Special concerns:

- Lactation, pregnancy, children less than 5 years old, hepatic dysfunction.

Side effects:

- Nausea, vomiting, diarrhea, cramps, heartburn ataxia, jaundice, increase SGOT & SGPT., Hemolytic anemia.
- Urine, saliva, sputum, sweat, teeth, and tears to turn a reddish-orange to reddish-brown color.

Dose:

- T.B. 600 mg single dose daily.
- Meningitis 600 daily for 4 days.

Aminosalicylate: PAS " para-aminosalicylic acid".

Class: 2nd line antitubercular.

Action:

- Interfere with folic acid synthesis of susceptible tubercle.
- Bacteriostatic - absorbed from GIT, 50% metabolized + execrated in urine.

Uses: with Isoniazide or streptomycin in treatment of T.B.

Contraindication: Hypersensitivity.

Special concerns: Pregnancy, impaired liver & kidney function, peptic ulcer, congestive heart failure.

Side effect: nausea, vomiting diarrhea, thrombocytopnea, leukopnea, hemolytic anemia.

Antifungal Agents

Antifungal agents can be classified according to route of administration

1. Systemic (oral or parenteral) for systemic infections:
   - Amphotericin B
   - Flucytosine
   - Azoles (Ketoconazole, itraconazole, Fluconazole and Posaconazole).
   - Echinocandines (Caspofungin)
2. Systemic antifungal for mucocutaneus infections:
   - Griseofulvin -Terbinafine
3. Topical antifungal therapy:
   - Nystatin - Topical azoles (Clotrimazole & Miconazole)
   - Topical allylamines (terbinafine)

An antifungal agent is a drug that used to

- Treat and prevent superficial mycoses such as athlete's foot, ringworm, candidiasis (thrush),
- Serious systemic infections such as cryptococcal meningitis, and others.

NB. An accurate diagnosis of the infection before therapy is most important for the choice of the therapeutic agents. It is important that drug therapy be continued until the infectious agent has been completely eradicated to avoid the emergence of resistant strains.

1. Amphotericin B:

Trade name: Fungizone, Amphotec

Class: antifungal.

Action:

- It binds to fungal cell membrane in presence of ergosterol and alter cell membrane and causing loss of cellular macromolecules and ions producing irreversible damage.
- Fungistatic or fungicidal “depending on the dose”. It is poorly absorbed from the GIT, it can be administered I.V., intrathecal or topical.

Uses:

- The drug is toxic and should be used only for patients under close medical supervision with a relatively certain diagnosis of deep mycotic infections.

- I.V. administration is usually reserved for life threatening diseases because it is toxic.
- Systemic fungal infections, e.g. histoplasmosis, disseminated candidiasis & monilial over growth resulting from oral antibiotic therapy.
- Topical coetaneous & mucocutaneous infections of candida “monilia” infections.

Contraindications: Hypersensitivity, lactation

Side effects:

- Topical use: pruritis, dry skin, and irritation, rash.
- IV use: nausea, vomiting, diarrhea, melena, abdominal cramps.
- Fever, headache, thrombophlebitis, bone marrow depression, pancytopnea. anaphylaxis.

2. Nystatin:

Trade name: Mycostatin Mycostatin.

Class: antifungal.

Action: similar to Amphotricin B.

Uses: candida infections of the skin, mucous membranes, GI tract, vagina & mouth (thrush).

Side effects: nausea, vomiting, diarrhea with oral adminesration.

Dosage forms:

- Suspension: contains 100,000 units\ ml “12 ml bottle with dropper”.
- Ointment: contains 100,000 units\g “15 g tube”.
- Vaginal tab.: Contains 100,000 units.

Dosage:

- Suspension: For adult, 2 ml, 4 times daily.
- Pediatric: 1 ml 4 times daily.
- Vaginal cream or tab., 1 tab. inserted into vagina once or twice each day for 2 weeks.

Nursing considerations:

- Don’t mix oral suspension in foods since the mediation will be inactivated.
- Apply cream/ointment with a swab.
- Instruct client to keep medication in mouth as long as possible before swallowing.
- Insert vaginal tab. high in vagina with an applicator.
- Treatment should be continued for at least 48 hr after clinical cure has been achieved to prevent relapse.
- Discontinue topical or vaginal administration if rash or sensitivity occurs.

3. Tolnaftate:

Trade name: pitrex.

Class: Topical antifungal.

Action: The exact mechanism is not known although the drug is thought to stunt mycelial growth causing antifugal effect.

Uses:

- Tinea pedis , tinea cruris, tinea corpuris & tinea versicolor.
- Fungal infections of moist skin areas.

Contraindications: Scalp & nail infections " avoid getting into eyes"

Side effects: mild skin irritation.

Dosage form: cream tubes of 10 g. Solution Bottles of 10 ml.

Dosage: Apply bid. For 2-3 weeks.

lmidazoles: Old

- Ketoconazole
- Clotrimazole
- Miconazole

They lack selectivity, they inhibit human gonadal and steroid synthesis leading to decrease testosterone & cortisol production. Also,inhibit human P-450 hepaticenzyme.

- Fluconazole
- Itraconazole
- Posaconazole
- Voriconazole

Theyare:

- Selective
- Resistant to degradation
- Causing less endocrine disturbance

4. Miconazole:

Trade name: Daktarin, Dactazol.

Class: antifungal agent.

Actions: it alter fungal cell membrane permeability by blocking biosynthesis of fungal lipids especially ergosterol in cell membrane

Uses:

- Systemically: fungal infection e.g. mucocutaneous candiasis.
- Topically: Skin, nail, hair infections, tinea pedis (athlete's foot) and tinea corpuris.
- Oral thrush (oral gel).

Contraindication: Hypersensitivity.

Side effects:

- Topical: skin rashes, headache, burning, and irritation.
- Systemic: nausea, vomiting, diarrhea, anorexia fever, thrombocytopnea.

Dosage forms:

- Cream: tubes of 15g or 30g.
- Vaginal Ovule 50, 150mg
- Oral gel: tubes of 40 g.

Dosage:

- Topical: apply to cover affected areas in morning & evening.
- Vaginal cream\supp. One supp. daily at bedtime for 7 days.

5. Ketoconazole:

Trade name: Nizoral, ketoral.

Class.: Broad- spectrum antifungal.

Action: similar to miconazole.

Uses: candidiasis, oral thrush.

Contraindications: Hypersensitivity, Lactation, Ophthalmic use.

Side effects:

- nausea, vomiting, diarrhea,
- fever, headache, photophobia,
- Hepatotoxicity,
- Gynecomastia, impotence,

- thrombocytopenia.

Dosage:

- tab. oral suspension:
- 200 – 400 mg. As a single dose\day.

6. Clotrimazole

Trade name: Canesten**,** Mycoten.

Class.: Broad-spectrum fungistatic agent. Topical drug for vaginal and oral candidiasis.

Action: similar to miconazole.

Uses:

- Local treatment of oropharyngeal candidiasis
- topically for fungal infection in otitis externa
- Vaginal candidiasis.
- Skin infections: for athlete's foot and candidal intertrigo

Contraindications: Hypersensitivity

Side effects:

- Nausea, vomiting, diarrhea,
- Fever, headache, photophobia,
- Hepatotoxicity

7. Griseofulvin:

MOA: Inhibits mitosis by interactions with microtubules.

Uses:

- Systemic for cutaneous infections (oral ly).so it uses for superficial dermatophytosis.

Side efect:

- GI disturbances & photosensitivity.
- Binds to androgen receptors.

Anthelmintic

The term anthelmintic is restricted to drugs acting locally to expel parasites from gastro intestinal tract. There are several types of worms which penetrate other tissues.

All family members should be examined since infestations are transmitted by sharing bathrooms (hygiene is very important).

Worms divided into 3 groups:

A. Cestodes:

Tape worm e.g. taenia saginata (beef), taenia solium (pork).

B. Nematodes:

1. Filaria: filariasis
2. Hook worm: ancylostoma.
3. Pinworm: oxyuriasis (common in school age).
4. Round worm: ascaris (can cause GI & respiratory obstruction).
5. Threadworm: strongoloid.
6. Whipworm.

C. Trematodes: Bilharziasis

Considerations:

- Instruct responsible family member how to prevent infestation with pinworm by :
    - Washing hands after toiling and before meals.
    - Keep nails short.
    - Applying antipruritic ointment to anal area to reduce scratching which transfers pinworms.
    - Alert all family members to be examined for pinworms.
    - Emphasize the need for follow-up examinations to check the results of treatment.

1. Mebendazole:

Trade name: Vermox, Vermacare

Class: anthelmintic.

Action: By blocking the glucose uptake of the organism which leads to worm death.

Uses: whipworm, pinworm, roundworm.

Contraindications: Hypersensitivity.

Special concerns: Pregnancy, children less than 2 years.

Side effects: nausea, vomiting, transient abdominal pain, diarrhea.

Dosage: for pinworm, whipworm, roundworm, hookworm 1 tablet (100 mg) in the morning and evening on 3 consecutive days.

Note: All treatments can be repeated after 2-3 weeks.

2. Niclosamide: Yomesan

Class: anthelmintic.

Action:

- Inhibit oxidative phosphorylation in the mitochondria of the worm.
- The drug is not absorbed from GIT and execrated in feces.

Uses: For tape worn only.

Special concerns: pregnancy, lactation, children under 2 years.

Side effects: nausea, vomiting, anorexia, bad month odor, shin rash, alopecia, dizziness, fever, sweating.

Dose: 2g as a single dose (4 tabs X 0.5 g).

3. Piperazine: vermizine

Class: anthelmintic

Action:

- Paralyze the muscle of the parasite, which leads to dislodge & excretion of parasite.
- Absorbed via GIT, metabolize in liver & excreted in urine.

Uses: Pinworm (oxyuriasis), round worm (ascariasis).

- Recommended for pediatric use.

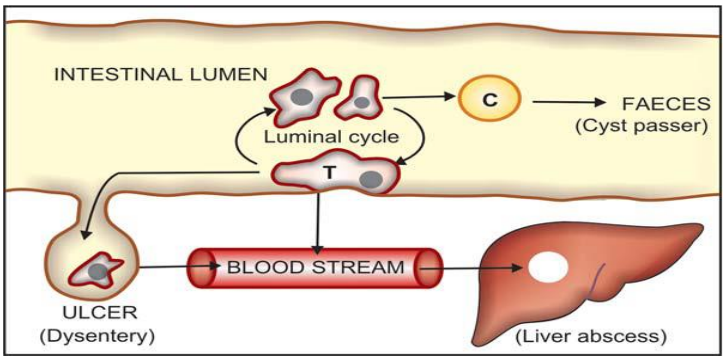
Contraindications: impaired liver & kidney function, seizures, and hypersensitivity.

Side effects: nausea, vomiting, diarrhea, tremors, ataxia, blurring of vision, cataract, bronchospasm, rash.

Dosage: 75mg/kg orally for two successive days.

Amebicides drugs

- **Antiamoebic drugs:** these are drugs useful in infection caused by the protozoa entamoeba histolytica.
- Amoebiasis has a worldwide distribution (over 40 million people are infected), but it is endemic in other developing countries.
- Poor environmental sanitation and low socio-economic status are important factors in the spread of the disease, which occurs by faecal contamination of food and water.



Classification

1. Tissue amoebicides

a) For both intestinal and extraintestinal amoebiasis:
   - Nitroimidazoles: Metronidazole, Tinidazole, Secnidazole, Ornidazole, Satranidazole
   - Alkaloids: Emetine, Dehydroemetine

b) For extraintestinal amoebiasis only: Chloroquine

2. Luminal amoebicides

a) Amide : Diloxanide furoate, Nitazoxanide

b) 8-Hydroxyquinolines: Quiniodochlor (Iodochlorohydroxyquin, Clioquinol), Diiodohydroxyquin (Iodoquinol)

c) Antibiotics: Tetracyclines

1- **Metronidazole:** Flagyl

Class: systemic trichomonocide, amebicide.

Action:

- Effective against anaerobic bacteria & protozoa.
- Inhibit growth of trichomona & amebae by binding to DNA & inhibit nucleic acid synthesis; resulting in cell death.
- Well absorbed from GIT & widely distributed in tissues.
- Eliminated in urine (primarily), 20% unchanged, resulting in red, brown discoloration in urine following P.O. or I.V. use.

Uses:

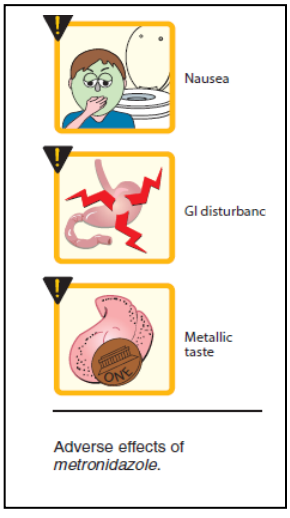
1- Amebiasis, trichomoniasis.
2- Amebic dysentery.
3- Amebic liver abscess.
4- Septicemia
5- Endocarditis
6- Giardiasis
7- To control anaerobic infections of the abdomen following colorectal surgery, emergency appendectomy.

Contraindications:

- Active organic disease of CNS.
- Blood disorders
- Lactation.
- 1st trimester of pregnancy.
- Topical Hypersensitivity.

Side effects:

- Headache, dizziness, confusion, irritability, restlessness, ataxia, depression, fatigue, drowsiness, insomnia, paresthesia, peripheral neuropathy, seizures.
- Nausea, vomiting, diarrhea, epigastric distress, anorexia, constipation, abdominal cramps.
- Blurred vision, sore throat, retinal edema, dry mouth, metallic taste, furry tongue, glossitis, stomatitis, photophobia, optic neuritis.
- Darkened urine, vaginal dryness, polyuria, albuminuria, dysuria,



Adverse effects of *metronidazole.*

cystitis, decreased libido, nephrotoxicity, incontinence, dyspareunia, candidiasis, increased urinary frequency .

- Leukopenia, bone marrow, depression, aplasia, thrombocytopenia
- Rash, pruritus, urticaria, flushing, Stevens-Johnson syndrome, phlebitis at injection site, toxic epidermal necrolysis

Dose: 500-750 mg, 3 times daily for 5-10 days.

Nursing considerations:

- If used IV, drug should not be given by IV bolus.
- If a primary IV fluid setup is used, discontinue the primary solution during infusion of metronidazole.
- Report any symptoms of CNS toxicity immediately, e.g. ataxia or tremor, which necessitate withdrawal of drug.
- The drug may turn urine brown.
- Explain for the male partner, the necessity to have therapy.

2- Tinidazole:

Trade name: Tinogyn

It is an equally efficacious congener of metronidazole, similar to it in every way except:

- Metabolism is slower; t½ is ~12 hr;
- Duration of action is longer; dosage schedules are simpler. Thus, it is more suited for single dose or once daily therapy.
- Some comparative trials in amoebiasis have reported higher cure rates.
- It appears to be better tolerated; the incidence of side effects is lower: metallic taste (2%), nausea (1%), rash (0.2%).
- Dosage form: 300, 500, 1000 mg tabs; 800 mg/400 ml i.v. infusion
- Anaerobic infections: prophylactic—2 g single dose before colorectal surgery;
- H. pylori: 500 mg BD for 2 weeks in triple combination.

3- Diloxanide furoate:

Class: amebicide

Action: probably directly amebicidal but exact mechanism is unknown.

Uses:

- It is a highly effective luminal amoebicide: directly kills trophozoites responsible for production of cysts.
- Diloxanide furoate exerts no antibacterial action.
- It is less effective in invasive amoebic dysentery, because of poor tissue amoebicidal action. However, a single course produces high (80–90%) cure rate in mild intestinal amoebiasis and in asymptomatic cyst passers.
- It is primarily metabolized by glucuronidation and is excreted in urine.

Side effects:

- GIT: flatulence, nausea, vomiting, diarrhea, dryness of mouth, abdominal cramps.
- Renal: proteinuria.
- Skin: urticarial, tingling sensation.

Contraindications: hypersensitivity, pregnancy and children under two years.

Dose: 500mg 3 time daily with meals for 10 days.

Miscellaneous drugs:

Gentian Violet

Class: Topical & vaginal anti-infective.

Action: Rosaniline dye is effective against some gram positive bacteria, many fungi & many strains of candida. Treatment should continue until symptoms subside.

Uses: Topically for treatment of cautoneous and mucocautaneous condida albicans as thrush, Vulvovaginal candidiasis, Candidiasis in general.

Contraindications: Hypersensitivity, Ulcerative lesions of face and vagina.

Side effects: Irritation, ulceration of mucus membranes, laryngitis, esophagitis, laryngial obstruction, & vaginal irritation.

Dose:

- Vaginal; one tampon (5 mg), inserted for 3 hours once or twice daily for 12 consecutive days.
- Topical solution: 1% - 2% solution to affected area 2-3 times for 3 days.

Povidone iodine: Betadine

Class: Antiseptic, germicide.

Action:

- Is a non-stinging, non-staining iodine complex with all of the antiseptic properties of iodine but without skin & mucus membrane irritation.
- Bactericidal for gram positive & negative bacteria, antibiotic resistant organisms, fungi, viruses, protozoa & yeast .

Uses:

- Topical dressing
- Antiseptic for wounds & burns.
- Degerming of skin
- Preoperatively.
- Treatment of dandruff.

Contraindications: skin hypersensitivity.

Dose: on full strength (solution or ointment) only once.

Cardiovascular system drugs

Components of the cardiovascular system include the heart, arteries, capillaries, veins, and lymphatics. These structures transport life-supporting oxygen and nutrients to cells, remove metabolic waste products, and carry hormones from one part of the body to another. Because this system performs such vital functions, any problem with the heart or blood vessels can seriously affect a person's health.

Types of drugs used to improve cardiovascular function include:

1. Inotropic drugs, 2. Antiarrhythmic drugs, 3. Antianginal drugs, 4. Antihypertensive drugs, 5. Diuretics, 6. Antilipemic drugs.

Diuretics

A **diuretic** is any drug that elevates the rate of urination and thus provides a means of forced diuresis. Diuretic drugs include several classes of diuretics that reduce blood pressure by increasing urine output and decreasing edema, circulating blood volume, and cardiac output.

Action of diuretics:

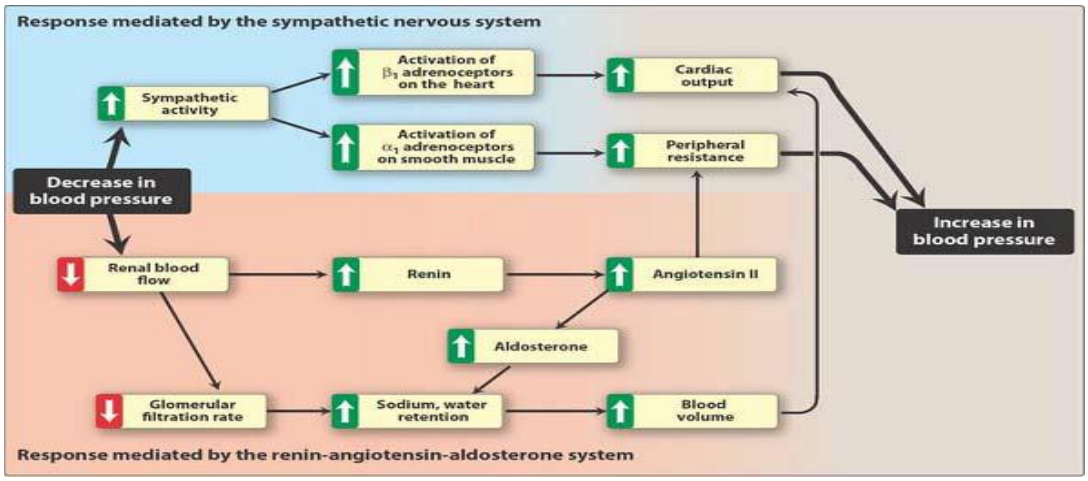
It increases the urinary output of water and sodium "prevention or treatment of edema" through one of the following mechanisms:

1- Increasing the glomerular filtration rate.

2- Decreasing the rate at which sodium is reabsorbed from the glomerular filtrate by the renal tubules, therefore water is excreted along with sodium.

3- Promoting the excretion of sodium & therefore water by the kidney.

Uses: Congestive heart failure, hypertension, edema.



1- Carbonic Anhydrase Inhibitor Diuretics:

Generic name: Acetazolamide (Diamox, uramox)

Class. : anticonvulsant, diuretic.

Therapeutic actions:

- In the kidneys, carbonic anhydrase inhibitors decrease the availability of hydrogen ions, which blocks the sodium-hydrogen exchange mechanisms. As a result, urinary excretion of sodium, potassium, bicarbonate, and water increases.
- Inhibition of carbonic anhydrase in the eye results in decreased secretion of aqueous humor results in lowering of intraocular pressure
- CNS inhibition of carbonic anhydrase and resultant diuresis may decrease abnormal neuronal firing

Indications

- Lowering of intraocular pressure in the treatment of glaucoma
- Management of acute altitude sickness
- Diuretic
- Adjunct to the treatment of refractory seizures
- Prevention of renal calculi composed of uric acid or cystine

Contraindications:

- Low serum level of sodium & potassium.
- Renal & hepatic dysfunction.
- Adrenal insufficiency.
- Hypersensitivity to thiazide diuretics.

Side effects: anorexia, polyuria, drowsiness, confusion, & acidosis.

Dosage forms: Tablets, 125 mg, 250 mg, ER capsules: 500 mg Injection: 500 mg/vial

Dose: Tab. 4-30 mg\kg\day in divided doses.

2- Loop Diuretics:

Generic name: Furosemide (Fused, Lasix)

Class: Loop diuretic.

Action:

- Inhibits sodium and chloride reabsorption in the ascending loop of Henle, thus increasing renal excretion of sodium, chloride, and water

- Increases excretion of potassium (like thiazide diuretics)
- Produces greater maximum diuresis and electrolyte loss than a thiazide diuretic

Indications

- Oral, IV: Edema associated with CHF, cirrhosis, renal disease
- IV: Acute pulmonary edema
- Oral: Hypertension.

Contraindications

- Allergy to furosemide, sulfonamides.
- Anuria
- Severe renal failure and Hepatic coma
- Pregnancy and Lactation.

Side effects:

- Dehydration, hypovolemia, Hypokalemia, hyperglycemia, Hyponatremia
- Nausea, vomiting, diarrhea, anorexia.
- Tinnitus, blurring of vision, headache, orthostatic hypotension, rashes & photosensitivity.
- Thrombophlebitis & cardiac arrest after I.V use.
- Pain at injection site after I.M use.

Forms: tablets 40 mg, Ampules 20 mg /2ml.

Dose:

- Oral: 20-80 mg as a single dose.
- I.V: 20-40 mg as a single dose.
- For hypertensive crisis:100-200 mg.



Nursing considerations:

1- Observe client for signs of dehydration & circulatory collapse.
2- Monitor pulse & blood pressure.
3- Advise the client to take medication in the morning to avoid interruption of sleep.
4- Monitor for adverse effects, such as pancreatitis, hematologic disorders, and electrolyte imbalances (especially hypokalemia).
5- Signs of hypokalemia may include leg cramps and muscle aches and discuss the need for a diet high in potassium.

3- Osmotic Diuretics

Generic name: Mannitol (Osmitrol)

Class: Osmotic diuretic.

Action:

- Increase the osmolarity of the glomerular filtrate, which decrease the reabsorption of water while increasing the excretion of sodium and chloride.
- It also increases the osmolarity of the plasma which cause increase flow of water from tissues to the interstitial fluid & plasma, thus cerebral edema, increased intracranial pressure & CSF volume & pressure are reduced.

Uses:

- Acute renal failure
- Cerebral edema
- To decrease intracranial pressure
- Glaucoma

Contraindications:

- Anemia, Dehydration, pulmonary edema

Side effects:

Hypernatremia, electrolyte imbalance, acidosis, Dehydration, dry mouth, thirst, edema, hypotension, blurring of vision, headache, dizziness.

Dose: 50 ml of 25%, 75 ml of 20 % or 100 ml of 15% solution infused over 3-5 minutes.

Nursing considerations:

1- Mannitol should not be added to other I.V. solutions nor should it be mixed with other medications.
2- If blood is to be administered at the sometime, add 20 mEq of sodium chloride to each liter of mannitol to prevent pseudoagglutination.
3- Monitor & record vital signs.
4- Observe for signs of electrolyte imbalance or dehydration.
5- Observe for signs, & symptoms of pulmonary edema (dyspnea,cyanosis, frothy sputum). "Slow the rate & notify the physician".
6- Assess for signs of orthostatic blood pressure changes, which could lead to falls and injuries.

4- Potassium – Sparing Diuretics:

Generic name: Spironolactone (Aldactone)

Class: Diuretic – potassium sparing.

Action:

- Is a mild diuretic that acts on the distal tubule to inhibit sodium exchange for potassium which results in increased secretion of sodium and water & conservation of potassium
- It is also aldosterone antagonist.
- It has slight antihypertensive effect.
- It also interferes with syntheses of testosterone & may increase formation of estradiol from testosterone thus leading to endocrine abnormalities.

Uses:

- Edema due to congestive heart failure
- Liver cirrhosis.
- Nephrotic syndrome.
- Essential hypertension.
- Primary hyperaldosteronism.
- Hypokalemia (as in CHF).

Contraindications:

- Acute renal insufficiency.
- Progressive renal failure.
- Patients receiving potassium supplement.
- Hyperkalemia.



Side effects:

- Hyperkalemia, hyponatremia (dry mouth, lethargy, thirst & easy fatigability).
- Vomiting, diarrhea, cramps.
- Menstrual irregularities, gynecomastia, hirsutism & deeping of voice, impotence.
- Skin rashes & breast carcinoma.

Dose: 100 mg (1 tab) per day.

Nursing consideration:

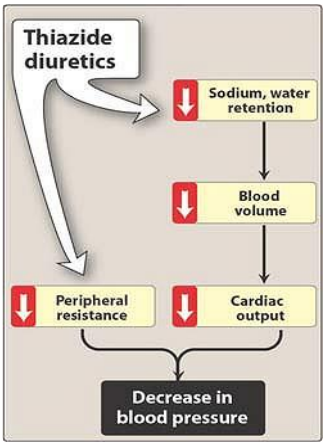
- Protect drug from light.
- Food may increase absorption of aldactone.
- Obtain serum electrolyte levels prior to starting therapy.
- Record vital signs, intake & output & body weight.
- Advise the client to avoid food high in potassium.

5- Thiazides & related diuretics:

Generic name: Hydrochlorothiazide (Esidrex.)®

Action:

- Interferes with sodium transport across tubules of the cortical diluting segment of the nephron
- Increases renal excretion of sodium, chloride, water, potassium, and calcium
- Increases bicarbonate, magnesium, phosphate, bromide, and iodide excretion
- Decreases excretion of ammonia, causing increased serum ammonia levels



N.B.:

- It is chemically related to sulfonamides.
- It has antihypertensive effect by direct dilation of arterioles as decrease total blood volume.

Uses:

- Edema due to congestive heart failure, nephrosis , liver cirrhosis & renal failure.
- Hypertension

Contraindications:

- Hypersensitivity
- Impaired renal function
- Edema due to toxemia of pregnancy (adverse of effect on the newborn) .
- Hypokalemia.

Side effects:

- Hypokalemia (cardiac arrhythmias).
- Hyponatremia (nausea, vomiting, lethargy, epigastric distress).
- Dry mouth, diarrhea, and easy fatigability.
- Skin rashes, muscle cramps.
- Hyperglycemia.

Nursing considerations: See Furosemide (lasix) .

Potassium Chloride

Trade name: KCL for IV preparation, Slow K for oral preparations.

Classification: Electrolyte, mineral.

Uses:

- Patients receiving high doses of potent diuretics.
- Patients having secondary hyperaldosteronism.
- When there is excessive loss of potassium in feces.
- When potassium intake is not adequate (especially in elderly).

N.B. maintenance of adequate potassium level is necessary for patients receiving digoxin to avoid digitalis toxicity.

Dose:

- Available in slow release tablets of 600mg/tablets and in ampules of 20 mmol/10 ML.
- Vary according to patients' needs
- Maintenance dose for patients who receive normal diet is 2-4 gm daily in divided doses.
- IV: 20 mmol is added to 500 ML of IV solution (D5W or NS) and to infuse slowly over 2-3 hours.

Contraindications:

- Severe renal failure.

- If serum potassium level is over 5 mmol/liter.

Side effects:

- Hyperkalemia
- Phlebitis at injection site.
- Heart toxicity if infused rapidly.
- With oral tablet: Dyspepsia, nausea and vomiting, esophageal or bowel ulceration.

Nursing considerations:

- Mix solution well.
- Administer solution slowly (500 ML of solution over 2-3 hours) Monitor serum potassium level.
- Assess insertion site for signs of phlebitis. Give oral preparation on full stomach.

B: drugs that alter sympathetic nervous system.

1. **Beta-adrenergic blocking agents:** discussed before
2. **Centrally acting agents:** Used for treatment of chronic hypertension, since it affects CNS.

Methyldopa: (Aldomin)

Classification: Antihypertensive, centrally acting antiadrenergic agent.

Action:

- Converted to alpha methyl norepinephrine, which is stored in adrenergic nerve granules where it replaces norepinephrine and is released by nerve stimulation to interact with presynaptic central alph2 adrenoceptors and decrease sympathetic outflow and decrease arterial pressure.
- Inhibit dopa decarboxylase lead to decrease stores of norepinephrine in the sympathetic nervous system and decrease BP.

Uses: Hypertension & hypertension crises (parenteral).

Side effects:-

- Headache, dizziness, general weakness, depression and sedation.
- Bradycardia, orthostatic hypotension.
- Dry mouth, nausea, vomiting, sore (black) tongue.
- Jaundice, liver disorders.
- Hemolytic anemia & leukopenia.
- Male impotence.

Contraindications: Hypersensitivity, Active hepatic diseases, Pheochromocytoma.

Dosage: Tabs. : initially 250 mg bid or tid.

Nursing considerations:-

1. Dose increases should be made with the evening dose to minimize drowsiness
2. Avoid activities that need mental awareness such as driving.
3. Note any evidence of jaundice and do liver function test on intervals.
4. Advise pt. to rise from the bed slowly.
5. Instruct pt. about reportable Signs & Symptoms.
6. Explain to pt. that urine rarely may be turn into dark / red-black color.
7. Advise patient to consult health care professional before taking any Rx, OTC, or Herbal products, especially cough, cold, or allergy remedies.

C: Direct acting vasodilators.

- Vasodilation refers to the widening of blood vessels resulting from relaxation of smooth muscle cells within the vessel walls, particularly in the large arteries, smaller arterioles and large veins.
- The process is essentially the opposite of vasoconstriction, or the narrowing of blood vessels. When vessels dilate, the flow of blood is increased due to a decrease in vascular resistance. Therefore, dilation of arterial blood vessels (mainly arterioles) leads to a decrease in blood pressure.

Hydralazine hydrochloride: (Apresoline)

Classification: Antihypertensive, direct action on vascular smooth muscles.

Action: directly affect smooth muscles results in vasodilation and increase acardiac output and blood flow to the brain and kidneys.

Uses:

- Management of moderate to severe hypertension.
- Management of congestive heart failure.
- Management of hypertension secondary to pre-eclampsia/eclampsia.
- Given parenterally in hypertension emergencies.

Contraindications:

- Angina pectoris.
- Rheumatic heart disease.
- Chronic glomerulonephritis.
- Systemic lupupus erthrmatosis (S.L.E.)

Side effects:

- Orthostatic hypotension, tachycardia, nausea, vomiting.
- Headache, dizziness, constipation and male impotence.

Available forms: Injection, solution, as hydrochloride: 20 mg/mL (1 mL), Tablet as hydrochloride: 10 mg, 25 mg, 50 mg, 100 mg

Dose:

- Tabs. : Initially 10 mg qid for 2-4 days, then 25 mg bid
- I.V, I.M:-50 mg (IV slowly) repeated as necessary. (May decrease Bp in 5 minutes.)

Other drugs: diazoxide, Minoxidil

D: Angiotensin-converting enzyme inhibitors (ACE-Inhibitor):

Captopril: (Capotin, inhabace.)

Class: antihypertensive, inhibitor of angiotensin synthesis**.**

Action:

- Captopril is a highly specific competitive inhibitor of angiotensin I converting enzyme. The enzyme is responsible for the conversion of angiotensin I to angiotensin II which decrease BP.
- Reduce peripheral arterial resistance.
- Decrease aldosterone secretion which works to increase level of serum potassium.

Indications:

- Hypertension.
- In combination with diuretics and digitalis in the treatment of CHF.

Contraindication: Hypersensitivity, renovascular disease, pregnancy and lactation.

Side effects:

- Skin rash, loss of taste, neutropnea, nausea, vomiting, hypotension, proteinuria, renal failure and hyperkalemia.

Dosage:

- Tablets: 12.5 mg 2-3 time per day. If there is no response, after 1-2 weeks, increase dose to 25 mg 2-3 time per day.

Nursing considerations:

1. In case of overdose, give normal saline to restore BP.
2. Should not be discontinued without medical order.
3. Obtain baseline hematological studies, liver & renal functions tests prior to beginning the treatment.
4. Determine client's understanding of the therapy and if he/she takes other medications.
5. Observe client closely for hypotension 3 hours after the initial dose.
6. Withhold potassium sparing diuretics and consult with physician (hyperkalemia may occur).
7. Take captopril 1 hour before meal or on an empty stomach.
8. Report skin rash, heartburn, and chest pain to physician.
9. Explain to client that he may develop loss of taste for 2-3 months, if it persist, notify the physician.

Enalapril maleate

Trade name: convertin, anapril

Class: ACE-inhibitor

Uses: Hypertension

Side effects:

- CNS: headache, dizziness, fatigue, vertigo, syncope.
- CV: hypotension, chest pain, angina pectoris.
- GI: diarrhea, nausea, abdominal pain, vomiting.
- GU: decreased renal function (in patients with bilateral renal artery stenosis or heart failure).
- Hematologic: bone marrow depression.
- Respiratory: dyspnea, dry, persistent, nonproductive cough. Skin: rash.
- Other: angioedema.

Interactions

- Diuretics: May excessively reduce blood pressure. Use together cautiously.
- Insulin, oral antidiabetics: May cause hypoglycemia, especially at start of enalapril therapy. Monitor patient closely.
- Lithium: May cause lithium toxicity. Monitor lithium level.
- NSAIDs: May reduce antihypertensive effect. Monitor blood pressure.
- Potassium-sparing diuretics, potassium supplements: May cause hyperkalemia. Avoid using together unless hypokalemia is confirmed.

Contraindications & cautions

- Contraindicated in patients hypersensitive to drug and in those with a history of angioedema related to previous treatment with an ACE inhibitor.
- Use cautiously in patients with renal impairment or those with aortic stenosis or hypertrophic cardiomyopathy.

Dose:

- Initially, 5 mg P.O. once daily; then adjusted based on response. Usual dosage range is 10 to 40 mg daily as a single dose or two divided doses. Or, 1.25 mg I.V. infusion over 5 minutes q 6 hours.

- Adjust-a-dose: If patient is taking diuretics or creatinine clearance is 30 ml/minute or less, initially, 2.5 mg P.O. once daily.

Nursing considerations

1. Closely monitor blood pressure response to drug.
2. Monitor CBC with differential counts before and during therapy.
3. Diabetic patients, those with impaired renal function or heart failure, and those receiving drugs that can increase potassium level may develop hyperkalemia. Monitor potassium intake and potassium level.
4. Instruct patient to report breathing difficulty or swelling of face, eyes, lips, or tongue. Swelling of the face and throat (including swelling of the larynx) may occur, especially after first dose.
5. Advice patient to report signs of infection, such as fever and sore throat.
6. Inform patient that light-headedness can occur, especially during first few days of therapy. Tell him to rise slowly to minimize this effect and to notify doctor if symptoms develop. If he faints, he should stop taking drug and call prescriber immediately.
7. Tell patient to use caution in hot weather and during exercise. Inadequate fluid intake, vomiting, diarrhea, and excessive perspiration can lead to light-headedness and fainting.
8. Tell woman of childbearing age to notify doctor if pregnancy occurs. Drug will need to be stopped.

E: Calcium channel blocking agents:

Action: for contraction of cardiac and smooth muscle to occur, extracellular calcium must move into the cell through openings called calcium channels. These agents inhibit the influx of calcium through the cell membrane resulting in a depression of automatically and conduction velocity in both smooth and cardiac muscles leading to:

1. Decrease myocardial contractility.
2. Dilatation of coronary arteries.
3. Decrease peripheral vasodilatation and peripheral resistance.
4. Decrease S. A. node automatically and conduction and decrease heart rate.

Diltazem Hcl:

Trade name: Cardiazem, dilatam.

Class: calcium channel blocking agent (anti-angina, antihypertensive).

Uses: vasospastic angina, essential hypertension.

Contraindications: hypotension, pulmonary congestion, and MI.

Side effects: AV block, bradycardia, CHF. Hypotension.

Dosage: 30 mg qid before meals and at bedtime.

Nifedipine:

Trade name: Adalat

Class: calcium channel blocking agent (anti-angina, antihypertensive).

Uses: vasospastic angina, essential hypertension.

Contraindications: hypersensitivity, lactation.

Side effects: pulmonary and peripheral edema, MI, hypotension, dizziness, light- headedness, palpitation, headache, muscle cramps.

Dosage: 10- 30 mg tid. In hypertensive emergencies: 10-20 mg given orally or sublingually by puncturing the capsule and squeezing contents under the tongue.

Amlodipine besylate (Amicore)

Class: calcium channel blocking agent (anti-angina, antihypertensive).

Use: Angina and hypertension.

Adverse reactions

- CNS: headache, fatigue, dizziness, light-headedness, paresthesia.
- CV: edema, flushing, palpitations.
- GI: nausea, abdominal pain.
- Musculoskeletal: muscle pain.
- Respiratory: dyspnea.
- Skin: rash, pruritus.

Contraindications & cautions

- Contraindicated in patients hypersensitive to drug.
- Use cautiously in patients receiving other peripheral vasodilators, especially those with severe aortic stenosis, and in those with heart failure.

- Because drug is metabolized by the liver, use cautiously and in reduced dosage in patients with severe hepatic disease.

Dose: Initially, 2.5 to 5 mg P.O. daily. Dosage adjusted according to patient response and tolerance. Maximum daily dose is 10 mg.

Nursing considerations

1. Monitor blood pressure frequently during initiation of therapy. Because drug-induced vasodilation has a gradual onset, acute hypotension is rare.
2. Notify doctor if signs of heart failure occur, such as swelling of hands and feet or shortness of breath.
3. Teach patient to continue taking drug, even when feeling better.
4. Grapefruit juice may increase drug level and adverse reactions. Discourage use together.

Verapamil: (Ikacor)

Class: calcium channel blocking agent (anti-angina, antihypertensive, antiarrhythmia).

Uses:

- **P.O:**
    - Angina pectoris.
    - Arrhythmia (atrial fibrillation, and flutter).
    - Essential hypertension.
- **IV:** supraventricular tachycardia.

Contraindications: hypotension, cardiac shock, and MI.

Side effects: AV block, bradycardia, headache, dizziness, abdominal cramps, blurring of vision, and edema.

Dosage: Initial 80-120 mg tid then 240-480 mg /day.

Nursing considerations for calcium channel blocking agents:

1. Discuss with the patient/family the goals of therapy.
2. Teach them how to take pulse and blood pressure. Hold the medication in case of hypotension or bradycardia and consult the treating physician.
3. Instruct the client to report any untoward sings as dizziness.
4. In case of postural hypotension, advise the client to change position slowly.
5. Advise client to sit down immediately if fainting occurs.
6. Calcium antagonists should be taken with meals to decrease GI irritation.

Cardiac Glycoside

Components of the cardiovascular system include the heart, arteries, capillaries, veins, and lymphatics. These structures transport life-supporting oxygen and nutrients to cells, remove metabolic waste products, and carry hormones from one part of the body to another. Because this system performs such vital functions, any problem with the heart or blood vessels can seriously affect a person's health.

Types of drugs used to improve cardiovascular function include: Inotropic drugs, Antiarrhythmic drugs, Antianginal drugs, Antihypertensive drugs, Diuretics, Antilipemic drugs.

- Cardiac glycosides are a group of drugs derived from digitalis, a substance that occurs naturally in foxglove plants. The most commonly used cardiac glycoside is digoxin.
- Cardiac glycosides are the most effective drugs for treatment of C.H.F.
- They increase myocardial contractions, which will increase blood supply to all organs including the kidneys therefore causing diuresis, which will decrease the edema.
- They are used to treat cardiac arrhythmia because they decrease heart rate.

Action:

- They increase the force of myocardial contractions (positive inotropic).
- It increases the contractility of the heart muscle by minimizing the movement of Na+ and K+ ions and increasing the release of Ca++ ions in the myocardial cells.
- It decreases the heart rate due to increase in parasympathetic nervous system and decrease in the sympathetic tone.
- They are primarily excreted through the kidneys.
- The initial dose is the larger dose (the loading or digitalizing dose), the subsequent doses are referred to as (Maintenance doses).

Results:

- Decrease in venous pressure, Coronary dilatation, Reduce heart size, Marked diuresis and decreasing edema.

Indications:

- Congestive heart failure (C.H.F), Cardiac arrhythmia (Supraventricular tachycardia, atrial fibrillation, atrial flutter and sinus tachycardia).

Contraindication:

- Hypersensitivity, Angina pectoris in absence of CHF, Given with caution for elderly and people who have kidney failure.

Side effects:

1. They are extremely toxic and may cause death.
2. There is a narrow margin of safety between the therapeutic dose and the toxic dose.
3. Could cause overdose by cumulative effects of the drug, so frequent assessment of the serum level is essential.
4. May cause cardiac arrhythmia such as bradycardia (below than 60 beat/ minute), ventricular fibrillation (which may lead to cardiac arrest and death), and Bigeminal rhythm.
5. Nausea, vomiting, and diarrhea.
6. Headache, malaise and muscle weakness.
7. Skin rashes, blurring of vision, diplopia and white halos.

Note:

- Digitalization: If there is no urgent need for a desired effect, an oral digitalizing dose is first administered and then the maintenance dose is adjusted on the bases of clinical and laboratory assessment.
- Patients suffering from digitalis intoxication should be admitted to the ICU for continuous monitoring of ECG. Administration of digitalis should be stopped.
- If serum potassium is below normal, administer K+ salts and give antiarrhythmic drugs as Lidocain as ordered by physician.

Predisposing factors for digitalis toxicity:

1. K+ loss (hypokalemia) which results from: diuretics, NPO, gastric suction, and poor K+ intake.

1. Pathological conditions;
    a. Liver disease: they decrease metabolism and therefore increase digitalis level.
    b. Kidney disease: they decrease the excretion of drug and therefore increase digitalis levels.

Implementation

- Keep in mind that patients with hypothyroidism are extremely sensitive to cardiac glycosides and may need lower doses. Reduce the dosage in patients with impaired renal function.
- Before giving a loading dose, obtain baseline data (heart rate and rhythm, blood pressure, and electrolyte levels) and question the patient about recent use of cardiac glycosides (within the previous 3 weeks). The loading dose is always divided over the first 24 hours unless the clinical situation indicates otherwise.

Nursing considerations:

1. Monitor the patient for adverse effects, such as fatigue, agitation, hallucinations, arrhythmias, anorexia, nausea, and diarrhea.
2. Withhold the drug if the apical pulse is less than 60 beats/minute, and notify the prescriber.
3. Periodically monitor serum potassium and digoxin levels.
4. Assess renal function because digoxin is excreted by the kidneys.
5. Check doctor's order, medication record and bottle label accurately.
6. Weigh the patient in daily basis.
7. Pulse should be checked by 2 nurses.
8. Provide the client with food high in potassium as banana, orange.
9. Monitor serum digoxin level.
10. Elderly people should be assessed for early signs of toxicity.
11. Teach patients that bradyacadia, nausea, vomiting, diarrhea, appetite loss, and visual disturbances could be early signs of toxicity.
12. Have digoxin antidote available (digoxin immune FAB).

Drugs:

1 . Digitoxin:

Trade name: crystodigin

Class: cardiac glycoside

Uses: Heart failure (CHF)**,** Atrial fibrillation flutter**,** Supraventricular tachycardia

Dose:

- Digitalizing dose is 0.6 mg in 4-6 hours.
- Maintenance: 0.05 – 0.3 mg/day.

2. Digoxin:

Trade name: Lanoxin, digoxin zori

Class: cardiac glycoside.

It is the drug of choice for CHF because of:

1. It has rapid onset.
2. It has short duration.
3. It can be administered P.O. or IV.

Dose: Digitalization dose = 0.4 – 0.6 mg followed by 0.05 – 0.35 mg once or twice daily.

3. Digoxin Immune FAB:

Trade name: (Ovine)

Class: digoxin antidote.

Action: antibodies bind to digoxin and excreted through the kidneys.

Uses: life threatening digitalis toxicity or overdose.

Note: cardiac arrest can be expected if an adult ingests 10 mg or if a child ingests 4 mg.

Drug interactions

Many drugs can interact with digoxin:

- Rifampin, barbiturates, cholestyramine, **antacids**, kaolin and pectin, sulfasalazine, neomycin, and metoclopramide **reduce** the therapeutic effects of digoxin.
- Calcium preparations, quinidine, verapamil, cyclosporine, tetracycline, nefazodone, clarithromycin, propafenone, amiodarone, spironolactone, hydroxychloroquine, erythromycin, itraconazole, and omeprazole **increase** the risk of digoxin toxicity.
- Amphotericin B, potassium-wasting diuretics, and steroids taken with digoxin may cause hypokalemia (low potassium levels) and **increase** the risk of digoxin toxicity.

Coronary vasodilators

Antianginal drugs

Angina pectoris: is a clinical syndrome characterized by paroxysm of pain in the anterior chest caused by insufficient coronary blood flow and/or inadequate oxygen supply to the myocardial muscle.

Causes: Atherosclerosis, Vasospasm.

There are three groups of drugs used for treatment of angina: a) Nitrates/nitrites, b) Beta-adrenergic blocking agents, c) Calcium channel blocking agents.

Nitrates/nitrites:

Action:

- Direct relaxation of blood vessels and smooth muscles vasodilatation lead to decrease O_2 requirements.
- Relaxation of smooth muscles of coronary arteries lead to coronary vasodilatation and increase blood supply to the myocardium.
- Relaxation of arteries and veins lead to decrease BP and decrease workload in the heart.

Objectives of treatment:

1. Treatment of anginal attack and thus relief pain.
2. Prophylactic treatment to prevent or delay the occurrence of MI.
3. Prolongs intervals between attacks.

Indications:

- Prophylaxis and treatment of acute angina pectoris.
- Treatment of chronic angina pectoris.
- Treatment of hypertension associated with MI or CHF.
- Nitroglycerin ointment for treatment of Raynaud's disease.

Contraindications:

- Sensitivity to nitrates (Hypotension).
- Severe anemia.
- Hypotension.
- Head trauma.
- Cerebral hemorrhage.

Side effects:

- Headache, syncope, dizziness.
- Postural hypotension, transient flushing, and palpitation.
- Topical application may lead to dermatitis.

Drug interaction: Antihypertensive agents, Beta-adrenergic blocking agents, and calcium-channel blocking agent (they may lead to additive hypotension).

Dosage: there are several forms available:

- Sublingual: Cordil 5 mg PRN.
- PO: Isotard 20 – 40 mg twice a day.
- Topical: available as patches or ointment.
- Parental (IV infusion).

Nursing considerations:

1. Medications should be taken on an empty stomach.
2. Carry sublingual tablets in a glass bottle, tightly capped.
3. If anginal pain is not relieved in 5 minutes by first sublingual tablet, to take up to 2 more tablets at 5 minutes interval. If pain has not subsided 5 minutes after the 3rd tablet, client should be taken to the emergency room as this case could be an infarction (MI) and not angina.
4. Take sublingual tablets 5-15 minutes prior to any situation likely to cause anginal pain such as climbing stairs.
5. Take sublingual tablets while sitting to avoid postural hypotension.

Isosorbide dinitrate:

Present in the forms of capsules chewable, sublingual, tablets.

Trade names: Isoral, Cordil, Isotard.

Class: coronary vasodilator.

Dosage forms: caps 20-40 mg, tabs 20-40 mg.

Uses:

- Tabs for only prophylaxis of anginal pain.
- Chewable, sublingual to terminate acute attack and relieve acute pain.
- Esophageal spasm.

Side effects: Headache, hypotension.

Dosage:

- Sublingual: acute attack 2.5-5 mg Q 2-3 hrs.
- Oral caps/tabs: 5-20 mg Q 6 hrs.
- Extended release tabs: 20 –80 mg Q 8-12 hrs.

Note: Isosorbide mononitrate given for patients with liver impairments.

Peripheral vasodilators:

Isoxsuprine:

Trade name: Vasodin.

Class: peripheral vasodilator.

Action: Direct relaxation of vascular smooth muscles increasing peripheral blood flow, the drug also has a cardiac stimulation and uterine relaxation effect "Alpha receptor blocking and Beta receptor stimulation.

Uses:

1. Symptomatic treatment of cardiovascular insufficiency.
2. Buerger's disease.
3. Raynaud's disease.

Contraindications: postpartum period, arterial bleeding.

Side effects: tachycardia, hypotension, chest pain, nausea, vomiting, rash, dizziness.

Dosage: 10-20 mg 3-4 time daily.

Papaverine:

Class: peripheral vasodilator.

Action: direct spasmolytic effect on smooth muscle and vascular system, bronchial muscle, GI & urinary tract.

Uses:

1. Cerebral and peripheral ischemia.
2. Smooth muscle relaxant/
3. Paraenteral use for: Acute MI and ngina pectoris, Pulmonary embolism, and ureteral, biliary, & GI colic.

Side effects: Flushing of face, hypertension, tachycardia, constipation, dry mouth and throat, headache.

Antiarrhythmic agents:

Antiarrhythmic drugs are used to treat arrhythmias (disturbances in the normal heart rhythm). Unfortunately, many antiarrhythmic drugs can also worsen or cause the very arrhythmias they're supposed to treat. Therefore, the benefits of this therapy must always be weighed against its risks.

1- Amiodarone hydrochloride: (Procor)

Classification:- Antiarrhythmic

Action: it blocks Na^+ channel and increases the duration of the myocardial cell action potential as well as alpha & beta antiadrenergic effect.

Indications: Should be reserved for life threatening ventricular arrhythmias which don't respond to other therapy.

Side effects:

- CVS: Bradycardia, CHF, Fatigue, tremors.
- CNS: Visual disturbances, photophobia, dry eyes.
- Blood: Hemolytic or aplastic anemia.
- Liver: Hepatotoxicity.

Contraindications:

- Sensitivity.
- Sinus bradycardia
- AV block
- Thyroid dysfunction

Dose:

- Tabs 200 mg.
    - Maintenance dose: 200-400 mg daily.
    - IV infusion: 5 mg/kg over 20-120 minutes.

2- Lidocaine Hydrochloride:

Trade name: xylocaine – Esracain

Class: antiarrhythmic.

Action: it blocks Na^+ channel and increases the duration of the myocardial cell action potential (shortens the refractory period & suppresses the automatically of ectopic foci without affecting conduction of impulses through cardiac tissue).

Indications: acute ventricular arrhythmias as which follow MI or cardiac surgeries and digitalis toxicity.

Contraindications:

- Hypersensitivity
- Heart block.

Side effects: hypotension, bradycardia, dyspnea, dizziness.

Dosage: available in ampules of a concentration of 1% (100mg), 2% (200mg)

Dose:

- IV. Bolus 50-100 mg at rate of 25-50 mg\min
- Infusion: 20-50 mg\kg at a rate of 1-4 mg\m.

Nursing considerations:

1. Do not add lidocain to blood transfusion assembly.
2. Make certain that vials state "for cardiac arrhythmias".
3. Use 5% dextrose solution to prepare drug (stable for 24 hours).
4. Assess for history of hypersensitivity.
5. Use electronic infusion device to regulate the infusion of the drug.
6. Obtain B.P., pulse, respiratory rate to use as baseline data to evaluate response to treatment.
7. Drug should be given in a monitored environment.
8. Assess B.P. frequently during administration.
9. Assess for respiratory depression.
10. If adverse reactions occur, discontinue infusion & prepare for emergency management.

3- Phenytoin (Epanutin, Dilantin)

Class: anticonvulsant, antiarrhythmic.

Action:

- Effects are very similar to lidocaine
- Increase conduction through damaged purkinje fibers.
- Acts in the motor cortex of the brain to reduce the spread of electrical discharges from the rapidly firing epileptic foci in this area. In addition, activity of centers in the brain stem responsible for the tonic phase of grand mal seizures.

Uses: Chronic epilepsy, Premature ventricular contractions, Tachycardia.

Contraindications: Hypersensitivity.

Side effects: Drowsiness, ataxia, dizziness, measles-like rash, gingival hyperplasia, Hirsutism (excessive hair growth) , hypoglycemia.

Note: rapid I.V. administration may result in hypotension & arrhythmia.

Dose for arrhythmias:

- Tabs 200-400 mg daily.
- I.V. 100 mg q 5 minutes up to a maximum of 1g.

Nursing Considerations:

1. I.V. phenytoin may forms precipitate, so flush tubing by saline (not dextrose) before & after administration.
2. Assess for hypersensitivity.
3. If a pregnant woman takes this drug, tell her not to breastfeed her baby.
4. Obtain liver & kidney function studies.
5. Monitor serum drug levels on a routine basis.
6. During I.V. therapy, monitor B.P. for signs of hypotension.
7. Take with food to minimize GI upset.
8. If the patient is diabetic, monitor for signs of hypoglycemia.
9. Oral hygiene to minimize bleeding from the gum.

4- Procainamide:

Trade name: procan

Class: antiarrhythmic

Action: produce a direct cardiac effect to prolong the refractory period of the heart & depress the conduction of the cardiac impulse.

Uses: Ventricular tachycardia, Atrial fibrillation, Digitalis intoxication.

Contraindication: Hypersensitivity, complete AV heart block.

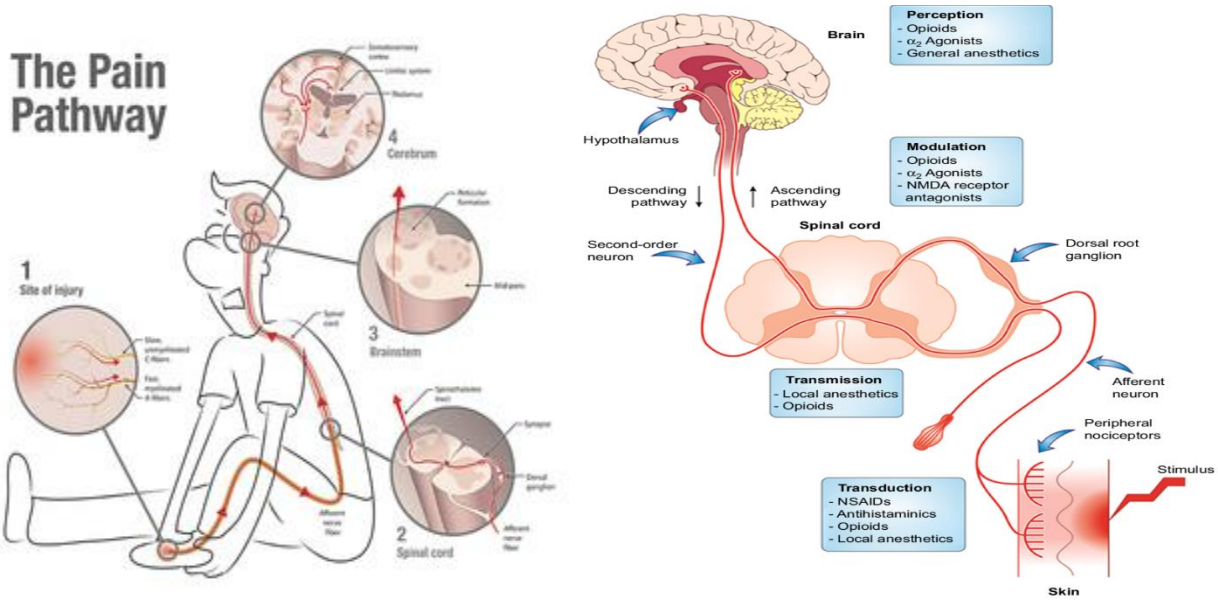
Side effects: Hypotension, abdominal pain, bitter taste.

5- Propranolol

6- Verapamil

Nonopioid analgesics, antipyretics, and NSAIDS

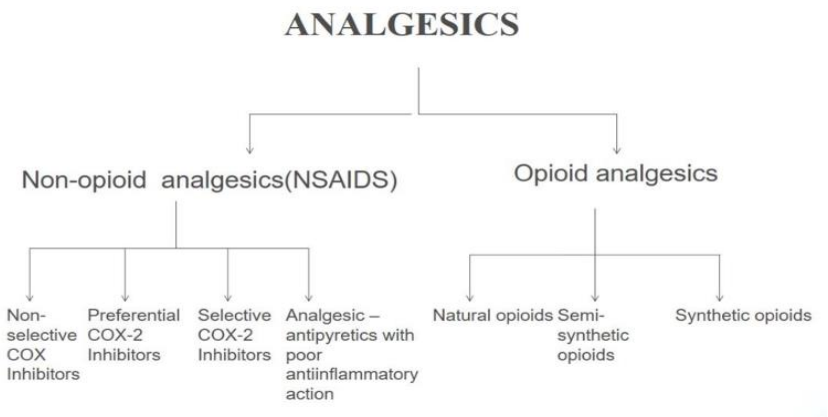
The pain pathway:

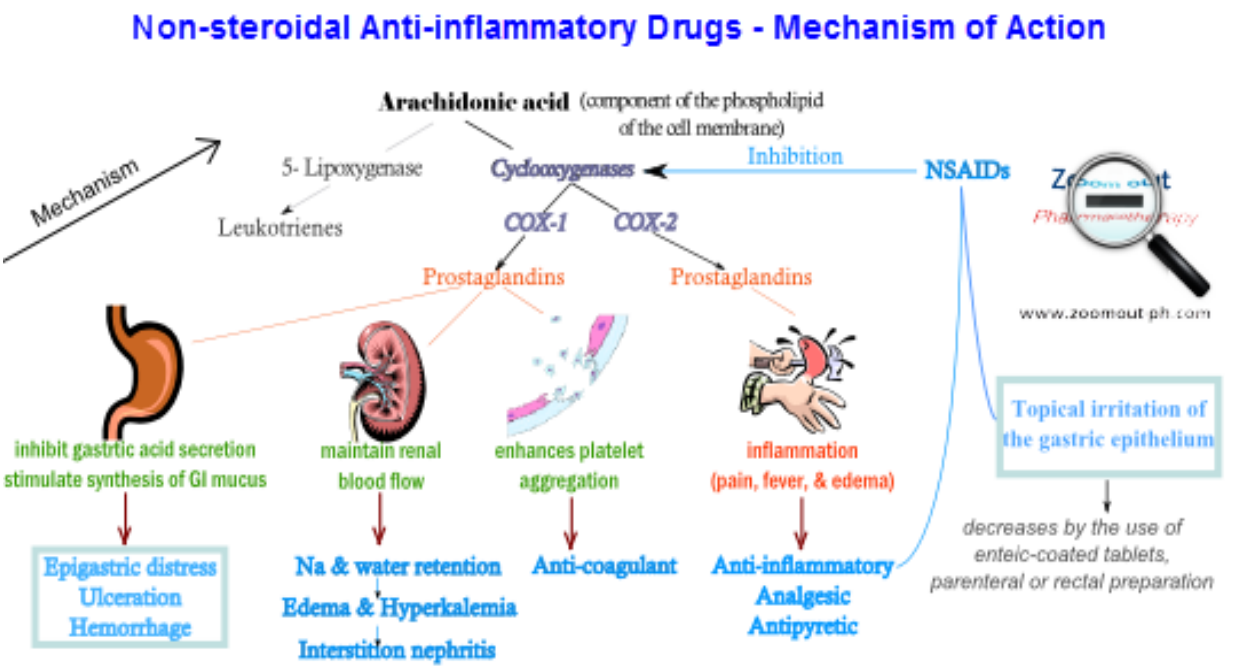


Non-opioid analgesics, antipyretics, and NSAIDs are a broad group of pain medications. In addition to pain control, they produce antipyretic (fever control) and anti-inflammatory effects. They can be used alone or as adjuvant medications. These drugs have a ceiling effect, and no physical dependence is associated with them. They are not steroids and exhibited anti-inflammatory actions, they have analgesic, antipyretic, and platelet-inhibitory actions.

The drug classes included in this group are:

- Salicylates (especially aspirin), which are widely used
- Acetaminophen, a para-aminophenol derivative
- NSAIDs (nonselective and selective)
- Phenazopyridine hydrochloride, a urinary tract analgesic.





Salicylates:

Acetylsalicylic Acid:

Trade name: Aspirin

Classification:

Non-narcotic analgesic, antipyretic, anti-inflammatory, anti-rhumatic, antiplatelet

Action:

- The antipyretic effect is due to an action on the hypothalamus that results in heat loss by vasodilation of peripheral blood vessels & promoting sweating.
- The anti-inflammatory effects probably by decreasing prostaglandin synthesis & other mediators of the pain response.
- The analgesic action is not fully known but may be due to improvement of the inflammatory condition.
- One salicylate, aspirin, inhibits platelet aggregation (the clumping of platelets to form a clot) by interfering with the production of thromboxane A2, a substance necessary for platelet aggregation. Unlike aspirin, NSAIDs' effects on platelet aggregation are temporary. As a result, aspirin can be used to enhance blood flow during myocardial infarction (MI) and to prevent an event such as recurrent MI.

Indications

- Mild to moderate pain, Fever
- Inflammatory conditions—rheumatic fever, rheumatoid arthritis, osteoarthritis
- Reduction of risk of recurrent TIAs or stroke in males with history of TIA due to fibrin platelet emboli
- Reduction of risk of death or nonfatal MI in patients with history of infarction or unstable angina pectoris
- MI prophylaxis

Side effects

The most common adverse reactions to salicylates include gastric distress, nausea, vomiting, and bleeding tendencies.

Other adverse reactions include:

hearing loss (when taken for prolonged periods), diarrhea, thirst, tinnitus, confusion, dizziness, impaired vision, hyperventilation (rapid breathing), Reye's syndrome (when given to children with chickenpox or flulike symptoms).

Contraindications:

Allergy to salicylates, NSAID and tartrazine, Hemophilia, bleeding ulcers, Hemorrhagic states, Blood coagulation defects, Hypoprothrombinemia, Vitamin K deficiency (increased risk of bleeding), Pregnancy, lactation, during period of menstruation, Before and after surgery (tooth extraction), asthma.

Dose:

- In minor conditions: 325-600 mg Q 4 hours.
- May reach up to 6 grams /day in divided doses in arthritis and rheumatic conditions.

Salicylate toxicity

- **Acute aspirin poisoning:** Respiratory alkalosis, hyperpnea, tachpnea, hemorrhage, confusion, pulmonary edema, convulsion, tetany, metabolic acidosis.

Drug interactions:

- Risk for bleeding increase if taken with other anticoagulants.
- Risk of GI bleeding increase if taken with steroids, alcohol, or other NSAIDs.
- Increased risk for salicylate toxicity if taken with frusimide (lasix)
- Hypotension may occur if taken with nitroglycerns.

Nursing considerations:

1. Take drug with or after food or with milk to decrease GI irritation.
2. Assess for history of asthma and history of hypersensitivity.
3. Do not use with other anticoagulants.
4. Note any history of peptic ulcer.
5. Report signs of side effect e.g. gastric irritation if occurs.
6. Aspirin is not given 1 week before & after surgery to prevent bleeding.
7. If patient is diabetic, discuss the possibility of hypoglycemia occurring to patients, and should monitor their blood glucose level frequently.
8. Teaches patient about the toxic symptoms (ringing in the ears dizziness, mental confusion-etc) and ask him/her to report it to physician.
9. Institute emergency procedures if overdose occurs: Gastric lavage, induction of emesis, activated charcoal, supportive therapy.

Salicylate warning

Before administering salicylates to a patient, be aware of its risks to special populations:

- Children and teenagers: Avoid the use of aspirin and salicylates to treat flulike symptoms or chickenpox because Reye's syndrome may develop.
- Surgical patients: If possible, discontinue aspirin 1 week before surgery because of the risk of postoperative bleeding.
- Asthmatics: Be aware that these patients are more likely to develop bronchospasm, urticaria, Angioedema, or shock when salicylates are administered.

Acetaminophen: "paracetamol" (Acamol, Panadol)

Acetaminophen is an OTC drug that produces analgesic and antipyretic effects. It appears in many products designed to relieve pain and symptoms associated with colds and influenza.

Class. : non-narcotic analgesic, para-aminophenol type.

Action:

- Acetaminophen decrease fever by an effect on hypothalamus leading to sweating & vasodilation. It also inhibits the effect of pyrogens on the heat-regulating center on the hypothalamus. It may cause analgesia by inhibiting CNS prostaglandin syntheses (it has no anti-inflammatory effect). It doesn't manifest any anticoagulant effect or any ulceration of GIT.

Uses:

- Pain due to Headache, dysmenorrhea, arthralgia, myalgia, muscluoskletal pain, immunization, teething, tonsillectomy.
- To reduce fever due to bacterial & viral infection .
- As a substitute for aspirin when contraindicated .

Contraindications: renal insufficiency, anemia.

Side effects:

- Most patients tolerate acetaminophen well. Unlike the salicylates, acetaminophen rarely causes gastric irritation or bleeding tendencies. However, it may cause liver toxicity, and the total daily dose should be monitored.

Other adverse reactions include: Skin rash**,** Hypoglycemia (with overdose)**,** Neutropenia.

Symptoms of over dosage:

- Hepatic toxicity (general malaise, delirium, depression seizures, coma & death, nausea, vomiting, fever, and vascular collapse).

Treatment of overdose:

- Induction of emesis.
- Gastric lavage.
- Activated charcoal.
- Oral N-acetyleystine (mucomyst) is said to reduce or prevent hepatic damage by inactivating acetaminophen metabolites which cause liver effects.

Dose: Tab. 500 mg Q 4 hrs or up to 1g Q 6 hrs.

Nursing considerations:

1- Suppositories should be stored below 27c.
2- Liver function studies for long term therapy.
3- Note signs of met-hemoglobinemia: bluish discoloration of gum & fingernails.
4- Have mucomyst available for signs of toxicity.
5- Teach patient signs of toxicity to be reported immediately.

Nonselective nonsteroidal anti-inflammatory drugs (NSAIDs)

As their name suggests, nonsteroidal anti-inflammatory drugs, or NSAIDs, are typically used to combat inflammation. Their anti-inflammatory action equals that of aspirin. They also have analgesic and antipyretic effects.

Nonselective NSAIDs inhibit prostaglandin synthesis by blocking two enzymes known as cyclooxygenase-1 (COX-1) and cyclooxygenase- 2 (COX-2). These drugs (called *COX-1* and *COX-2 inhibitors*) include indomethacin, ibuprofen, diclofenac, etodolac, fenoprofen, flurbiprofen, ketoprofen, ketorolac, mefenamic acid, meloxicam, nabumetone, naproxen, oxaprozin, and sulindac.

Selective NSAIDs selectively block COX-2 enzymes, thereby inhibiting prostaglandin synthesis. This selective inhibition of COX-2 produces the analgesic and anti-inflammatory effects without causing the adverse GI effects associated with COX-1 inhibition by nonselective NSAIDs.

As in aspirin, the therapeutic actions of these substances are believed to result from the inhibition of the enzyme cyclo-oxygenase which results in decreased prostaglandin synthesis so it is effective in:

- Reducing joint swelling, pain & morning stiffness.
- Increasing the mobility in arthritic patients.
- Antipyretic action due to decreased production of prostaglandin from the hypothalamus.
- Having irritating effect on the GIT.

Uses:

- Rheumatoid arthritis
- Osteorthritis.
- Gout
- Other musclоskletal diseases.
- Dental pain
- Strains & sprains.

Contraindications:

- Children less than14 years of age.
- Lactation.
- Hypersensitivity (asthma, rashes, rhinitis).
- Uses with caution in patients with a history of GI disease & reduced renal functions.

Side effects:

- Peptic, duodenal ulcer, GI bleeding nausea, vomiting, dyspepsia.
- Dizziness, drowsiness. Hypo + hyperglycemia.
- Bronchospasm, rhinitis. Blurring of vision.
- Tinnitus, loss of hearing.
- Bone marrow depression C.H.F.

Nursing considerations:

- Note any history of allergic responses to aspirin or nonsteroidal anti-inflammatory agents.
- Note the age of the client.
- Determine if patient is taking oral hypoglycemic or insulin and document it.
- Take these agents with milk or meal or antacids as prescribed.
- Encourage patient to take drug regularly.
- Report signs of GI irritation.
- Instruct client to report signs of bleeding, blurring of vision, tinnitus rashes – etc.
- If the client has Diabetes Mellitus, explain the possible in increasing hypoglycemic effect of the drugs, to test urine & blood for glucose. To adjust dose of these agents.

1. Diclofenac Sodium:

Trade name: Voltaren , Rufenal

Class. : Non steroidal anti-inflammatory analgesic.

Dose: Suppositories, tabs or injection of 150-200 mg daily in 2-4 divided doses.

Nursing considerations:

1. Give on full stomach to avoid GIT irritation.
2. When given IM, Give it deep into a large muscle because drug is very irritant.

2. Indomethacin: Trade name : Indocid.

Class. : Anti-inflammatory, analgesic, antipyretic.

Dose: suppositories & caps. 25mg – 50 mg bid-tid.

3. Naproxen:

Trade name: Naprex.

Class.: Non- steroidal anti-inflammatory analgesic.

Dose: 500 mg bid.

4. Ibuprofen: (Trufen, Isofen, Brufen, artofen**).**

Class.: nonsteroidal anti-inflammatory analgesie.

Dose: tablet, suspension, gel**.** 200, 400 mg bid.

5. Generic name : Celecoxib (Celex , Celebrex)

Class: NSAID, Analgesic (non-opioid)**,** Specific COX-2 enzyme blocker

Therapeutic actions :

- Analgesic and anti-inflammatory activities related to inhibition of the COX-2 enzyme, which is activated in inflammation to cause the signs and symptoms associated with inflammation; does not affect the COX-1 enzyme, which protects the lining of the GI tract and has blood clotting and renal functions.

Indications

- Acute and long-term treatment of signs and symptoms of rheumatoid arthritis and osteoarthritis
- Reduction of the number of colorectal polyps in familial adenomatous polyposis (FAP)
- Management of acute pain
- Treatment of primary dysmenorrhea
- Relief of signs and symptoms of anklylosing spondylitis

Contraindications and cautions

- Contraindicated with allergies to sulfonamides, celecoxib, NSAIDs, or aspirin; significant renal impairment; pregnancy (third trimester); lactation.
- Use cautiously with impaired hearing, hepatic, and CV conditions.

Dosage forms: Capsules 100, 200 mg

Dose: Initially, 100 mg PO bid; may increase to 200 mg/day PO bid as needed.

Nursing considerations

- Administer drug with food or after meals if GI upset occurs.
- Establish safety measures if CNS, visual disturbances occur.
- Arrange for periodic ophthalmologic examination during long-term therapy.
- If overdose occurs, institute emergency procedures—gastric lavage, induction of emesis, supportive therapy.
- Provide further comfort measures to reduce pain (eg, positioning, environmental control), and to reduce inflammation (eg, warmth, positioning, rest).

Narcotic Analgesics & Antagonists

- The word opioid refers to any derivative of the opium plant or any synthetic drug that imitates natural narcotics. Opioid agonists (also called narcotic agonists) include opium derivatives and synthetic drugs with similar properties.
- They're used to relieve or decrease pain without causing the person to lose consciousness. Some opioid agonists also have antitussive and antidiarrheal effects.
- It includes opium such as morphine, codeine & opium derivatives such as Meperidine.
- These substances have similar pharmacological properties.
- Following prolonged use, patients develop tolerance to most opioid effects. Constipation and miosis are notable exceptions,
- The relative activity of all narcotic analgesics in measured against morphine.

Dependence & Tolerance:

- Drugs of this group may lead to addiction.
- Psychological & physical dependence & tolerance develop even when using clinical doses.
- Tolerance usually develops because the patient requires shorter periods between doses or larger doses for relief of pain.

Opioid agonists include:

- Codeine
- Fentanyl citrate
- Hydrocodone
- Hydromorphone hydrochloride
- Levorphanol tartrate
- Meperidine hydrochloride
- Methadone hydrochloride
- Morphine sulfate (including morphine sulfate sustained-release tablets and oral solution)
- Oxycodone
- Oxymorphone
- Propoxyphene
- Remifentanil
- Tramadol.

Effects of narcotic analgesics:

Central nervous system effects:

- **Analgesia:** Related to their ability to change both pain perception and effectively raise the threshold for pain.
- **Euphoria:** A pleasant floating sensation and freedom from anxiety and distress.
- **Sedation:** Drowsiness and clouding of mentation are frequent concomitants action.
- **Respiratory depression:** By inhibiting brain, stem respiratory mechanisms.
- **Cough suppression:** Suppression of the cough reflex is a well-recognized action
- **Miosis:** Constriction of the pupils is seen with virtually all of the opioid agonists.
- **Nausea and vomiting:** it can activate the brain stem chemoreceptor trigger zone to produce nausea and vomiting.

Peripheral Effects:

- **Cardiovascular system:** cerebral vascular dilation and concomitant decrease in cerebral vascular resistance, increase in cerebral blood flow, and increase in intracranial pressure.
- **Gastrointestinal tract:** Constipation, decrease motility, decreased gastric. In the large intestine delays passage of the fecal mass and allows increased absorption of water, which leads to constipation (management of diarrhea).
- **Billiary tract:** The opioids constrict biliary smooth muscle, which may result in biliary colic.
- **Genitourinary tract:** Renal function is depressed by opioids. It is believed that in humans this is chiefly due to decreased renal plasma flow.
- **Uterus:** The opioid analgesics may prolong labor.
- **Miscellaneous:** Flushing and warming of the skin accompanied sometimes by sweating and itching; central effects and histamine release may be responsible for these reactions.

Acute toxicity: Characterized by respiratory depression, deep sleep, stupor, coma, pinpoint pupil, cyanosis, hypotension, decreased urinary output, decreased temperature, clammy skin, and finally Death (due to Respiratory failure).

Treatment of acute overdose:

1. Induce vomiting or gastric lavage.
2. Artificial respiration.
3. Give narcotic antagonist (Narcan).

Signs & symptoms:

- Constricted pupil, constipation, skin infections, needle scare abscesses & itching on the anterior surface of the body.
- Withdrawal signs appear when drugs are withheld for 4-12 hrs. & characterized by intense craving for the drug, insomnia, yawning, sneezing, vomiting, diarrhea, tremors, sweating, mental depression, muscular aches, pain, chills & anxiety. (They are rarely life-threatening).

Action of narcotic analgesics:

- Narcotic analgesics attach to specific receptor in the CNS resulting in analgesia- action
- Action exactly is unknown but may be by decreasing cell membrane permeability to sodium transmission of pain impulses.

Uses:

- Sever pain
- Hepatic & renal colic.
- Pre-anesthetic medication
- Post-surgical pain.
- Diarrhea & dysentery
- Pain from MI, carcinoma.
- Postpartum pain & burns.
- Antitussive.

Contraindications:

- Asthmatic conditions
- Emphysema
- Sever obesity
- Convulsions
- Diabetic acidosis
- Myxedema
- Addisson's disease
- Hepatic cirrhosis
- Children less than age of 6 months.

Side effects: Respiratory depression, apnea, dizziness, euphoria headache, mental clouding, insomnia, nausea, vomiting, constipation, dry mouth, skin rashes, laryngospasm, urinary retention, and decreased libido.

Nursing considerations:

- Use supportive nursing measures as relaxation techniques to relieve pain before using narcotics.
- Explore the source of pain, use non-narcotic analgesia if possible.
- Administer the medication when needed, prolonging the medication administration will decrease the effect of the medication.
- Monitor vital signs & mental status.
- Monitor respiratory rate (drug may lead to respiratory depression).
- Monitor blood pressure (hypotension may occur)
- Monitor pulse rare (if 60\m withhold the drug).
- Watch for constricted pupils. Document it and notify the physician.
- Monitor bowel function, since drug may cause constipation.
- Encourage client to empty bladder every 3-4 hrs (since drug may cause urinary retention).
- If client is bed ridden, use side rails.
- Inform the client\family that the drug may become habit forming and leading to addiction.
- Document any history of asthma or other contraindications.
- Have emergency equipment and narcotic antagonist available.

1. Codeine sulfate:

Trade name: Codivis, Rekod , (Tussosedan)

Class. : Narcotic analgesic, morphine type.

Action:

- Resembles morphine pharmacologically but produce less effect on respiratory system, less nausea & less vomiting.
- In high doses (more than 60 mg), it will irritate the cough center, but in lower doses, it is a potent antitussive and is an ingredient in many cough syrups.

Uses: Relief of mild to moderate pain, Antitussive.

Dose: Analgesic: 15-60 mg 4-6 hrs, Antitussive: 10-20 mg 4-6 hrs.

2. Meperidine Hydrochloride "Pethedine Hydrochloride":

Trade name: Demerol

Class. : Synthetic opioid agonist analgesic.

Action: Similar to opiates.

- It has no antitussive effect.
- The duration of action is less than that of opium.

Uses:

- Sever pain.
- Renal & hepatic colic.
- Obstetric pre-anasthetic medication.
- In minor surgeries.
- Spasm of GI tract, uterus.
- Prior some diagnostic procedures e.g. cystoscopy.
- Post operative pain.

Additional Contraindications: Hypersensitivity, convulsive states, children less than 6 months, head injuries, diabetic acidosis.

Additional Side effects: Transient hallucinations, hypotension.

Dose:

- Drug can is available in the form of tablets, syrup, I.M, S.C. Dose is 50-100 mg Q 4-6 hr.
- It can be given as I.V. continuous infusion on a concentration of 1 mg\ml.
- It also can be given IV slowly, and should be diluted in a concentration of 10mg/ml.

3. Methadone Hydrochloride:

Class.: Narcotic analgesic, morphine type.

Action:

- Produce only mild euphoria and used as a heroin withdrawal substitute.
- It produces physical dependence but the abstinence syndrome develops more slowly upon termination of the therapy.
- Withdrawal symptoms are less intense but more prolonged than those associated with morphine.
- It is not effective for preoperative or obstetric anesthesia.

Uses: Sever pain, Drug withdrawal.

Additional Contraindications:

- Pregnancy since it depresses respiration of neonate.
- I.V. use.
- Liver disease.

Additional side effects:

- Constipation and pulmonary edema.

Dose: Can be given oral, I.M., S.C. at a dose of 2.5 – 10 mg Q 3-4 hrs.

4. morphine Sulfate:

Class.: Narcotic analgesic, morphine type.

Action: See narcotic analgesic.

Uses:

- Intrathecally, orally or I.V. infusion for acute or chronic pain.
- Preoperative medication.
- To facilitate induction of anesthesia or to decrease the dose of anesthesia.

N.B.: It is given in lower doses for continuous pain & in higher doses in sharp intermittent & all kinds of pain.

Additional contraindications:

- It is not given intrathecal, if infection is present at injection site.
- In patients on anticoagulant therapy.
- Bleeding disorders.
- If patients have received parenteral corticosteroids within the past 2 weeks.

Dose:

- Oral: 10-30 mg Q 4 hr.
- I.M.: 5-20 mg\70 kg Q 4 hr as needed.
- I.V.: bolus of 2.5-15 mg for a person of average weight of 70 kg over 4-5 minutes (slowly).
- Continuous infusion: 0.1-1 mg\ ml in 5% dextrose in water by a controlled infusion pump.

5. Tramadol hydrochloride

Trade name: Ultram, Tramal

Class. : Narcotic analgesic, synthetic.

Action: Unknown. A centrally acting synthetic analgesic compound not chemically related to opiates. Thought to bind to opioid receptors and inhibit reuptake of norepinephrine and serotonin.

Indications & dosages

- Moderate to moderately severe pain
- Adults: Initially, 25 mg P.O. Adjust by 25 mg q 3 days to 100 mg/day (on divided doses).

Adverse reactions

- **CNS:** dizziness, vertigo, headache, CNS stimulation, anxiety, confusion, euphoria, nervousness, sleep disorder, seizures, malaise, visual disturbances.
- **CV:** vasodilation.
- **GI:** nausea, vomiting, constipation, dyspepsia, dry mouth, diarrhea, abdominal pain, anorexia, flatulence.
- **GU:** urine retention, urinary frequency, menopausal symptoms, proteinuria.
- **Respiratory:** respiratory depression.
- **Skin:** pruritus, diaphoresis, rash.

Contraindications & cautions

- Hypersensitive to drug or other opioids,
- In breast-feeding women
- Acute intoxication from alcohol, hypnotics, centrally acting analgesics, opioids, or psychotropic drugs.
- Use cautiously in patients at risk for seizures or respiratory depression; in patients with increased intracranial pressure or head injury, acute abdominal conditions, or renal or hepatic impairment; or in patients with physical dependence on opioids.

Nursing considerations

- Reassess patient's level of pain at least 30 minutes after administration.
- Monitor CV and respiratory status. Withhold dose and notify doctor if respirations decrease or rate is below 12 breaths/minute.
- Monitor bowel and bladder function. Anticipate need for laxative.

- For better analgesic effect, give drug before onset of intense pain.
- Monitor patients at risk for seizures. Drug may reduce seizure threshold. In the case of an overdose, naloxone may also increase risk of seizures.
- Monitor patient for drug dependence. Drug can produce dependence similar to that of codeine and thus has potential for abuse.
- Withdrawal symptoms may occur if drug is stopped abruptly. Reduce dosage gradually.
- Caution ambulatory patient to be careful when rising and walking. Warn outpatient to avoid driving and other potentially hazardous activities that require mental alertness until drug's CNS effects are known.

Narcotic Antagonists:

The narcotic antagonists are able to prevent or reverse many of the pharmacological actions of morphine-type analgesics & meperedine as respiratory depression induced by these drugs within minutes.

Naloxone Hydrochloride:

Trade name: Narcan.

Class.: Narcotic antagonist

Action:

- Block the action of narcotic analgesic by displacing previously given narcotics from their receptor sites or preventing them from attaching to opiate receptors.
- The duration of action of naloxone is shorter than that of the narcotic analgesic so the respiratory depression may return when the narcotic antagonist has washed off the body.

Uses:

- Respiratory depression induced by narcotics.
- Drug of choice when the depressant drug is unknown.
- Diagnosis of acute opiate overdose.

Note: Naloxone is not effective when respiratory depression is induced by hypnotic, sedative or other narcotic drugs.

Contraindications: Sensitivity to drug, Narcotic addicts since it will cause severe withdrawal symptoms, Neonates.

Side effects:

- Nausea, vomiting, sweating, hypertension, tremors.
- If used postoperatively: tachycardia, pulmonary edema, hypo or hypertension.

Dose: 0.4-2 mg I.V , S.C. or I.M.

Nursing considerations:

1. Determine the etiology of respiratory depression.
2. Assess & obtain baseline vital signs.
3. Monitor respiration closely after the duration of action.
4. Have emergency drugs & equipment available.
5. If the patient is comatosed, turn him to his side to avoid aspiration.
6. Maintain safe environment (side rails & soft support).

Adreno-corticosteroids and analogs

Corticosteroids suppress immune responses and reduce inflammation. They're available as natural or synthetic steroids.

Nature's bounty

Natural corticosteroids are hormones produced by the adrenal cortex; most corticosteroid drugs are synthetic forms of these hormones.

Natural and synthetic corticosteroids are classified according to their biological activities:

- *Glucocorticoids,* such as cortisone and dexamethasone, affect carbohydrate, fat, and protein metabolism.
- *Mineralocorticoids,* such as aldosterone and fludrocortisone acetate, regulate electrolyte and water balance.

Action:

- They are a group of natural hormones produced by the adrenal cortex.
- They are used for a variety of therapeutic purposes.
- Many slightly modified synthetic variants are available today.
- Some patients respond better to one substance than to another.
- These hormones influence many metabolic pathways & all organ systems & are essential for survival.
- The release of corticosteroids is controlled by hormones such as corticotropin- releasing factor produced by the hypothalamus & ACTH produced by the anterior pituitary.

Most glucocorticoids are synthetic analogues of hormones secreted by the adrenal cortex. They exert anti-inflammatory, metabolic, and immunosuppressant effects.

Drugs in this class include:

- Beclomethasone
- Betamethasone
- Cortisone
- Dexamethasone
- Hydrocortisone
- Methylprednisolone
- Prednisolone, Prednisone
- Triamcinolone.

Corticosteroids have the following effect:

1- **Promote normal intermediary metabolism:**

   Increase gluconeogenesis (Deposition of glucose as glycogen in the liver & conversion of glycogen to glucose when needed.(Gluconeogenesis).

2- **Protein metabolism:** The stimulation of protein loss from many organs.

3- **Fat metabolism:** The deposition of fatty tissue in facial , abdominal & shoulder regions.

4- **Water & electrolyte balance:** Alteration of glomerular filtration rate, increase sodium & fluid retention, also affect the excretion of potassium, calcium & phosphorus.

5- Have anti-inflammatory effect: they decrease prostaglandin synthesis.

6- The immunosuppressant effect: they decrease number of T-lymphocyte, monocytes, and eosinophils.

7- They help the organism to cope with stressful situations e.g. trauma & sever illness.

According to their chemical structure, they classified into two classes.

1. Glucocorticoids e.g. cortisone & hydrocortisone:- regulate the metabolism of glucose, protein & fat .
2. Mineralocorticoids e.g. Aldosteron & desoxycorticosterone.:- increase reabsorption of Na+ & excretion of potassium & hydrogen.

Uses:

Therapy with glucocorticolds is not curative & many situations should be considered as adjunctive rather than primary therapy:

1. Replacement therapy: adrenal insufficiency (Addison's disease) .
2. Rheumatic disorders: rheumatoid arthritis & osteoarthritis.
3. Collagen diseases: systemic lapus erythematosus, rheumatic cardiac.
4. Allergic diseases: drug hypersensitivity, urticarial transfusion reaction.
5. Respiratory diseases: bronchial asthma, rhinitis.
6. Ocular diseases: allergic & inflammatory conjunctivitis, keratitis.
7. Dermatological diseases: psoriasis, contact dermatitis, urticaria.
8. Diseases of the GIT: ulcerative colitis.
9. Nervous system: Myasthenia gravis.
10. Malignancies: leukemia, lymphoma.

11. Nephrotic syndrome.
12. Hematologic diseases: hemolytic anemia, thrombocytopenic purpura.
13. Miscellaneous: septic shock, liver cirrhosis, stimulation of surfactant production.

Contraindications:

1. If infection is suspected (Mask signs & symptoms).
2. Peptic ulcer.
3. Acute glomerulonephritis.
4. Cushing's syndrome.
5. Congestive heart failure.
6. Hypertension.
7. Hyperlipidemia.

Side effects:

Glucocorticoids affect almost all body systems. Their widespread adverse effects are more common in long-term or high-dose use and include:

- Insomnia
- Increased sodium and water retention
- Increased potassium excretion
- Suppressed immune and inflammatory responses
- Osteoporosis
- Intestinal perforation
- Peptic ulcers

Prolonged therapy may cause Cushing-like syndrome & atrophy of the adrenal cortex & subsequent adrenocortical insufficiency.

N.B: steroid withdrawal syndrome may lead to: anorexia, nausea, vomiting, weight loss, headache, myalgia & hypotension. It is most important that therapy not be discontinued abruptly.

Dose: Highly individualized according to the condition & response of the patient.

Nursing Considerations:

1- Administer oral forms with food to minimize ulcerogenic effect.
2- For chronic use, give the smallest dose possible.
3- Corticosteroids should be discontinued gradually if used chronically.

4- Document baseline weight , B.P. , Pulse & temperature.
5- Frequently take BP, monitor body weight (signs of Na+ & H2O retention).
6- Periodic serum electrolytes, blood sugar monitoring.
7- Report signs & symptoms of side effects (cushing-like syndrome).
8- Discuss with female client potentials of menstrual difficulties.
9- Instruct the client to take diet high in protein & potassium.
10- Instruct the client to avoid falls & accidents (osteoporosis causes pathological fracture).
11- Remind the client to carry a card identifying the drug being used.
12- Stress the need for regular medical supervision.
13- Advice the client to delay any vaccination while taking these medications (weakened immunity).
14- Explain the need to maintain general hygiene & cleanliness to prevent infection.

1. Betamethasone:

Trade name: betacort

Class: Adrenocorticosteroid, synthetic, glucocorticoid type.

Additional Uses: prevention of respiratory distress syndrome in premature infants

2. Dexamethasone:

Trade name : dexacort.

Class: adrenocorticosteroid –synthetic, glucocorticoid type.

Dosage: Tablets 0.5 mg and ampule 4 mg, 20 mg.

3. Hydrocortisone:

Trade name: solu –cortef , hydrocortone.

Class: adrenocorticosteroid , naturally occurring,

glucocorticoied.

Forms: Vials 100 mg, 500 mg.

4. Prednisone :

Trade name : deltasone.

Class: adrenocorticosteroid , synthetic.

Forms: Tablets 5mg, 20mg.

5. **Methylprednisone**

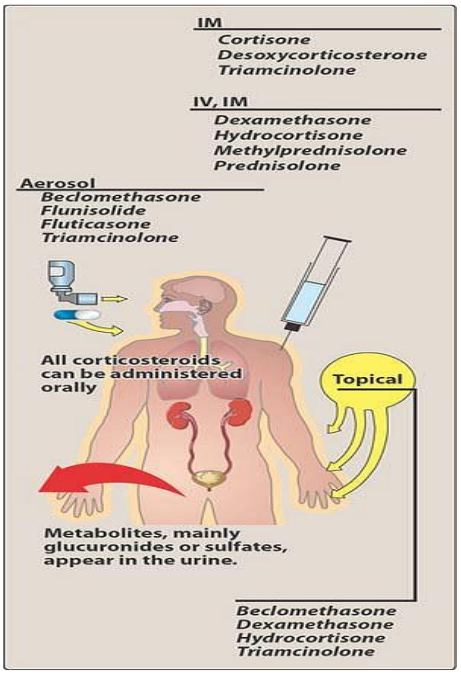
   Trade name : Depo medrol.

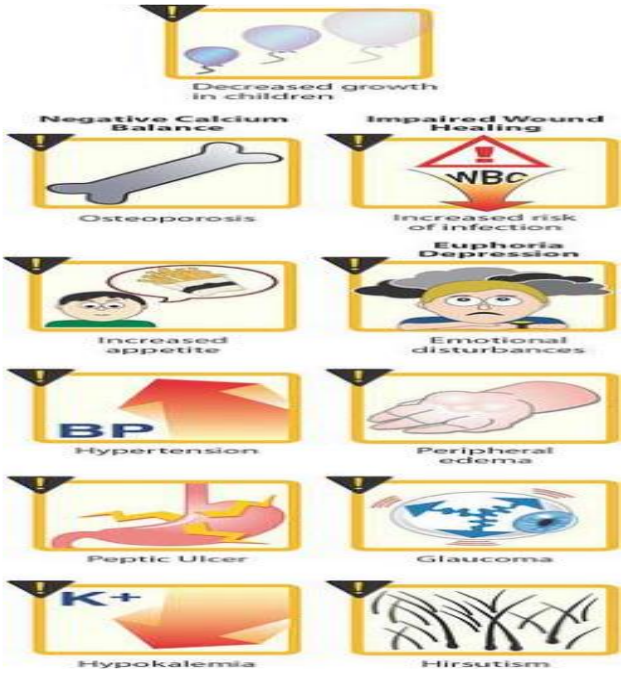
   Class: adrenocorticosteroid , synthetic

   Forms: 1,2,5 ml × 40ml /ml

6. **Betamethasone ointment:**

   Trade name : Betacortin





Antihistamine “H 1 Blockers”

- Antihistamines primarily act to block histamine effects that occur in an immediate (type I) hypersensitivity reaction, commonly called an allergic reaction. They’re available alone or in combination products and may be obtained by prescription or over the counter.
- The term antihistamine refers to drugs that act as histamine-1 (H1)-receptor antagonists; that is, they compete with histamine for H1-receptor sites throughout the body. However, they don’t displace histamines already bound to receptors.
- Appropriate stimuli including: tissue injury, antigen- antibody (allergic) reactions, and extreme cold trigger the release of histamine from its storage sites into the vascular system where it induces the following responses:
    1. Dilation & increased permeability of the small arterioles & capillaries results in increasing permeability to fluid leading to hypotension & edema nasal congestion & laryngeal edema “associated with allergies”.
    2. Contraction of some smooth muscles such as those of bronchioles leading to bronchoconstriction “the role of histamine plays in bronchial asthma”. & uterine contraction.
    3. Stimulation of acid secretion in the stomach & salivary, bronchial & intestinal secretions.
    4. Dilation of cerebral vessels (Headache).
    5. Pain & itching because it stimulates the sensory nerve endings.

Action:

- The effect of histamines may be reversed either by drugs that block histamine receptors or by drugs that have effects opposite to those of histamine e.g. epinephrine.
- Antihistamines used for the treatment of allergic conditions are referred to as H1-receptor blockers while those used for treatment of GI disorders as peptic ulcer are referred as H2-receptor blockers.
- They don’t prevent the release of histamine
- They prevent or reduce increased permeability that results in decreasing edema & itching, & bronchospasm.
- H1-blockers manifest varying degrees of CNS depression, anti- cholinergic & antiemetic effect.

Uses:

- Treatment of seasonal allergic rhinitis, allergic conjunctivitis.
- Treatment of urticarial transfusion reactions.
- Treatment of topic dermatitis.
- Treatment of insect bites.
- Sneezing & rhinorrhea due to common cold.
- Prophylaxis & treatment of motion sickness "nausea & vomiting".
- Night – time sleep aid.

Contraindications:

- Hypersensitivity.
- Pregnancy.
- Glaucoma
- Prostatic hypertrophy
- CNS depression (phenothiazine type).
- Bone marrow depression
- Comatose patients.

Side effects:

- Sedation, deep sleep, Dizziness, Headache, muscle weakness, disturbed coordination, epigastric distress, dry mouth, nausea, vomiting, urinary frequency, anemia (pancytopnea).
- Paradoxical excitation (especially in children & elderly) including restlessness, irritability, insomnia, hysteria, tremors euphoria, nervousness, hallucinations, disorientation & convulsion.
- Usually caused by overdose (acute toxicity).

Treatment of overdose:

- Symptomatic & supportive.
- Vomiting is induced with syrup of ipecac.
- Gastric lavage.
- Vasopressors (to treat hypotension) –e.g. Dopamine , adrenaline.
- Phenytoin for treatment of convulsion.

N.B.: Do not use CNS depressants including diazepam.

Nursing Considerations:

- Inject I.M. preparations deep into muscles.
- Oral preparations may cause gastric irritation, so give drug with meals.
- Note if the client has any medical history of ulcer, glaucoma & if the client is pregnant.
- Obtain a baseline B.P., Pulse & respiration.
- Note signs of CNS depression (signs of overdose so induce vomiting).
- If in hospital, use side rails (safety measures).
- Advice client to report signs of side effects immediately.
- Instruct client to avoid undue exposure to sun.
- If the drug is being used for motion sickness, it should be taken 30 minutes before transporting. Caution the client not to drive a car or operate other machinery.

1- Astemizole:

Trade name: Hismanal

Class.: antihistamine

Action: It has no sedative, antiemetic or anticholenergic effects.

2- Chlorpheniramine maleate:

Trade name: Anaphyl, Ahistone

Class: Antihistamine

Action: sedation is less pronounced.

Dose: syrup each 5ml contains 5mg, 2 teaspoonful 3-4 times daily.

3- Promethazine Hcl:

Trade name: phenergan, prothiazine.

Class:

- It is aphenothiazine derivative. It is a potent antihistamine with prolonged action. It may cause severe drowsiness. It also provides antiemetic effect (it chemoreceptor trigger zone).
- It also has a sedative action, effective in vertigo and decrease vestibular apparatus

Uses:

- Motion sickness.
- Nausea & vomiting due to anesthesia.

Forms:

- Syrup: 5ml contains 5 mg , 25 mg.
- Ampule: 50 mg \2ml.

Dose:

- Antihistamine: 125 mg 4 times daily.
- Sedative: 25 mg – 50 mg.
- Antivertigo: 25 mg 2 times daily.

4- Loratadine

Trade name: lorastine

Class: Antihistamine

Action: Non-sedating antihistamines

Dose:

- Tablet 10 mg
- Syrup each 5ml contains 5mg, 1 teaspoonful 1-2 times daily.

Anti-peptic ulcer drugs

Drugs classification:

1. Antacids
2. Gastric anti secretory drugs
    1. H_2 receptor antagonists (cimetidine, famotidine, nizatidine, ranitidine).
    2. Anti muscarinics
    3. Proton pump inhibitors (omeprazole, lansoprazole)
3. Mucosal protective agents (sucralfate, colloidal bismuth compound, prostaglandine analogues e.g. misoprostol)

A. Antacids

Antacids are over-the-counter (OTC) medications used in combination with other drugs to treat peptic ulcers. Types of antacids include:

- aluminum carbonate gel
- calcium carbonate
- magaldrate (aluminum-magnesium complex)
- magnesium hydroxide and aluminum hydroxide
- simethicone.

Action:

- They are weak bases that neutralizes gastric acidity by reacting with gastric HCL to form salt and water.
- Properties of ideal antacids
    - Immediate and prolonged neutralization of acid without increasing PH above 4.
    - Action should be confined to GIT.
    - No constipating or laxative effects
    - should not be absorbed systemically
    - Palatable to patients

Uses:

- Treatment of hyperacidity. (Heart- burns).
- Peptic ulcer
- Duodenal ulcer.
- Gastro-esophageal reflux.

Contraindications:

- Sodium containing products are contraindicated in C.H.F., hypertension, and other conditions requiring low sodium diet.
- Pregnancy
- Children less than 6 years of age.
- N.B.: chronic use of aluminum containing antacids may contribute to development of Alzheimer's disease.

Nursing considerations:

- It is recommended that most antacids be taken at 3 hours after meals & at bed – time.
- Tablets should be thoroughly chewed before swallowing & followed by a glass of milk or water.
- Shake liquid suspensions thoroughly before pouring the medication.
- Clients' taking aluminum or calcium containing antacids should take 2500-3000 cc of fluids to prevent constipation.
- Advise clients to report persistent diarrhea or constipation physician.

1. Aluminum Hydroxide Gel:

Class.: Antacid

Action: it reacts with intestinal secretions to produce insoluble salts esp. phosphates that are eliminated in feces prevent phosphates urinary stones.

Additional uses: prevention of urinary stones, hypophosphatemia..

Side-effects: constipation, intestinal obstruction, bone pain, muscle weakness.

Drug interaction: reduce bioavailability of some drugs e.g. tetracycline, atropine.

Dose: 500-1800 mg 3-6 times daily after meals, between meals & at bed time.

2. Calcium carbonate:

Class: antacid.

Action: Non-systemic, since calcium carbonate is constipating it is often alternated or mixed with Mg^{+2} salts.

Additional uses: calcuim deficiency .

Side effects: constipation, flatulence, eructation, intestinal obstruction, metabolic alkalosis, hypercalcemia , rebound hyperacidity.

Milk alkali syndrome: occur with concomitant heavy milk drinking and characterized by:

Headache, weakness, anorexia, nausea, vomiting, abdominal pain, constipation, thirst, polyuria.

Dose: 0.5 – 1 g as necessary.

3. Maalox:

Class. : Antacid.

Additional uses: hiatus hernia

Dosage:

- Tablet: antacids Al(OH)3 200 mg + Mg (OH)2 200 mg.
- Suspension: antacids 225 mg of Al(OH)3 /5ml + 200mg of Mg(OH)2.

Dose:

- 2-4 tabs - 20-60 m after meals & at bedtime.
- Suspension: 10- 20 ml aid- 20-60 m after meals & at bedtime.

4. Magnesium Hydroxide (magnesia) :

Class: antacid , laxative

Action:

- Acts as an antacid by neutralizing Hcl, it does not produce alkalosis & has a demulcent effect.
- As a laxative, it increases the bulk of the stools by attracting & holding large amounts of fluids. The increased bulk results in the mechanical stimulation of peristalsis.

Uses:

- Antacid
- Laxative to empty the bowel prior to diagnostic or surgical procedures .

Contraindications: - poor renal function.

Side effects: Diarrhea, abdominal pain, nausea, vomiting, hypermagnesemia .

Gastric anti secretory drugs

1. H_2 receptor antagonists:

Cimetidine:

Trade name: Tagamet

Class: Histamine H_2-receptor blocking agent.

Action: decreases the acidity of the stomach by blocking the action of histamine, which involved in triggering gastric acid secretion. It blocks the action of histamine by competitively occupying the histamine H_2-receptors in the gastric mucosa leading to decrease secretion of HCl.

Uses:

- Short-term (up to 8 wks) & maintenance treatment of duodenal ulcer & treatment of benign gastric ulcer.
- Management of hypersecretion of gastric acid.
- Reflux esophagitis.

Contraindications:

- Children under 16 years.
- Lactation
- Impaired renal & hepatic function.

Side effects:

- Diarrhea, hepatic fibrosis, Hepatitis, Pancreatitis ,
- Hallucinations, Dizziness, Headache, confusion, ataxia, double vision.
- Hypotension, Arrhythmias following I.V. administration. .
- Aplastic anemia, thrombocytopenea.

Dose: 300 mg, 4 times daily with meals & at bedtime.

Nursing considerations:

- Administer oral medication with meals, if I.V. dilute as prescribed.
- Note the general condition of the patient (If take chemotherapy or radiation)
- Note signs of infection.
- For diarrhea, maintain adequate hydration.
- Monitor renal function.
- Be alert for mold swings that may occur.

2. Proton pump inhibitors (omeprazole, lansoprazole)

Proton pump inhibitors disrupt chemical binding in stomach cells to reduce acid production, lessening irritation and allowing peptic ulcers to better heal.

They include:

- Esomeprazole
- Lansoprazole
- Omeprazole
- Pantoprazole
- Rabeprazole.

Omeprazole

Trade name: Losec, Pepticum, Mepral

Uses:

- Gastroesophageal reflux disease
- Zollinger-Ellison syndrome
- Duodenal ulcer
- Gastric ulcer
- Helicobacter pylori infection

Action

- Inhibits activity of acid (proton) pump and binds to hydrogen-potassium adenosine triphosphatase at secretory surface of gastric parietal cells to block formation of gastric acid.

Dose: 20-40 mg daily for 4-8 weeks

Side effects:

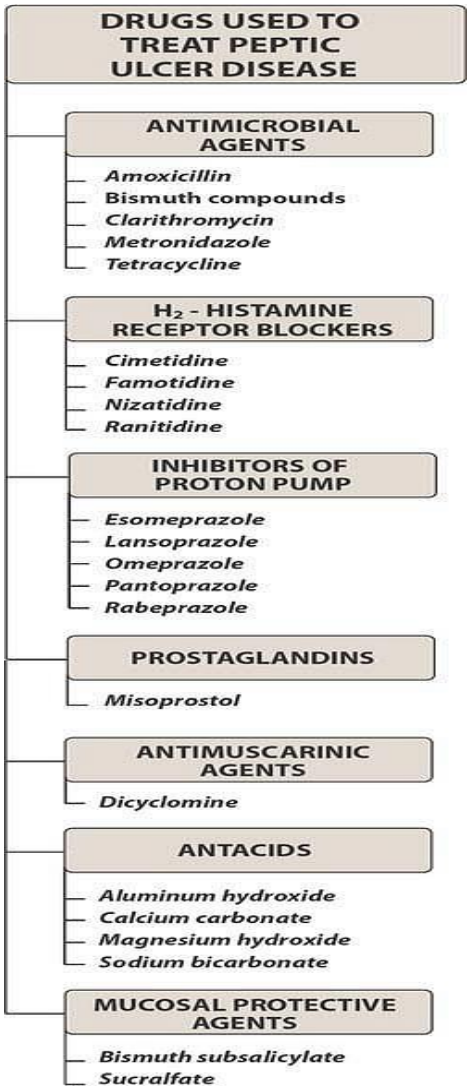
- CNS: headache, dizziness.
- GI: diarrhea, abdominal pain, nausea, vomiting, constipation, flatulence. Musculoskeletal: back pain.
- Respiratory: Cough upper respiratory tract infection. Skin: rash.

Contraindications & cautions

- Contraindicated in patients hypersensitive to drug or its components. Use cautiously in patients with hypokalemia and respiratory alkalosis.

Nursing considerations

1. Dosage adjustments may be necessary in patients with hepatic impairment.
2. Tell patient to swallow tablets or capsules whole and not to open, crush, or chew them.
3. Instruct patient to take drug 30 minutes before meals.
4. Caution patient to avoid hazardous activities if he gets dizzy.



Anti-emetics

Nausea & vomiting can be caused by a variety of conditions such as infections, drugs, motion, organic disease or psychological factors. The underlying cause of the symptoms must be elicited before emesis is corrected. The act of vomiting is complex. The vomiting center in the medulla responds to stimulation from many peripheral areas as well as stimuli from CNS itself, the CTZ in the medulla, the vestibular apparatus of the ear & the cerebral cortex.

The selection of antiemetic depends on the cause of the symptom as well as on the manner in which the vomiting is triggered.

Many drugs used for other conditions such as antihistamine, phenothiazines & barbiturates have antiemetic properties & can be so used.

Drug interaction:

Because of their antiemetic and antinauseant action, the antiemetics may mask overdose caused by other drugs.

Nursing considerations:

1. Take a complete history, if it is unusual occurrence or if it is a recurring phenomenon.
2. Assess for other untoward symptoms as increased intracranial pressure or intestinal obstruction (antiemetic may mask signs of underlying pathology)
3. Caution the client that drug tends to cause drowsiness & dizziness, advise him\her to avoid hazardous tasks.

Metoclopromide Hcl:

Trade name: Pramin

Class: Antiemetic

Action: It is dopamine receptor antagonist acts both centrally & peripherally, centrally due to the effect in the CTZ (inhibition), peripherally it stimulate the motility of the upper GIT without affecting gastric & biliary or pancreatic secretions. It relaxes the pyloric sphincter & increases the peristalsis of the duodenum resulting in accelerated gastric emptying & intestinal transit.

Indications:

a) Digestive disorders leading to relief GIT pain, dyspepsia & regurgitation in peptic ulcer, reflux esophagitis & post anesthetic vomiting.
b) Nausea & vomiting as in chemotherapy.
c) Facilitate diagnostic procedure e.g. barium meal.

Side effects:

GI disturbances, transient hypertension, supraventricular tachycardia, dizziness & extrapyramidal effect “convulsion”.

Forms:

- Ampule 50 mg\10ml
- Tablet 10mg
- Syrup 5mg\5ml
- Suppository. 20mg (adult) , 5mg (children).

Dose:

- 10 mg, 30 minutes before meal & at bed time .
- For chemotherapy 2 mg /kg, 30 minutes before chemotherapy.

Contraindications: Seizure (epilepsy) , Pheochromocytoma ,intestinal obstruction.

Nursing considerations:

1- Don’t give pramin to patients with epilepsy, pheochromocytomes or patients with intestinal obstruction.
2- Administer oral medication 30 minutes before meal & at bed time .
3- Administer I.V. injection slowly over 1-2 minutes.
4- Be aware of the extrapyramidal symptoms specially in children .

Drugs affecting The Central Nervous system

Sedative &Hypnotics

A. Barbiturates:

Phenobarbital:

Trade name: Luminal.

Class: sedative- anticonvulsant- barbiturate.

Action:

- Long-acting barbiturate- act as a sedative- hypnotic and anticonvulsant by producing CNS depression.
- It depresses the action of excitatory neurotransmitters

Uses:

- Anxiety
- Pre-anesthetic medication.
- Sedation
- Hypnotic
- Epilepsy (anticonvulsant).
- Hyperbilirubinemia and kernicterus in neonate (due to the ability of barbiturates to stimulate hepatic glucuronyl transferase.

Contraindication: Hypersensitivity, respiratory depression, bronchial asthma, shock, advanced renal and hepatic disease.

Side effects:

Headache, fever, megaloblastic anemia, dizziness, hypotension, nausea, vomiting epigastric pain.

Forms and Dose:

- Tablets 100mg, ampules 130mg in 1 cc Sedation: 30-120 mg daily in 2-3 divided doses. For adults: Hypnotic 100-320 mg at bed-time.
- Anticonvulsant: I.V. 100-320 mg, repeated as necessary.
- Preoperative sedation: I.M. only 130-200 mg – 60-90 minutes before surgery.

Over dose: Tachycardia, hypothermia, coma, respiratory depression, absent reflexes & circulatory collapse respiratory relaxation unseals vascular collapse and renal failure.

Treatment of overdose toxicity:

1- Maintain & assist respiration as indicated.
2- Support circulation by vasopressor & I.V. fluids as required.
3- Aspirate stomach content; take care to avoid pulmonary aspiration.
4- Diuretics may be given as ordered.
5- Intake & output measurement.
6- Dialysis if indicated.

Nursing considerations:

1- If given I.V, closely monitor the rate of flow because rapid administration may lead to respiratory depression.
2- Monitor the site of I.V. for soft of extravasation which cause sever pain, nerve damage & necrosis.
3- Instruct the client not to drive a car or operate other hazardous machinery after taking the medication .
4- Take the medication only as prescribed.
5- If used for hypnotic effect, give 30 min. before bedtime.
6- Teach patient about signs and symptoms of toxicity, and instruct patient to report them to treating physician.
7- If taken for 8 weeks or more, instruct patient not to stop it suddenly to avoid withdrawal symptoms as convulsion.
8- Keep the drug out of reach of the children.

Secobarbital:

Trade name: seconal

Class: sedative – hypnotic , barbiturate type.

Action: short acting barbiturate, (as luminal) .

Uses: short- term of insomnia.

- Sedative to relief anxiety.
- Preoperative sedation.
- Some times parenterally as anticonvulsant.

Dose: tab. 100 mg at bedtime.

B. Nonbarbiturate sedative- hypnotics:

Paraldehyde:

Trade name: paral

Class. : nonbarbiturate sedative- hypnotic.

Action: as luminal, is a bitter fasting liquid & has a strong unpleasant odor, in usual doses it has no or little effect on respiration & blood pressure.

Uses:

- Sedative & hypnotic (produce hypnosis in about 15 min., which lasts 4-8 hr).
- It is useful for patients with hepatic or renal disease (mainly eliminated by the lung).
- Emergency treatment of seizures.
- Delirium tremors.
- For patients undergoing withdrawal from alcohol.

Contraindication:

- Gastroenteritis, peptic ulcer
- Broncho-pulmonary disease.

Side effects: nausea, vomiting, skin rash, pain at injection site, bradycardia.

C. Anti-anxiety Agents

Diazepam, chlordiazepoxide, lorazepam, alprazolam

Diazepam

Trade name: Valium, assival

Class: antianxiety agent, benzodiazepine.

Action:

- The anxiolytic effect is believed to be mediated through the action of benzodiazepine to increase the inhibitory action of GABA “Gamma aminobutyric acid” inhibit CNS neurotransmitter.
- The drug is metabolized in the liver & excreted through urine.

Indications:

1- Anxiety & tension.

2- hypnosis

3- Alcohol withdrawal.

4- Skeletal muscle relaxant.
5- Anticonvulsive.
6- Preoperatively.
7- Before gastrescopy or esophagoscopy.
8- Treatment of status epilepticus .
9- Relief of facial muscle spasm.

Contraindications:

- Hypersensitivity.
- Acute narrow angle glaucoma.
- Pregnancy.
- Shock, coma.
- Alcoholic intoxication (to avoid respiratory of depression).

Side effects:

Drowsiness, fatigue, ataxia, hypotension, visual disturbances, headache, phlebites at injection site.

Dosage and forms:

- Ampules of 2 ml containing 10 mg. Tablets 2 mg, 5 mg or 10 mg.
- I.V. or I. M. 2-20mg deepening on the indication. Tablets 2-10 mg 2-4 times daily.

Nursing Considerations:

- Stress that drug may reduce patients ability to handle dangerous equipment.
- Avoid alcohol ingestion.
- Don't stop taking the medication suddenly, withdraw drug gradually.
- Monitor B.P. before & after administration.

Chlordiazepoxide:

Trade names: librium, lipoxide.

Class: antianxiety agent.

Action: Depress subcortical levels of CNS, particularly the limbic system. (cortex). It ihhibits sensory input in CNS.

Uses:

- Anxiety, Acute withdrawal symptoms of chronic alcoholic, Preoperatively.
- Sedative, hypnotic.

Side effects:

Drowsiness, fatigue, ataxia, dizziness jaundice, hepatic dysfunction.

Dose: P.O. 5-25 mg tid or qid.

Contraindications: coma, shock, lactation, pregnancy.

Nursing considerations:

- Monitor vital signs.
- Don't mix it with any medication.
- Administer oral medications with food.
- When given I.V., should be administered slowly.
- When given I.M or I.V., dilute with water, don't shake the vial to mix it.
- Inject deep into large muscle.

Antipsychotic Drugs

Chlorpromazine: (largactil)

Class: Antipsychotic, phenothiazine.

Action: Act by blocking dopamine receptors. It has significant antiemetic effect, hypotensive, sedative & anticholenergic effect.

Uses:

- Acute & chronic psychodsis (such as schizophrenia, mania & manic depression.
- Preanasthetic .
- Intractable hiccups.
- Nausea & vomiting.

Contraindication:

- Severe depression, coma.
- Bone marrow depression.
- Patients with history of seizures & on anticonvulsant therapy.
- Hepatic & renal diseases.
- Prostatic hypertrophy.
- Dehydration
- Glaucoma, measles.

Side effects:

Depression, dizziness, seizures, gynecomastia, orthostatic hypotension, bronchospasm , larlynyospasm tardive dyskinesia, photosensitivity, leukopnea, aplastic anemia, and dry mouth.

Dose:

- Tablets 10-25 mg 2-4 times a day.
- I.M. 25-50 mg repeated after 1 hour if needed.

Nursing considerations:

- Shouldn't be used to treat nausea & vomiting in children less than 6 months of age.
- Should avoid getting solution on hands or clothing (it will cause dermatitis).
- Solutions with marked discoloration should be discarded.
- Note any history of seizures.
- Take liver & kidney function test periodically.
- Document & rotate injection sites.
- Report side effects immediately.
- Determine age of male patients & assess for prostatic hypertrophy.

Thioridazine: (mellaril**)**

Classification: Antipsychotic – phenothiazine.

Uses: Schizophrenia, depression, anxiety, alcohol withdrawal, intractable pain sleep disturbances.

Action: probably related to drug's antidopaminergic effects.

Dose: P.O.: 25 mg tid.

Side effects: Headache, tardive dyskinesia, blurring of vision, dry mouth, skin rashes, and photosensitivity.

Contraindications:

- Comatose patients.
- Parkinson's disease.
- Sever hypertension or hypotension.

Nursing considerations:

- Monitor vital signs before beginning therapy & at regular intervals.
- Administer medication with meals to minimize GI upset.

Haloperidol: (Haldol)

Class.: antipsychotic.

Action:

- Block dopamine receptors (sedation) .
- Alpha-adrenergic blockade (decrease the release of growth hormone & the release of prolactin).
- Anticholinergic effect (sedation & orthostatic hypotension).

Uses:

- Psychotic disorders such as mania
- Drug induced psychoses.
- Schizophrenia.
- Aggressive & agitated patients. (mental retatdal)
- Short-term of hyperactive children.
- Treating symptoms of dementia in elderly peoples.

Dose:

- P.O. 3-5 mg bid –tid.
- I.M. 2-5 mg q 4-8 hrs.
- In sever situations: I.V. 2-25 mg every 30 minutes.

Side effects:

Headache, tardive dyskinesia, hypotension, dry mouth photosensitivity, skin rashes, nausea, vomiting, and constipation.

Contraindications:

- Pregnancy, Lactation, Shock, Bone marrow depression, Hepatic & renal insufficiency.

Nursing considerations:

- Monitor vital signs before & during therapy.
- Administer oral medication with food.
- Administer IM deep into large muscle.
- Don’t mix it with any other drugs.
- If concentrated drug is spilled on skin, wash it immediately to prevent dermatitis.

Lithium carbonate:

Trade name: lithium.

Class: antipsychotic , anti-manic.

Action:

Action is not known. Theories trying to explain the action of this drug include effectiveness to an alteration in Na ion metabolism within nerve & muscle cells "K^+ ion & ATP ase" lead to decrease in catecholamine neurotransmitter levels results in hyperactivity.

Uses:

- Control of manic and hypomanic episodes in manic depression patients.
- Prophylactic of bipolar depression.
- Excited phase in schizo-affective disorders.

Dose: P.O. 600 mg tid or qid .

Side effects:

- Drowsiness, dizziness, hand tremors, lethargy.
- Hypothyrsidism, ECG changes,
- anorexia, dry mouth, nausea, vomiting, polyuria,
- leukocytosis slurred speech .

In case of toxication (blood level over 2.0 mmol/L): hyper-reflexia and hyperextension of limbs, convulsions, toxic psychosis, syncope, oliguria, circulatory failure, and coma.

Contraindications:

- Cardiovascular, renal diseases.
- Brain damage.
- Pregnancy & lactation.
- Dehydration.
- Patients receiving diuretics.
- Sodium depletion.

Nursing considerations:

- Monitor serum level of lithium every 1-2 weeks to prevent toxicity (normal level is 0.4 - 1.0 mmol/L and toxic level is above 1.5 mmol/L).
- Monitor for pulse irregularities & changes in B.P.
- Provide diet adequate in sodium.

- Monitor for signs & symptoms of toxicity.
- Avoid factors that enhance toxicity: dehydration, renal failure, infection, co-administration of diuretics, and sodium depletion (may occur with diuretics).
- Maintain adequate fluid and sodium levels.
- Withdrawal (stopping) drug should be gradual (over weeks)

Antidepressants

Depression is a psychiatric syndrome consisting of dejected mood, psychomotor retardation, insomnia and weight loss. It is alteration of mood characterized by sadness, worry, anxiety, loses of weight, libido and enthusiasm.

1. Monoamine oxidase (MAO) inhibitors :

Action:

- MAO is one of the enzymes that break down biogenic amines (Norepinephrine, epinephrine & serotonin).
- These drugs prevents this process therefore, amines accumulate in the presynaptic granules and increase the concentration of neurotransmitters results in nerve stimulation and antidepressant effect.
- They are highly toxic, prescribed only if triecyclic compounds are ineffective. They also may interfere with detoxification mechanisms, which occur in the liver.

Uses: individualized.

Contraindications:

- Hypersensitivity.
- History of liver disease.
- Pheochrcmocytoma and Glaucoma.
- Impaired renal function.
- Hypertension.
- Epilepsy.
- Hyperthyroidism.

Side effects:

- Headache, dizziness, ataxia, euphoria, agitation, hyperreflexia.
- Urinary retention, jaundice, skin rashes, glaucoma.
- Constipation, diarrhea, nausea, CHF and hypertension.

Nursing considerations:

- Teach patient to avoid food containing tyramine (dairy products, meat, fish, liver, some fruits (such as avocado, fig, and banana), chocolate, and yeast extracts. If patient ate these foods, the tyramine will break down in the GI tract (in the presence of MAO inhibitors) and release voasopressers will lead to hypertensive crisis (severe elevation in blood pressure).
- Teach patient to avoid alcohol and other sleep inducing drugs.

Phenelazine sulfate:

Trade name: Nardil.

Class.: MAO inhibitor , antidepressant.

Uses; Major depression with or without melancholia.

Dose: P.O. 15 mg tid.

2. Tricyclic antidepressants:

Clomipramine Hydrochloride:

Trade name: anafranil.

Class.: antidepressant, tricyclic.

Action: prevent the presynaptic re-uptake of the neurotransmitters (norepinephrine and serotonine) which will increase their concentration at the synaptic area and alleviate depression.

Dose: P.O. 75-150 mg /day in 1-3 divided doses.

Uses:

- Treatment of obsessive- compulsive neurosis.
- Panic disorders
- Phobic disorders.

Contraindications: pregnancy, lactation, shock, bone marrow depression.

Side effects:

Hyperthermia, seizures, anemia, muscle weakness, drowsiness, ataxia, blurring of vision orthoslatic hypotension, dry mouth, constipation.

Nursing considerations:

- Monitor vital signs before & during therapy.
- Take with foods to decrease GI upset.

Imipramine Hydrochloride:

Trade name: Tofranil.

Class.: antidepressant, tricyclic.

Action; as anafranil.

Uses:

- Relief symptoms of depression.
- Enuressis in children.

Dose:

- For treatment of depression P.O: 50 mg bid or tid .
- For treatment of children enureses (6 years or older): 25 mg\day 1 hr before bedtime.

Oral Anti-diabetic

- Diabetes mellitus is a clinical syndrome characterized by hyperglycemia due to absolute or relative deficiency of insulin. It is classified as insulin dependent (type 1 or juvenile-onset) & noninsulin dependent (type II or maturity –onset) . It can be treated successfully by the administration of insulin isolated from the pancreas of cattle or hogs or of human insulin made either semi-synthetically or derived from recombinant DNA technology .
- The structure of insulin from pork - sources more closely resembles human insulin than that from beef sources.
- Several oral anti-diabetic agents are available for patients with noninsulin dependent diabetes.
- Oral hypoglycemic agents are classified as either first or second generation.
- Generation refers to structural changes in the basic molecule.
- Second–generation oral hypoglycemic agents are more lipophilic & have greater hypoglycemic potency (200 times) than first generation.

Oral hypoglycemic classification:

1. **Sulfonylureas**
    a. **First – generation sulfonylureas compounds such as:**
        - Tolbutamide (orinase).
        - Chlorpromide (diabenase).
    b. **Second-generation sulfonylureas compounds such as:**
        - Glyburide (Micronase).
        - Glibenclamide (Daonil).
        - Gliclazide
2. **Biguanides**
        - metformin

Action of oral anti-diabetic agents:

1. Stimulate insulin secretion by pancreatic β cells.
2. Potentiates the action of insulin on its target tissues.
3. Decrease serum glucagon level

Indication:

- Non- insulin dependent diabetes mellitus (NIDDM) (type II).
- Patients should be subjected to a 7 day therapeutic trial.

Contraindications:

- Type I of D.M.
- Renal & liver disease.

Side effects:

- Hypoglycemia (most common) .
- Nausea, vomiting, heartburn, diarrhea
- Headache, dizziness, general weakness.
- Pancytopenia, hemolytic anemia.
- Chronic use increases risk of cardiovascular mortality.

Nursing considerations:

- Explain the necessity for close regular medical supervision.
- Explain to patient how to test the urine for sugar & acetone.
- Explain the use & care of equipment & the storage of medication.
- Explain the importance of exercise & adhering to the prescribed diet.
- Explain the importance of carrying candy or sugar at all times to counteract hypoglycemia should it occur.
- Drugs may be taken with food to minimize GI upset.
- Stop the medication if signs of side-effects or ketoacidosis appear.

1. Chlorpromide:

Trade name: Diabenase.

Class: first generation sulfonylurea.

Dose:

- Initial 250 mg daily as a single or divided doses.
- Maintenance 100-250 mg daily as a single or divided doses.

Doses: More than 750 mg are not recommended.

2. Glyburide:

Trade name : Micronase.

Class: Second-generation sulfonylurea .

Dose: Initial 2.5-5 mg daily given with breakfast (or the first main meal) then increased by 2.5 mg weekly to achieve the desired response.

3- Tolbutamide :

Trade name : Orinase.

Class: First-generation sulfomylurea.

Dose:

- Initial 0.5 – 2 g daily , so adjust the dose depending on response
- (Maintenance 0.25 –3 g daily) (not exceed 3g) .

4- Glibenclamide : (daonil.**)**

Class: second generation snlfonylurea .

Dose: ½ -1 tablet (5mg) daily , increased by 2.5 – 5 mg weekly to achieve the desired response.

Insulin:

Patients with type 1 diabetes require an external source of insulin to control blood glucose levels. Insulin may also be given to patients with type 2 diabetes in certain situations.

Two main hormones are secreted from the pancreas:

1. Insulin which is secreted by β -cells of islets of langerhans & stored in the pancreas (β cells) as a large protein known as **proinsulin** .
2. Glucagon which is thought to oppose the action of insulin. It is secreted by the α cells of islets of langerhans, it converts glycogen to glucose & elevates blood glucose level.

Types of insulin include:

- Rapid-acting, such as lispro
- Short-acting, such as regular insulin
- Intermediate-acting, such as nph
- Long-acting, such as glargine.

Action:

1- Facilitates the transport of glucose into cardiac & skeletal muscles & adipose tissue.

2- Increases synthesis of glycogen in the liver.

3- Stimulates protein synthesis & lipogenesis.

4- Inhibits lipolysis & release of free fatty acids from fat cells.

5- Causes intracellular shifts of potassium.

Note:

- Insulin is a protein so it is destroyed in the GIT thus it must be administered parenterally. It is metabolized mainly in the liver.

Uses:

- Replacement therapy in type I diabetes.
- Indicated in type II diabetes when other measures have failed or with surgery , trauma, infection, fever , endocrine dysfunction , pregnancy , gangrene , kidneys or liver disease .
- Regular insulin is used in I.V. hyperalimentation.
- Regular insulin is used in I.V. dextrose to treat sever hyperkalemia.

Contraindications:

Hypersensitivity to insulin.

Side effects:

1- Hypoglycemia due to overdose , decreased food intake or hard exercise, " Hunger, weakness, fatigue, nervousness pallor or flushing , profuse sweating , headache , numbness of mouth , tingling in the fingers, blurred vision, hypothermia & loss of consciousness. "Sever prolonged hypoglycemia may cause brain damage."

2- Allergic urticaria , lymphadenopathy. " Use human Insulin product".

3- At the site of injection: developing of swelling, itching, atrophy or hypertrophy of S.C. fat tissue so rotate site of injection to minimize the problem.

4- Insulin resistance caused by obesity, infection, trauma ,surgery ….etc.

5- Hyperglycemic rebound (somogyi effect) in patients who receive chronic overdose.

6- Diabetic coma is usually precipitated by the patient's failure to take insulin.

Treatment of diabetic coma:

- 20 – 30 units of insulin, then 20 units every 30 minutes.
- To avoid hypoglycemia give 1 g dextrose for each unit of insulin is administered with supplemental electrolytes (K+) & fluids .
- Monitor vital signs.
- Urine samples for analysis.

Treatment of hypoglycemia:

- Mild hypoglycemia: relieved by oral administration of orange juice.
- In comatose patients: administer 10 –30 ml of 50% dextrose solution I.V.

Dose:

- Usually administered S.C.
- Regular insulin is the ONLY preparation that may be administered. I.V This route should be used only for patients with sever ketoacidosis or diabetic coma.
- Always expressed in units.
- Dosage is individualized, it is established & monitored by blood glucose, urine glucose & acetone test.

Insulin antagonists:

- Growth hormone elevates glucose level & decreases glycogen synthesis.
- Glucocorticoids enhance conversion of protein to glucose.
- Adrenaline decreases insulin release & enhance glycogenolysis.
- Glucagon.

Nursing considerations:

In Add. to oral hypoglycemic

1. Read the product information & any important notes inserted into the package.
2. Refrigerate stock supply of insulin but avoid freezing.
3. Follow the guidelines with respect to mixing the various types of insulin.
4. Invert the vial several times to mix before the material is withdrawn "avoid vigorous shaking".
5. Assist patient for self-administration of insulin.
6. Rotate the sites of S.C. injections to prevent the problem of hypertrophy or atrophy at injection site.
7. Allow insulin to remain at room temperature 1 hour before administration.
8. Apply pressure for 1 minute, don't massage since it may interfere with rate of absorption.
9. If breakfast must be delayed, delay the administration of morning dose of insulin.
10. Obtain a thorough nursing history from the client / family.
11. If the client has symptoms of hyperglycemia reaction:
    - Have regular insulin available for administration.
    - Monitor client closely after administration.
    - Check blood glucose, urine glucose, and acetone.

12. Check for early symptoms of hypoglycemia.
13. Provide the client & family with a printed chart explaining symptoms of hypoglycemia , hyperglycemia & instructions concerning what to do for each.
14. Instruct client that blurring of vision will subside within 6-8 weeks.
15. Advise client to check vials of insulin carefully before each dose. Regular insulin should be clear, where as other forms may be cloudy.

1. Human Insulin:

Class: Human insulin from semisynthetic or recombinant DNA sources.

Action:

Derived from recombinant DNA technology utilizes genetically modified E.Coli . These organisms synthesize each chain of insulin into the same aminoacid sequence as human insulin. The chains are then combined & purified to produce human insulin see information for insulin.

2. Insulin Injection (Regular, crystalline Zinc insulin)

Class: Rapid- acting insulin.

Kinetics:

Onset ½ -1 hr (S.C).

Uses: suitable for treatment of diabetic coma, acidosis (diabetic) or other emergency situations.

Dose: individualized, initial 5-10 units 15-30 minutes before meals & at bedtime.

Diabetic acidosis 0.1 unit / kg given by continuous I.V. infusion.

3. Isophane Insulin Injection (NPH) :

Class: Intermediate – acting insulin.

Kinetics: onset 3-4 hr, duration 18-28 hr. Peak 6-12 hr.

Dose: S.C. Individualized , initial 7-26 units as a single dose 30-60 minutes before breakfast .

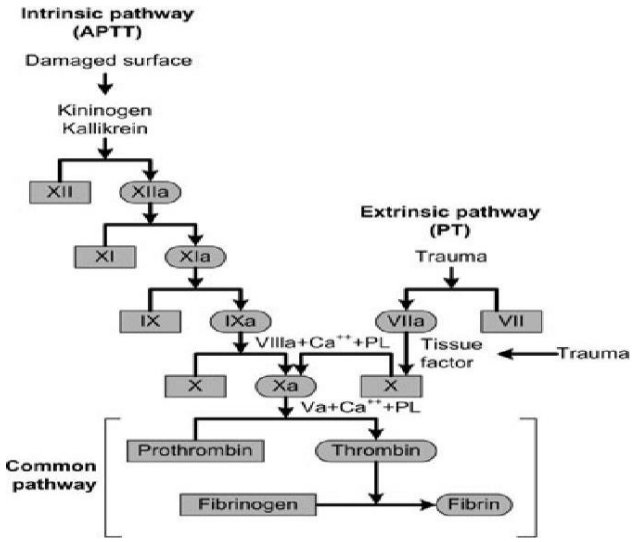
4- Protamine Zinc Insulin (PZI): Class: long –acting insulin

Kinetics: onset 4-6 hr , peak 14-24 hr , duration 36 hr.

Dose : see NPH.

Anticoagulants & Hemostasis

- Hemostasis is the spontaneous arrest of bleeding from a damaged vessel.
- Anticoagulants are a class of drugs that work to prevent the coagulation (clotting) of blood while, several factors participate in blood clotting manufactured by the liver as vitamin K. (Liver disease affect blood clotting)
- Many diseases lead to defect of coagulation (hemophilia, C.V. diseases).



Anticoagulants:

1. Warfarin sodium: coumadin

Class: anticoagulant.

Action: prevent the formation of factors II, VII, IX and X in the liver.

Uses:

- Prophylaxis and treatment of deep venous thrombosis.
- Thromboembolison
- Thrombophlebitis.
- Prophylaxis from myocardial infarction.

Contraindications:

- Hemorrhagic tendencies
- Blood disorders.
- Ulcerative lesion of GIT.
- Impaired renal and hepatic function.

- Severe hypertension.
- Thrombocytopnia.

Drug interaction:

- Antacids: decrease effect of anticoagulants (by decreasing its absorption)
- Salicylate: effect of anticoagulants.

Side effects: hemorrhagic accidents.

Antidote: vitamin K.

Nursing considerations:

- Daily monitoring of prothrombin time is recommended.
- Instruct clients to take the drug before meal.
- Remind clients to wear identification band that states that they are on anticoagulant therapy.
- Advise client to avoid activities that may cause injury.
- Vitamin K should be available
- Food rich in vitamin K should be avoided.

2. Heparin:

Class: anticoagulant

- Naturally occurring substance isolated from porcine intestinal mucosa or bovine lung tissue.
- Must be given parentally.
- Doesn't interfere with wound healing.

Action: Potentiate the inhibitory action of antithrombin III on various coagulation factors.

- Inactivate thrombin and prevent the conversion of fibrinogen to fibrin.

Uses:

- To prevent extension of clots.
- To prevent thrombi and emboli from recurring.
- Prophylactic from thromboembolic diseases.
- After some types of surgery (cardiac, orthopedic & vascular).
- Prevent clotting during hemodialysis.
- Treatment of DIC (disseminated intravasculor coagulation)
- Coronary occlusion after MI.

Contraindications:

- Blood disorders with bleeding tendencies (hemophilia).
- Suspected intracranial hemorrhage.
- Open wounds.
- During surgery of the eyes, brain and spinal cord.
- Menstruation and abortion.
- Any drugs or conditions affecting blood coagulation.

Side effects: Hemorrhage.

Overdose: Nose bleeding, hematouria, petechiae, tarry stool.

Antidote: protamine sulfate.

Dose: IV or S.C measured in units according to bleeding & clotting time.

Nursing considerations:

- Should not be administered IM, administer by deep sc to minimize local irritation and to prolong the action of drug.
- Don't massage before and after injection.
- Change site of administration to avoid ecchymosis.
- Instruct and stress the importance of reporting any signs of active bleeding.
- Use electric razor for shaving, soft bristle tooth brush to decrease gum irritation.
- Patient should be hospitalized for IV therapy.
- Clotting time, PTT should be done before the start of therapy each dose of drug then daily.

3. Enoxaparin sodium

Trade name: Clexane , Lovenox 3

Class*:* Anticoagulant**,** low-molecularweight heparin.

- Pregnancy risk category B

Action

- Inhibits thrombus and clot formation by blocking factor Xa and factor IIa. This inhibition accelerates formation of antithrombin III-thrombin complex (a coagulation inhibitor), thereby deactivating thrombin and preventing conversion of fibrinogen to fibrin.

Indications and dosages

- Patients at risk for thromboembolic complications (pulmonary embolism and deep-vein thrombosis) due to severely restricted mobility during acute illness or post a major surgery.
- Prevent blood clots.
- In some cases of pregnancy (sometime, a blood clot is formed in the umbilical cord due to antiphospholipid syndrome, which leads to miscourage).
- **Adults:** 40 mg subcutaneously daily

Contraindications

- Hypersensitivity to clexane or heparin.
- Thrombocytopenia
- Active major bleeding
- Major blood disorders
- Certain types of stroke
- Do not give Clexane to a child.

Use cautiously in:

- Severe hepatic or renal disease, retinopathy (hypertensive or diabetic), uncontrolled hypertension, hemorrhagic stroke, bacterial endocarditis, GI bleeding or other bleeding disorders
- Recent history of ulcer disease
- Pregnant or breastfeeding patients
- Children.

Adverse reactions

- **CNS:** dizziness, headache, insomnia, confusion, **cerebrovascular accident**
- **CV:** edema, chest pain, **atrial fibrillation, heart failure**
- **GI:** nausea, vomiting, constipation
- **GU:** urinary retention
- **Hematologic:** anemia, **bleeding tendency, thrombocytopenia, hemorrhage**
- **Skin:** bruising, pruritus, rash, urticaria, irritation, or erythema at injection site

Nursing considerations:

- Monitor CBC and platelet counts. Watch for signs and symptoms of bleeding or bruising.
- Instruct patient to promptly report irregular heartbeat, unusual bleeding or bruising, rash.
- Teach patient safety measures to avoid bruising or bleeding.

Phytonadione (vitamin K)

Class: Vitamin

Action: Promotes hepatic synthesis of active prothrombin.

Indications

- Hypoprothrombinemia caused by anticoagulant therapy
- Hypoprothrombinemia secondary to other causes
- Prevention and treatment of hemorrhagic disease of newborn
- Neonates: For prevention, 0.5 to 1 mg I.M. as a single dose within 1 hour of birth.

Contraindications and precautions

- Contraindicated in hypersensitivity to drug
- Use cautiously in pregnant or breastfeeding patients, and children.

Adverse reactions

- Hyperbilirubinemia (in infants); with parenteral administration—pain, swelling,
- Tenderness at injection site; itchy rash after repeated injections; transient flushing sensations; **anaphylactic reaction.**

Thrombolytic Agents

- Agents used to promote the dissolution(lysis) of the insoluble fibrin trapped in intravascular emboli and thrombi.
- The most serious complication is hemorrhage.
- Heparin therapy usually follows treatment with these agents.

1. Streptokinase:

Class: Thrombolytic agent.

Action:

Acts with plasminogen to produce an activator complex, which enhance the conversion of plasminogen to plasmin, which breaks down fibrinogen, fibrin clot & other plasma proteins.

Uses: Deep venous thrombosis (DVT)

Myocardial infarction (MI)

To clear occluded arteriovenous or IV canula.

Contraindication: Hemorrhage.

Side effects: Bleeding, nausea, and headache.

2. Urokinase:

As streptokinase,

Note: Thrombolytic agents should be given within 6 hours of clot formation.

Table XI-1-3. Antidotes

Antidote	Type of Poisoning
Acetylcysteine	Acetaminophen
Atropine + pralidoxime (for irreversible AChE inhibitors)	AChE inhibitors—physostigmine, neostigmine, and pyridostigmine; organophosphates, including insecticides, such as malathion and parathion
Deferoxamine	Iron and iron salts
Digoxin immune F(ab)	Digoxin
Dimercaprol (BAL)	Arsenic, gold, mercury, lead; oral succimer for milder lead and mercury toxicity
EDTA	Backup in lead poisoning, then for rarer toxicities (Cd, Cr, Co, Mn, Zn)
Esmolol	Theophylline, beta agonists
Ethanol, fomepizole	Methanol or ethylene glycol
Flumazenil	Benzodiazepines, zolpidem, zaleplon
Naloxone	Opioid analgesics
Oxygen	Carbon monoxide
Penicillamine	Copper (e.g., Wilson's disease), iron, lead, mercury
Physostigmine	Anticholinergics: atropine, antihistamine, antiparkinsonian—*not* tricyclics
Protamine	Heparins
Vitamin K	Warfarin and coumarin anticoagulants
Activated charcoal	Nonspecific: all oral poisonings except Fe, CN, Li, solvents, mineral acids, or corrosives

References:

- Durkin, M. T. (2020). Nursing Pharmacology Made Incredibly Easy 3rd ed.
- Whalen, K. (2018). Lippincott® illustrated reviews: pharmacology. Wolters kluwer india Pvt Ltd.
- Barber, P., & Robertson, D. (2020). Essentials of Pharmacology for Nurses, 4e. McGraw-Hill Education (UK).